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7

Hydrology and Run off Computations for the Design of Hydraulic Structures across Rivers and Streams

INTRODUCTION TO HYDROLOGY

7.1. Definition, History and Importance of Hydrology

Hydrology may be defined as the science that deals with the depletion and replenishment of our water resources. It deals with surface as well as ground waters, in as far as their occurrence, circulation, distribution, chemical and physical properties, reactions to environment and living beings, etc., are concerned. In fact, hydrology embraces the entire domain of water, its history and quantitative use on the earth.

Hydrology is comparatively a new science. Even in the pre-historic periods, it seems to have remained under-developed, although many other sciences, such as mathematics, astronomy etc., were fully developed. This science, perhaps, found its origin in the year 1930, and since that time, it has progressed considerably, giving us more and more confidence in predicting hydrological events, such as rainfall, runoff, ground water accretion, etc.

The basic knowledge of this science is a must for every civil engineer, particularly the one who is engaged in the design, planning or construction of irrigation structures, bridges and highway culverts, or flood control works, etc. Some of the various questions, which this science will be called upon to answer, while designing the above engineering feats, are :

(*i*) What maximum flows, after different years are expected at a spillway, or a highway culvert in a city drainage system ?

(*ii*) What minimum reservoir capacity will be sufficient to assure adequate water for irrigation or municipal water supply, during droughts ?

(*iii*) How much quantity of water will become available at a reservoir site, and when will it become available ? In other words, what will be the available possible supplies at the reservoir site ?

(*iv*) What effects will be there on the river water levels after the construction of proposed reservoirs, levees, or any other controlling structures ?

Generally, specialist hydrologists are rarely employed for answering such questions; and the civil engineers responsible for the planning, design and construction of hydraulic structures, are themselves required to answer them ; thereby making it imperative upon them to gain a thorough knowledge of Hydrology. Only then, they will be able to design efficiently and economically functioning hydraulic structures.

In the following pages of this chapter, we shall discuss the various hydrologic principles involved in this science, as well as, its applications to engineering problems.

7.2. The Hydrologic Cycle

Most of the earth's water sources, such as, rivers, lakes, oceans and underground sources, get their supplies from the rains, while the rain water in itself, is the evaporation

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from these sources. Water is lost to the atmosphere as vapour from the earth, which is then precipitated back in the form of rain, snow, hail, dew, sleet or frost, etc. This evaporation and precipitation continues for ever, and thereby, a balance is maintained between the two. This process is known as *hydrologic cycle*. It can be represented graphically as shown in Fig. 7.1.

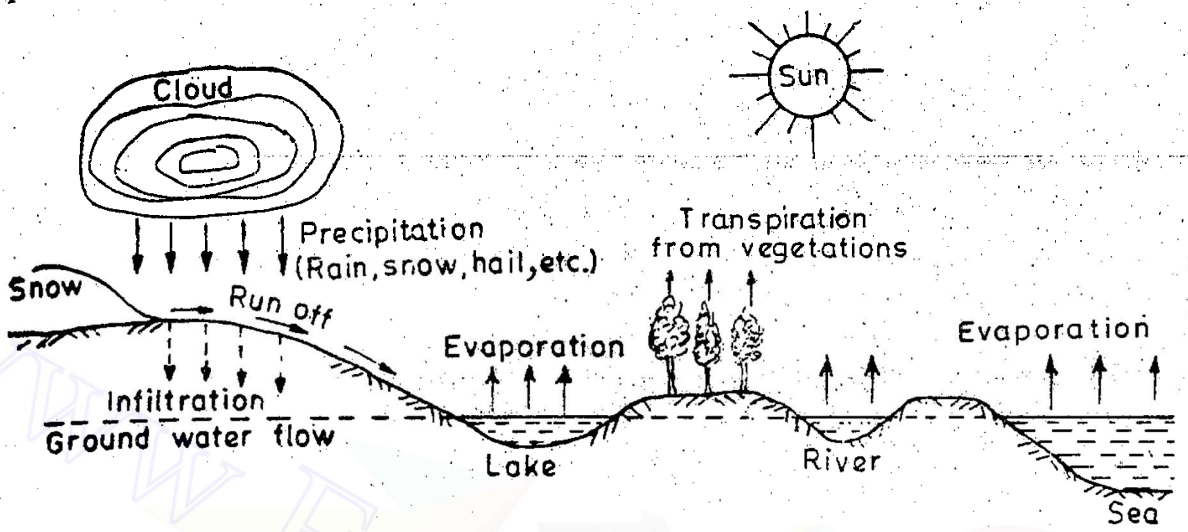

Fig. 7.1. Hydrologic cycle.

Sun and Coriolis force play important parts in completion of hydrologic cycle. *Sun* evaporates water, and *Coriolis force,* by causing and controlling winds, circulates the evaporated water vapour, and thus, helping in its reprecipitation at different places.

WEATHER AND ITS PRECIPITATION POTENTIAL

7.3. Definition of Precipitation

Water evaporates from exposed water surfaces like streams, rivers, oceans, ponds, and also from the lands and plants, in the form of water vapour. These water vapour get collected in the atmosphere and behave like a gas. Under a normal range of temperature and pressure, the water vapour obeys the various gas laws (*i.e.* Boyle's law, Charle's law, etc.). As the evaporation continues, the amount of atmospheric vapour goes on increasing. But since a space (atmosphere) can hold only a certain fixed amount of water vapour in the presence of a solid or a liquid surface, a state is reached when any further addition of vapour will get condensed on the surfaces. The vapour may get condensed in different forms, such as mist, rain, hail, snow, sleet etc. The evaporated water thus returns to the earth surface in any of these forms. *The water which comes back to the surface of the earth in its various forms, like rain, snow, hail, sleet, etc. is known as precipitation.* A major part of the precipitation occurs in the form of rain, and minor part occurs in the form of snow. Other forms of precipitation, such as *hail, sleet, mist,* etc., are negligibly small, and generally ignored in the design of most of the hydrological works; and, therefore, they are not of much importance to us.

7.4. Saturation Pressure

As stated above, water vapour are generally present in the atmosphere. Sometimes, they are alone and sometimes they are mixed with other gases. The pressure exerted by each component of such a mixture is known as *partial pressure.* The pressure exerted by the vapour present in the air is known as *vapour pressure,* and if the air is fully saturated with these vapour, it is then known as *saturation vapour pressure or* **saturation pressure.**

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If the barometric pressure is kept constant and the temperature is reduced; *i.e.* if the air is cooled at constant barometric pressure, a stage will come when the air will become saturated with the same amount of vapour. This temperature is known as **dew point.** If the cooling is continued, the vapour will get condensed on the contact surfaces. This condensation will be in the form of *dew,* if the dew point is greater then 0°C; and it will be in the form of *frost* if the dew point is less than 0°C.

7.5. Lapse Rate

In order to understand climate and rain-formation, it is necessary to understand the changes of temperature with height, in the atmosphere. In fact, not only temperature, but also density and pressure of the atmosphere go on decreasing, as the height above the surface of Earth (altitude) increases, as shown in Fig. 7.2.

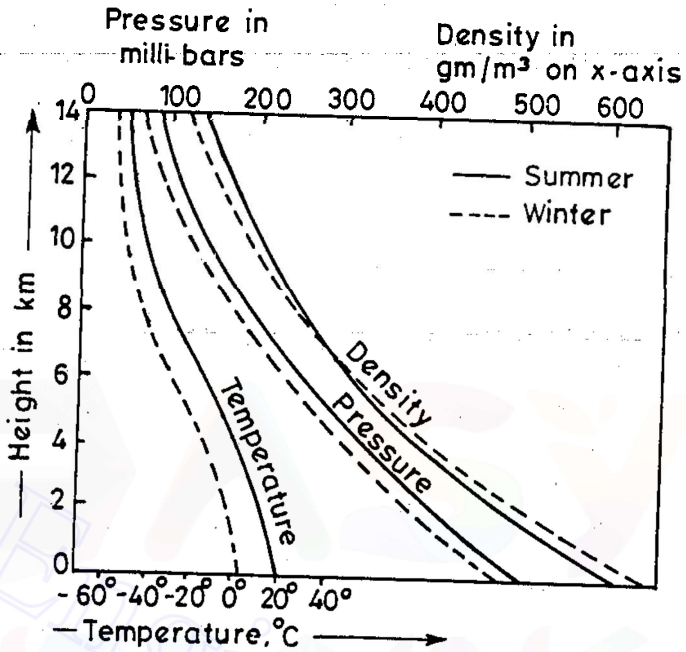

Fig. 7.2. Variation of temperature, pressure and density of atmosphere with altitude.

The decrease in temperature of the undisturbed free atmospheric air, with altitude, is called *Lapse rate* or *Temperature lapse rate.* The general variation of temperature with altitude, through different zones of atmosphere (such as *troposphere, stratosphere, ionosphere*) is shown in Fig. 7.3

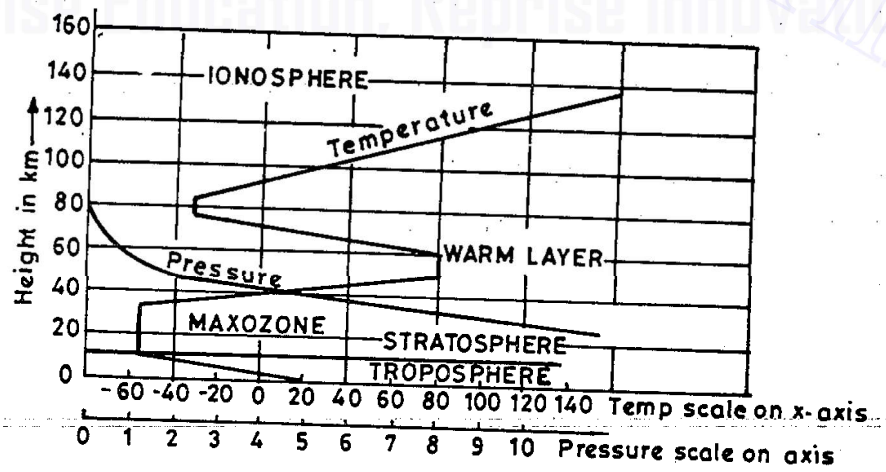

Fig. 7.3. Variation of temperature and pressure through different zones of atmosphere below the exosphere.

7.5.1. Dry Adiabatic Lapse Rate. When air expands, its given volume will become rarer and lighter ; and when it contracts its given volume will become dense and heavier. The process of expansion, as the light air rises ; and the compression as the heavier air sinks, involves no transfer of heat to or from the surrounding air, and is hence

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called an *adiabatic process*. Although, there is no loss of heat, yet there is a definite change in temperature, solely due to the process of expansion and contraction. Expanding air will *drop* its temperature at the rate of 10°C for every km of ascent. This *fall* in temperature is much more than the lapse rate of free air under normal condition (*i.e.* 6.5° per km) ; and is called the *dry adiabatic lapse rate*. This is shown in Fig. 7.4. The temperature rise in compression is equivalent to the temperature fall in expansion.

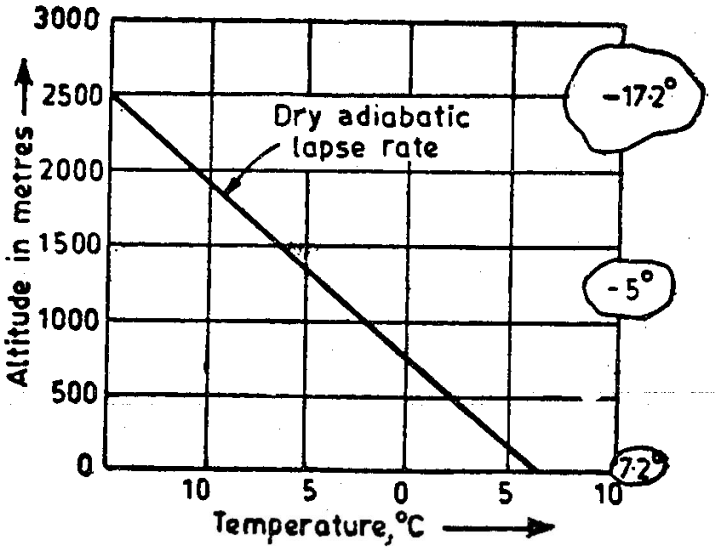

Fig. 7.4. Dry adiabatic lapse rate.

As air is cooled adiabatically, it eventually reaches in altitude where its decreased temperature corresponds to the 'saturation pressure; in other words, the temperature cannot fall further without condensing some of its water vapour. The altitude where this condensation begins, is called the **condensation level,** and the corresponding temperature as the *dew point.*

7.5.2. Adiabatic Saturation Lapse Rate. As the air continues to expand, the process of condensation also continues, and the fall in temperature becomes less rapid. This temperature change is now called *adiabatic saturation lapse rate.* This rate of change of temperature during the moist adiabatic process (when condensation is taking place) ranges from 3 to 10°C per km of altitude rise, being smaller for warm saturated air and larger for cold air.

7.6. Humidity and Relative Humidity

The term humidity is used in order to obtain an idea of the amount of moisture present in the air. The amount of moisture present in the air, when expressed as mass per unit of volume, is known as *Absolute humidity.*

Absolute humidity at a given temperature

= Mass of moisture present in a unit volume of air at the given temperature.

Relative Humidity is defined as the ratio of the actual vapour pressure to the saturation vapour pressure at the same temperature. Hence, it gives an idea of the extent to which the air is saturated.

Relative humidity (R.H.)

$$= \frac{\text{Actual vapour pressure at a given temperature } (e_a)}{\text{Saturation vapour pressure at the same temperature } (e_s)} \qquad ...(7.1)$$

Sometimes, relative humidity is defined as the mass per unit volume of the actual vapour present in the air, to that it could contain at the same temperature when fully saturated.

Humidity can be measured either by a *psychrometer* or by a *hygrometer. Absolute humidity* can also be measured by directly passing a known volume of air through a drying agent, which absorbs the moisture of air. The increase in mass of this drying agent divided by the volume of the air passed, will give the absolute humidity of the air.

Absolute humidity decreases rapidly, as we go at higher and higher altitudes. About one-half of the total moisture present in the atmosphere occurs only within a distance

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of one mile from the earth's surface. Humidity decreases with altitude, because at higher altitude, the convective currents which are responsible for carrying the water vapour into the air, are very much reduced.

7.7. Weather and its Role in Causing Precipitation

The prospective precipitation in an area is primarily dependent upon the *weather conditions* of the area. The weather, inturn, depends upon the *atmospheric pressure, temperature, humidity (moisture),* etc. The interaction between these variables, infact leads to formation of *winds* and *cyclones,* which cause precipitation.

The branch of science, which exclusively deals with the study of weather, is known as meteorology, and, hence, a hydrologist, or a water resources expert, is to a larger extent, dependent upon a meteorologist, because all rainfall predictions and *measurements* are done by a meteorologist.

Even in our country, *Indian Meteorological Department* (IMD) is responsible for all weather and rainfall predictions, and this Department controls the measurement of rainfall and snowfall, although most of the *rain gauges* in our country are maintained by the State Governments, but under the technical control of IMD. In the National Capital Territory of Delhi, however, IMD itself is maintaining the rain gauges.

7.7.1. Formation of Clouds. We have stated earlier that the condensation of water vapour in the atmosphere, as caused by the cooling of the air mass, causes precipitation. Infact, this is a simplified statement, because such condensation does not necessarily cause precipitation, but it certainly forms **clouds** or **fogs.** *It is the formation of clouds, which ultimately causes precipitation.*

A cloud is, thus, formed by the cooling and condensation of the rising vapour (air) into some visible aggregates of minute particles. *A cloud, therefore, generally contains ice particles and small droplets of water.* The nature of the cloud formed, largely depends upon the type of the ascending motion, to which the air gets subjected to, as discussed below :

(*i*) *Prolonged wide spread slow ascent of air,* as in a low pressure area, or *stirring motion,* gives rise to stratification in clouds. Such clouds are formed in layers, and may be called **stratiform clouds.** Such clouds lead to *cyclonic precipitation.*

(*ii*) *Convective or violent ascent of larger bubbles of air,* resulting from **insolution** (*i.e.* cold air coming over a warm region), may lead to formation of *heap* type of clouds, called **cumuliform clouds.** Such clouds lead to *convective precipitation.*

(*iii*) When air is forced to ascend a hill or a mountain barrier, typical **orographic clouds** are formed, which lead to *orographic precipitation.*

Conventional classification of clouds

Depending upon the elevation at which the clouds are formed, they may be classified as :

(*i*) *high clouds ;*

(*ii*) *medium clouds ;*

(*iii*) *low clouds; and*

(*iv*) *vertically developing clouds.*

The levels at which these clouds form, may differ in *equatorial regions (tropics), temperate latitude regions,* and *polar regions* to some extent, as shown in Table 7.1

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Table 7.1. Formation of Clouds at Different Levels

Sl. No.	*Type of cloud (basic group)*	*Level of occurrence in different regions*		
		Tropical regions	*Middle latitude regions*	*Polar regions*
(*i*)	High clouds	6–18 km	5–12 km	3–8 km
(*ii*)	Medium clouds	2–8 km	2–7 km	2–4 km
(*iii*)	Low clouds	0–2 km	0–2 km	0–2km
(*iv*)	Vertically developing clouds	(*a*) base 0.5–2 km (*b*) tops* 18–20 km	0.5–2 km 10–12 km	0.5–2 km 5–8 km

7.7.2. Basic Forms of Clouds. There are three basic forms of clouds: (*i*) *stratiform* (**stratus**) ; (*ii*) *cumuliform* (**cumulus**) ; and (*iii*) *fibrous* (**cirrus**). Despite the infinite variety of forms, it is possible to classify clouds into the *basic groups* or *genera*. These can further be sub divided into *species* according to their shape and structure.

The makes of the clouds and the distinguishing features of the genera, as given in Table 7.2, will help in their identification.

Table 7.2. Distinguishing Features of Clouds

Type of clouds	*Distinguishing features of genera*
(A) **High Clouds,**	
(*i*) *Cirrus* (Ci)	Occur in the form of detached white patches or filaments or narrow bands with fibrous silky appearance. They are composed of ice crystals at temperatures below 0°C.
(*ii*) *cirrocumulus (Cc)*	Occur in the form of thin white sheets or small patches or elements in the form of grains, ripples, etc., merged or separate, more or less regularly arranged. They are composed of ice crystals.
(*iii*) *Cirro-stratus* (Cs)	Transparent whitish cloud veil, either smooth or fibrous, often producing a halo. They are composed of ice crystals.
(B) **Medium Clouds,**	
(*iv*) *Alto-stratus* (As)	They are grey striated or fibrous veils, like Cs, but thicker, through which sun is either seen only as diffuse bright patch or not seen at all. They are mixed clouds of ice crystals and water droplets. They may give rise to corona.
(*v*) *Alto-cumulus* (Ac)	These clouds occur as white grey sheet or layer of rounded masses, rolls, etc. arranged in groups, lines or waves. They are composed predominantly of water droplets, sometimes with a few ice crystals.
(*vi*) *Nimbo-stratus* (Ns)	They are amorphous dark-grey clouds, sometimes reaching low, *rain or snow falls from such clouds*, and they are thick enough to obscure the sun. They are mixed clouds with ice crystals at top (when the top is very high) and water droplets below.
(C) **Low Clouds,**	
(*vii*) *Strato-cumulus* (Sc)	Such clouds occur as layer or patch, consisting of laminae or globules in groups, lines or waves with soft grey appearance. They are composed of water droplets at temperatures above 0°C.
(*viii*) *Stratus* (St)	They are uniform featureless layer clouds, akin to fog but not resting on ground. When broken into irregular ragged pieces, they are known as fracto-stratus. Composed of water droplets above 0°C.
(D) **Vertically developing Clouds**	
(*ix*) *Cumulus* (Cu)	They are detached dense clouds with marked vertical growth in the form of rising white domes or towers ; tops looking like cauliflowers. Base nearly horizontal. White tops when sunlit. Composed of liquid water with large droplets.
(*x*) *Cumulonimbus* (C_b)	They are heavy dense masses with large vertical growth, appearing like mountains or towers. Upper portion develops fibrous structure resembling an anvil or plume. They produce lightening, thunder, showers, hail, squall developing into a thunderstorm. They are composed of liquid droplets at bottom and ice crystals in the portion above.

* The tops reach higher in the tropics, because tropopause land is higher.

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7.7.3. Measuring the Clouds. The clouds are generally measured *visually*. The size of the patch, in which the clouds occur, is estimated mentally as a part fraction of the total sky. This amount is reported in *oktas**, or eighths of the sky covered by the clouds. When observing the clouds, the amount of each type of cloud present in the sky is recorded separately, as also the total coverage of the sky by all the clouds put together. Total cloud count equal to 8 oktas, would evidently mean a fully overcast sky. The height of the base of the clouds is estimated by optical instruments.

7.7.4. Formation of Rain Drops and Snow Flakes in Clouds. Evidently, a large quantity of water always remains suspended in a cloud. This water remains present in the form of *fine droplets,* which remain floating. These droplets do have a tendency to fall down under their own weight by gravity, but this tendency is opposed by the upward motion of the atmosphere. Hence, for a droplet to fall and reach the ground as precipitation, it must grow to a sufficiently large size, so as to acquire fall speed more than the up-current.

Depending upon the size of the falling droplets from the cloud, precipitation may occur in the form of drizzle, *rain* or *hail.* The smaller diameter droplets give rise to **drizzle ;** whereas, larger diameter droplets produce **hail stones.** *Medium sized droplets of the order of 300—4000 micron** in diameter, fall down as* **rain drops :** as shown in Table 7.3. The *intense showers* may still be having larger sized droplets, say up to 10,000 microns.

Table 7.3

S. No.	*Form of Precipitation*	*Droplet size diameter in microns*	*Fall speed in m/sec*
1.	Drizzle drops	40—500	1
2.	Rain drops	500—4,000	10
3.	Hail stones	10,000—50,000	20—50

Infact, a cloud initially contains small droplets of the size of 40 microns. Somehow or the other, the droplets grow in size to fall down as precipitation. This **growth of the droplets** have been directly related to : (*i*) cloud thickness ; and (*ii*) magnitude of up-currents. *The larger the cloud thickness, and stronger are the up currents, the bigger will be the size of the droplets formed.*

The mechanism responsible for the growth of the droplets in a cloud, has been related to the two different theories, which are applicable separately to *warm clouds* and *cold clouds,* as discussed below :

(1) *Coelescence theory for warm clouds.* In tropical regions, rains occur from clouds which are not cold enough to produce ice crystals. The growth of droplets in such warm clouds is considered to be under the action of *aggregation of smaller drops.* A droplet bigger than its neighbours, sweeps and catches a large number of droplets in a cone, on its downward journey. By collision and coelescence, the drops grow to limiting size, and breakup. The larger fragments fall to the ground as rain, while the smaller ones evaporate. *This is a self-sustaining chain reaction.*

The larger drops are formed in such clouds, because of the large magnitudes of the vertical up currents in the tropics. In the tropical maritime regions, showers can therefore occur even from clouds with lesser vertical development, compared to inland areas. This is explained to be due to availability of giant nuclei (initial droplets) in the form of *large sea spray particles,* which can easily form the larger drops, to initiate the chain reaction of further coelescence.

* 1 oktas = 1/8th of full sky.

** 1 micron = 10^{-6} m = 10^{-3} mm.

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This process excludes the ice-crystal phase, and explains the *heavy warm showers* of the tropics, caused by warm clouds, which do not rise up to the freezing level.

(2) *Ice crystal theory for cold clouds.* The clouds, which rise to higher levels, having temperatures of the order of −12°C, will normally contain a vast number of *super-cooled drops* at the top of the cloud, along with a small number of *ice crystals* below. Even dust particles can help in providing these initial ice crystals, called *ice nuclei.*

For such freezing temperatures, evidently the saturation vapour pressure is lower at an ice surface than at a water surface. The air in the cloud will therefore, have a vapour pressure somewhere between these two saturation pressures ; and hence, the water droplets will evaporate, and further condensation will occur on the ice particles, thereby increasing their sizes. This process leads to preferential growth of ice crystals by direct *sublimation* at the expense of super cooled water drops. This growth of ice crystals is most intense at temperatures around −12.5°C, when the difference in vapour pressure over ice surface and that over water surface, is maximum. The resultant **snow flakes,** so formed in such clouds, fall down to lower levels, and melt in the warm region below. If, however, the temperatures continue to be very low, **snowfall** occurs.

Another phenomenon, which may help in increasing the number of ice crystals in such clouds, is the **splintering process.** In this process, initially an *ice-shell* is formed on the super-cooled drop. As the ice-shell grows, and presses the drop inwards, the drop gets shattered, causing the inside water to eject out in minute jets, which immediately freeze into *spikes.* The spikes are finally splintered, scattering fine ice crystals in the cloud.

7.8. Scanning and Predicting Weather

The advent of modern machines, like RADARS, and more recently of SATELLITES, have enabled us these days to thoroughly scan the weather around the globe, and to predict the rain *storms, cylclones, winds, squall, thunderstorms,* etc., well in advance, by days and weeks.

Weather forecasts are thus issued to give advance warnings to the public in general, and *aviation authorities* in particular, regarding the possible atmospheric disturbances in their areas.

Short range forecasts are valid up to 48 hours, and are precisely worded ; whereas, **long range forecasts** are given between 1–2 weeks in advance, or over longer periods. Evidently, the **medium range forecasts** are valid for periods ranging between 2 to 7 days.

Short range forecasts, as are announced daily on our T.V. programme, cover the imminent possibilities of *atmospheric disturbances,* including information on winds, temperature, weather, clouds, thunderstorms, rainfall, snowfall, etc. They are considered to be the most important in all weather forecasts.

Long term forecasts are generally framed to give an idea of the *mean state of the atmosphere* or *trends.* Mean precipitation and temperature are the elements most commonly forecast for long periods. They are usually indicated as departures from the normal. *Seasonal rainfall forecasts,* issued in our country by IMD over the general outlook of the coming yearly monsoons, well before the commencement of the season, are covered in this type of forecasts.

7.8.1. Weather Radars. The word RADAR, as we all know, stands for '*Radio Detection and Ranging*'. In general, such machines are used in *aviation* for detecting and guiding aeroplanes, etc. besides their use in weather detection. Since the radars used in weather scanning differ from the ones used for aviation purposes, they are here called as *weather radars* or *meteorological radars.*

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A weather radar, infact, is a powerful tracking device, used for locating *thunderstorms, rain-storms, duststorms, cyclones,* etc. The radar, infact, locates and keeps a watch on the movements of such atmospheric disturbances, which together are called *metors**. A radar can, thus, help in estimating the stage of development of thunderstorms, dust storms, squall, rainfalls, snowfalls, etc. It can estimate the direction and speed of the movement of severe weather phenomena, and their formation, growth, and dissipation. It can also estimate the height of the bases and tops of the clouds, and freezing level. The quantitative estimate of precipitation is also possible (for which it is seldom used).

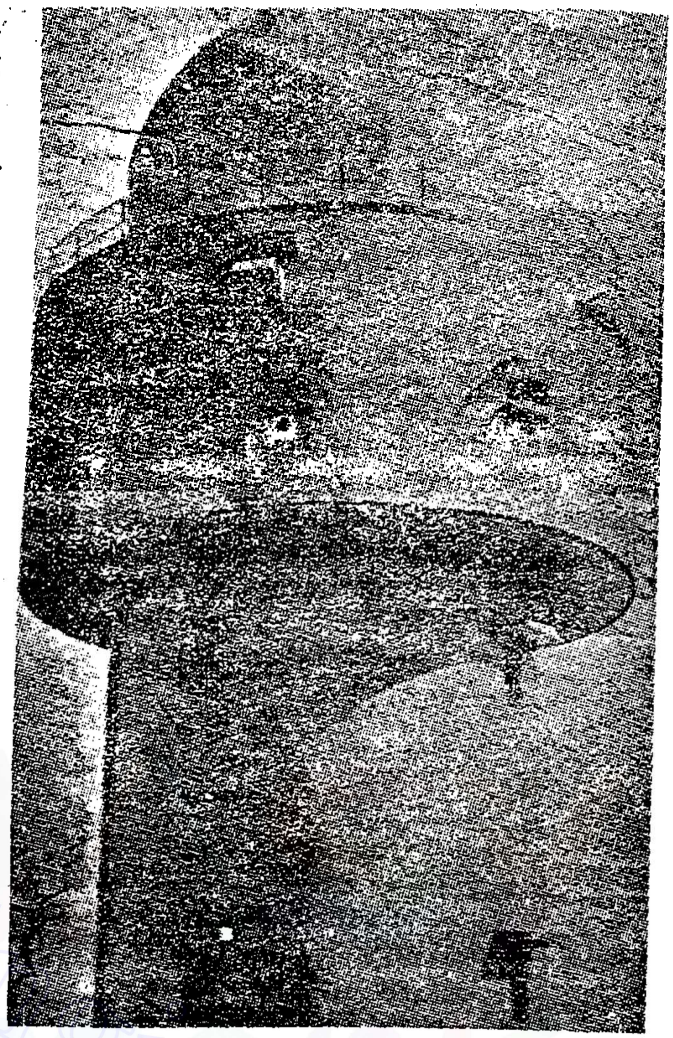

Fig. 7.5. A PHOTOVIEW of a radar, shielded by transparent dome, covers the west coast from Goa to Gulf of Cambay. It gives a bird's eye view of rain bearing clouds within an area of approximately 5,00,000 sq. km. around Bombay.

Weather radars are most widely used for locating and keeping track of cyclones, which are liable to cause wide spread misery in coastal areas. In supplementing rain gauge measurements, they are used for determining the *areal extent, orientation* and *movement of rain-storms.* Hence, in a unified rain measuring arrangement, the rain gauges are used for measuring the total amount as well as the intensity of a rainfall ; and the weather radar is used for determining the areal extent, orientation, and movement of such a rain, in addition to taking certain sample measurements of rain intensity and rainfall quantity for checking gauge readings. *A radar thus, helps in checking and supplementing rain gauge data,* in addition to its use in weather forecasting.

A radar, infact, releases a narrow beam, consisting of regular succession of pulses of electromagnetic radiation. This *radiated beam,* travels in a particular direction (being determined by the direction of the movable antenna) at the speed of light, and is partially reflected back, by clouds or precipitation particles. This reflected wave is received by the same antenna at the radar station. The electromagnetic energy of the reflected wave is lesser than the transmitted energy, as certain energy is lost in passing through the precipitation target. This reflected energy is represented on the radar screen, and is called an **echo**. The brightness and the shape of the echo formed, will indicate the type of the prevailing hydrometeor in the path of the transmitted beam.

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* **Meteors** consist of *precipitation, suspensions* or *deposits* of liquid or solid particles, optical or electrical manifestations. Depending upon the composition of particles and the physical processes involved, meteors may be classified as : *hydrometeors, lithometeors, photometeors* or *electrometeors.*

Hydrometeors include *haze, mist, fog, drizzle, rain, shower, snow, hail, dew, hoar-frost, rime* and *glaze* etc. **Lithometeors** include *smokes, dust-winds, sand storm,* etc. **Photometeors** include *mirage, glory, halo, corona, rainbow,* etc. **Electrometeors** include *lightening, thunder including thunderstorms, St. Elmos' fire, aurorae,* etc.

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Fig. 7.6. The ANTENNA of the Automatic Picture Transmission equipment in Bombay is operated by remote control. In the background is the observatory tower at the Regional Meteorological Centre, which is about 130 years old.

Different kinds of hydrometeors give rise to different types of echoes, having distinct characteristics. Even convective and stratiform clouds produce different echoes. Different weather phenomena like thunderstorms, dust storms, line squalls, hail, tornado, snowfall, etc. display different characteristic features, such as *bright bands, dry holes, hooks, protruberances,* etc. The brilliance of the echoes varies with the intensity and proximity of the phenomenon.

A mathematical equation, expressing the interception of raindrop by a radar beam is given by

$$P_r = C \cdot \frac{Z}{r^2} \qquad ...(7.2)$$

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where P_r = average echo power

Z = radar echo factor given by Eq. (7.3)

C = a constant.

The *radar echo factor* (Z) is related to intensity of rainfall in mm/h by the Eqn. (7.3), as :

$$Z = a \cdot I^b \qquad ...(7.3)$$

where a and b are constants for the given radar station, and are to be determined by calibration with the help of recording rain gauges. Typical values of a and b are 200 and 1.6 respectively, for some particular weather radar. The factor Z for this radar is given by :

$$Z = 200 \cdot I^{1.6} \qquad ...(7.3\ a)$$

Proper *attenuation* of the transmitted signal governs the selected radar frequency. Say for example, smaller wave length radars are required to provide better attenuation due to clouds and lighter rains, as compared to those required for attenuating heavy flood producing rains. Hence, radars of 3 or 5 cm wave lengths are generally used for detection of clouds, lighter rainfalls, or snowfalls ; whereas, 10 cm radars may suffice for detecting heavy rains and cyclone surveillance.

For studying the *micro structure of clouds* and detection of *clear air turbulences (CAT),* **laser probes** *(LIDAR)* are increasingly being employed these days. *On line processing of radar data on a computer, and applications of* **droppler type radars** for measuring the velocity and distribution of raindrops, are other advancements, being used in modern days in this field.

Limitations of weather radars. Inspite of their varsatality, weather radars suffer from certain inherent **limitations,** as pointed out below :

(*i*) Distant thunderstroms and precipitation areas may not get detected due to curvature of the earth. The hydrologic range of the radar is about 200 km.

(*ii*) A huge thunderstorm area near the radar station can obliterate the cells beyond it, due to attenuation. The shape of the echoes may get distorted, and may not represent the true extent, shape, and area of the phenomenon.

(*iii*) The intensity of the echo may not correspond to the severity of the weather features. Sometimes, spurious echoes, called **angels** may form due to extraneous factors, such as *hydrolapse, flocks of birds,* etc., and may be misleading. Similarly, echoes always appear from the ground features and topography, and they need to be suppressed by adjustments and intelligence, to highlight the real echoes caused by a particular weather phenomena. *Iso-echo contouring facilities* and other devices installed in the radars do help in suppressing extraneous echoes, and in identifying the particular weather phenomena with greater precision and clarity.

7.8.2. Weather Satellites. The first ever weather satellite of the world was orbitted by USA in the year 1963, and since then, several other improved versions, and *multi-purpose satellites,* have been launched, and put into orbits, by several countries. These satellites continuously provide the meteorological observations, through the process of **remote-sensing** consisting of *imaging* and *communication.* Several countries, including India, are now-a-days, having their exclusive satellites, for providing exclusive data.

A satellite, as you know, is a device or a vehicle, which is launched into the space through powerful rockets, with the injection velocity not less than 11 km/sec, so as to enable it to escape the Earth's gravity and become a **space probe.** The satellite after its

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launch, is placed in the *chosen orbit,* called the *parking orbit,* by manoeuvres from the ground control, in order to suit the particular requirements of the country, launching the satellite. Say for example, a satellite placed in a *polar orbit* (*i.e.* an orbit at right angles to the Earth's equator and passing through the polar axis), can scan the maximum area of the Earth's surface. Such *polar satellites* can cover the globe surface twice a day, and hence, such a satellite will send pictures of the same strip of Earth at fixed times, twice a day.

Geostationary or geosynchronous satellites are those, whose orbitting period is 24 hours. Such a satellite will, therefore, revolve round the Earth to complete its one revolution in 24 hours, and hence such a satellite will always remain overhead, *i.e.* in the same relative position with reference to a particular point on the globe, since Earth also takes 24 hours to rotate. The angular speeds of Earth and that of the satellite will be the same, in such a case. This is achieved by putting the satellite in a *geosynchronous circular orbit,* over the Earth's equator, at an altitude of 35786 km. When once the satellite has been put in the *right orbit,* it is placed in the *right slot* at the *decided longitude in the geosynchronous orbit.* Such **parking slots,** are limited, and an international body allots the slot, and your satellite will have to remain within 0.1 degree of the same. This is necessary to avoid interference and over-lapping between satellites. Placing your satellite in a given slot is known as *station acquisition ;* and maintaining it in that position, is known as *station keeping.*

The advantage offered by a geosynchronous satellite is that : the same area of the Earth's surface remains under the constant observation of the satellite all the time ; and this enables the evolution of cyclones, etc., to be studied without interruption.

Theoretically three such satellites, placed at appropriate longitudes over the equator, can watch the entire globe ; but in practice, polar regions will not get covered by such placements, and there will be some overlapping, causing distortions in pictures. Five such satellites, can however, provide practically full coverage of the entire Earth's surface. Indian satellites, INSATs, belong to this category of satellites, as they are *multipurpose geosynchronous satellites.*

A satellite provides us an *overview of the Earths' surface and that of the atmosphere* covering a vast geographical area at any given time, through taking and sending us the integrated pictures or photographs. Such continuous pictures obtained with the passing time, need to be carefully processed and analysed in comparison with the data obtained from other sources. Say for example, *cloud pictures,* transmitted by the satellite as *cloud imageries,* must be correlated with those obtained from other sources, such as *aircrafts.* Similarly, *cyclone formations,* as seen by the satellite from above, may have to be correlated with the ground realities, by *aircraft reconnaissance flights,* etc., to pinpoint the exact cyclone centre. The satellite data, thus, needs to be validated with the aid of ground truth, before it is put to actual use. The satellite observations, thus cannot replace the conventional observations, but can only be complementary to them.

Inspite of their data being only complimentary in character, such satellites provide a very versatile and powerful tool to the meteorologists, who can now have an instataneous panormic view of the entire globe on a continuous basis. Cyclonic storms can, thus, be detected and tracked right from their genesis and fronts, without fail. *Jet streams*, Easterly waves,* etc., are also pinpointed with great success. Satellites can also interrogate unmanned data collecting platforms, constant level baloons, etc., and transmit their data to the main *Data Acquisition Centre,* for analysis and processing.

* *Jet streams* are the high velocity winds, at high altitudes of 16000 m or so, flowing from West to East; helping fast moving air flights, such as from Gulf countries to Japan and Pacific. Reverse flights, against the winds, to be avoided at such heights, where Jet streams exist.

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The **body of a satellite** can be broadly divided into three components; *viz* :

(*i*) *pay load systems* (vol ≈ 15%)

(*ii*) *subsystems of the satellite* (vol ≈ 25%) ; and

(*iii*) *fuel systems* (vol ≈ 60%).

(*i*) and (*ii*) together are referred to as *dry mass* of the satellite.

Pay load system is that part of the satellite, which performs its *major service functions,* such as *communication, imaging* and *data relay.* **The subsystems** provide smooth and efficient functioning of the pay load systems by creating the right environment and giving necessary engineering support.

The pay load systems, as pointed out above, provides *communication and imaging.* Functionally, the *communication system* receives signal from the ground at one frequency, amplifies it, and transmits it on another frequency. The tranmission and reception is done at *super high frequencies* (*i.e. C* band, 4000—6000 megahertz) to increase the signal carrying capacity through special antennas. These antennas are made up of ultra high stiffness carbon fibres, and have to be oriented continuously towards the earth station on ground.

The *imaging system* consists of : (*i*) *Very High Resolution Radiometer* (VHRR), which generates the images ; and (*ii*) the *data relay transponders.* The *radio-meter* maps the Earth's surface in the visible and infra-red (I.R.) bands during the day ; and in the infra-red band alone at night, by means of highly sensitive detector elements. The *detectors* need to be kept 'cooled' at about minus 150°C, and, this is achieved by a passive cooler, which isolates the detector assembly, and radiates the heat of the cold space outside.

The reflected/emitted energy from the Earth's surface, is focussed through suitable optics on detectors, which convert it into electrical signals. These signals are then amplified, processed, and transmitted to ground as *digitized signals, making up a single frame, every 33 minutes.* As a detector forms a single picture element, a mirror with *two axis scanning mechanism,* is used to gather the energy from different portions of the Earth, to form a frame. The scanner can be commanded from the ground, to image either the full Earth disc (20°) or a small sector of it (0.5°).

The *cloud pictures* taken by the satellite, together with the information they provide about *winds in the upper atmosphere,* the *sea-surface temperature, precipitation* and *on-going long wave radiation,* are used by the concerned Meteorological Department for *routine weather forecasts,* as well as for developing *weather prediction computer models.*

The *fuel system* of the satellite is so designed as to tranfer and raise the satellite, through its propellant system, into the desired circular orbit of 35786 km over the equator, after initially putting it into some egg-shaped *transfer orbit* (say 200 km × 35786 km) through rocket launching. Say for example, India's satellite—INSAT–2A carried a *bipropellant system,* in the form of an *Apogee Boost Motor*(ABM), and four powerful *thrusters* for emergency. About 400 kg of mono-methyl hydrazine fuel, and 650 kg of nitrogen tetroxide (N_2O_4) oxidiser, was carried on board, in specially made titanium alloy *fuel tanks. High pressue helium gas,* which expels the fuel under zero gravity conditions, was also stored. When once the satellite was launched in a transfer orbit of 200 km × 35786 km, it was raised by *three firings* from ABM, for a total duration of 70 minutes, to kick the satellite from 200 km to 35786 km. *Small thrusters* on board were then activated to make the orbit exactly cicular, and to firmly anchor it in the assigned slot of 74° East. The four emergency thrusters were also provided to raise the satellite, in case the ABM failed. The use of these thrusters would, however, have depleted the fuel and reduce the **satellite life** from nine years to five years.

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This INSAT satellite is to draw power from the sun, using *solar cells** in the form of 5 solar panels, having a total area of 15 sq m. A chargeable nickel-cadmium battery system is also provided to store solar energy, to meet its requirement during the time, the sun becomes unavailable to the satellite.

7.8.2.1. India's INSATs. India launched its first ever service satellite, called INSAT–1A in August, 1983, which developed a series of snags, and hence could not be put to use. *INSAT–1B* launched a little later, however became fully operational, and served India's needs quite well for about 10 years. INSAT–1C again failed, because of the difficulty in opening its solar panel. *INSAT–1D* was launched quite late in June 1990, and became fully operational in the parking slot of 83° East. With the launching of INSAT–1D, which worked well till about the year 1998, the first phase of INSAT programme was completed.

The second phase of this programme was started with the successful launching of *INSAT–2A*** on 10.07.1992. *INSAT–2B* was then successfully launched on 23.07.1993, and placed in the old parking slot of INSAT–1B which was shifted and shut down. **INSAT–2C** was then successfully launched on 08.12.1995, and is co-located with INSAT–2B in the same parking slot at 93.5 degrees East.

INSAT-2D, which was subsequently launched in June 1997, however, had to be soon abandoned due to excess loss of power in its placement and manoeuvring. The compensation amount received from the Insurance Company on the failure of INSAT-2D, however, enabled the ISRO to purchase a replacement satellite *Arabsat-IC* (with 26 transponders), which was renamed as **INSAT-2DT**. This satellite was placed in parking slot of 55° East, and continued to work till Oct. 2003, when it was replaced by INSAT-3E.

INSAT-2E, after several design improvements (to take care of the causes leading to the failure of INSAT-2D), was then successfully launched on 03.04.1999, and was co-located with INSAT-ID, which was shut off completely. This satellite is placed at 83° East along with INSAT-1D & INSAT-2D, which are totally shut off. INSAT-2E is the last satellite of INSAT-2 series.

The first of the third generation INSAT-3*** series satellites, *i.e.* the **INSAT-3B** with 12 UXC band transponders, was successfully launched by Ariane-5 launch- vehical on 22.03.2000, well before the launching of INSAT-3A, due to severe demand and shortage of UXC band (upper extended C band) transponders required in television services. INSAT-3B has also been co-located with INSAT-2E at 83° East, and is working well today.

The second satellite of INSAT-3 series, *i.e.* **INSAT-3C**, weighing 2750 kg, was launched on 24.01-2002, successfully. This INSAT, with a designed mission life of 12 years, has replaced the aging INSAT-2C, to give impetus to India's telecommunication, broadcasting, business communication and mobile services.

The third setallite of this series **INSAT-3A** weighing 2950 kg, was, then launched on 10.4.2003, successfully by Arian-5 launch-vehicle from Korou in Franch Guyana, to replace old INSAT-2A, to boost country's communication and meteorological services.

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* Basic unit is a small rectangular *silicon chip* (2 cm × 4 cm) converting about 12-13% of the incident solar energy into electrical energy by using **photo voltaic effect.** The silicon solar cells used in most of our Earth satellites have, however, been replaced in INSAT-2E, by high efficiency gallium arsenide/germanium based solar cells, which have a solar power conversion efficiency of 18% as against the 12-13% of silicon cells.

** Truly speaking, INSAT-2A, was only notionally operational with none of its transponders being capable of providing any service due to depletion of on-board propellent, caused by some problem with its propellant tank design from April, 1998 onwards. It has now been replaced by INSAT-3A.

*** The figs-1, 2, 3..... etc. in INSATs usually refer to the satellite mass in tonnes. Thus, the Fig. 1 in INSAT-1' series refer to 1 tonne class of satellites ; and the Fig 2 in INSAT-2's refers to the 2-tonne class of satellites. The INSAT-2A thus had a mass of 2 tonnes, while INSAT-2E had a mass of 2.55 tonnes. INSAT-3 class, launched recently are heavier satellites, each having a mass of the order of 3 tonnes. The latest launched INSAT-3E, Infact, had a mass (lift of mass) of 2.75 tonnes. *Note : Heavier satellites need heavier cryogenic engines for launching.*

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The fourth satellite of INSAT-3 series, i.e. INSAT-3E, weighing 2,750 kg was launched on 28-09. 2003, with a designed mission life of 12 years, and to replace INSAT-2DT, to give a boost to telecommunication and television services. This INSAT was also launched by an European Ariane-5 rocket from the spaceport of Kourou in French Guyana and injected into the geosynchronous transfer orbit in 29 minutes of its flight in multistep intricate mission.

Most of our INSATs are multipurpose geosynchronous satellites rendering *three types* of services ; *viz :*

(1) **Meteorological services,** including the supply of weather maps, cyclone and tornado movements, cloud imageries, etc.; including *Relay* of meteorological, hydrological, and oceanographic data, (collected by *unmanned Data Collection Platforms* (DCPs) located in remote uninhabited places) to the *Meteorological Data Utilisation Centre* (MDU Centre) at New Delhi, for further processing.

INSATs are also being used to *activate Disaster Warning Systems* (DWS), installed in several cyclone-prone villages of Andhra Pradesh and Tamil Nadu coasts, regarding the approaching cyclones. Such a DWS system, on being activated by the *Cyclone Warning Centre of Madras, via,* INSAT, emits a siren for one minute, giving ample warning to the area residents, regarding the approaching cyclone.

As a matter of fact, such centres are purely the reception centres, receiving coded messages from the cyclone warning centre, through INSAT ; and are even provided with public address system, so that an announcer from the cyclone warning centre can inform the people, about the time as to when the storm is expected to hit them, and what action to take. Such a DWS, hardly costs around Rs. 1 lakh, but their installations have tremendeously helped in restricting loss of life during cyclones. We have already installed more than 100 such systems in Andhra and Tamil Nadu areas ; and we plan to further install such systems in West Bengal, Orissa, Gujarat, and Maharashtra's coastal areas. Even Bangladesh, where cyclones cause heavy human loss, may install such systems, as we have already offered this service to that country.

(2) **Telecommunication services**, including improved voice and quickly fructifying STD and international telephone calls ; and commissioning of many such new routes. INSAT system has als. provided a novel rural tele-graphic service in north-eastern inaccessible areas of the country, where the use of conventional wire telegraphy becomes impracticable, due to hilly terrain.

(3) **Television services,** including much wider coverage of transmitted T.V. programme throughout the country, as well as much wider reception from around the world.

INSAT–2A, besides giving the above services, had six additional transponders in expanded C–band to provide **business communication, satellite news gathering,** and **distance education,** using *talk-back system.*

Weather pictures were expected to be clearer from INSAT–2A as compared to those obtained from INSAT-1B, because of the enhanced resolution of the on-board radiometer*. One additional feature of INSAT–

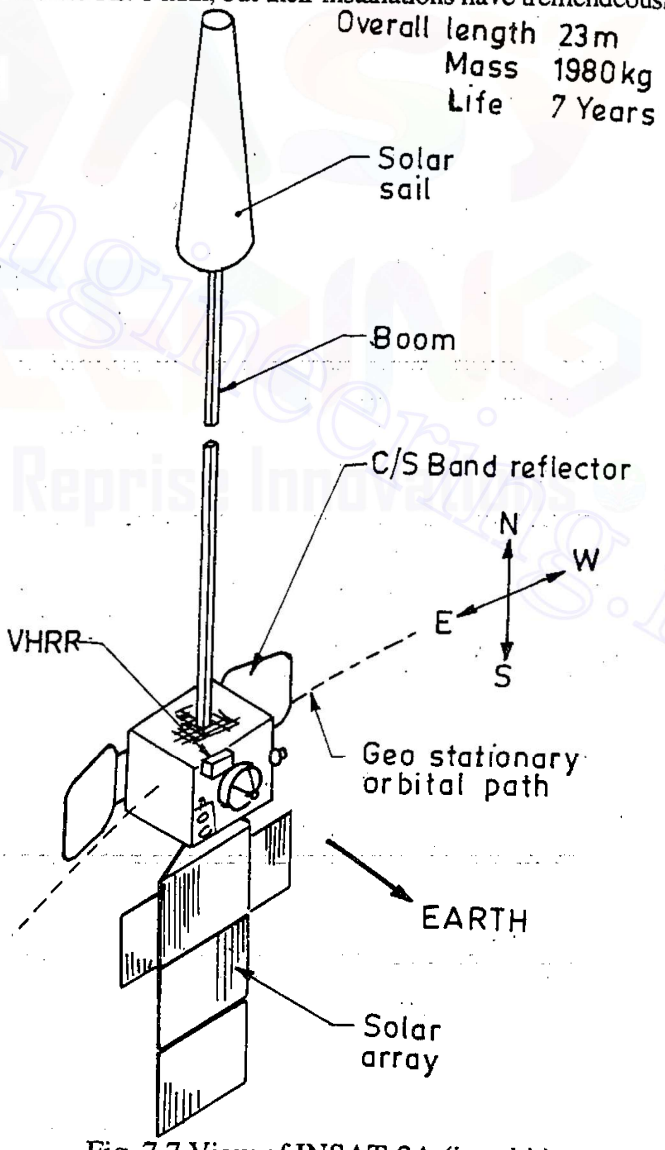

Fig. 7.7 View of INSAT-2A (in orbit).

* The resolution was improved in INSAT–2A to 2 km in the visible band (from the earlier one of 2.75 km in INSAT–1B) and to 8 km in infra-red band (from the earlier one of 11 km).

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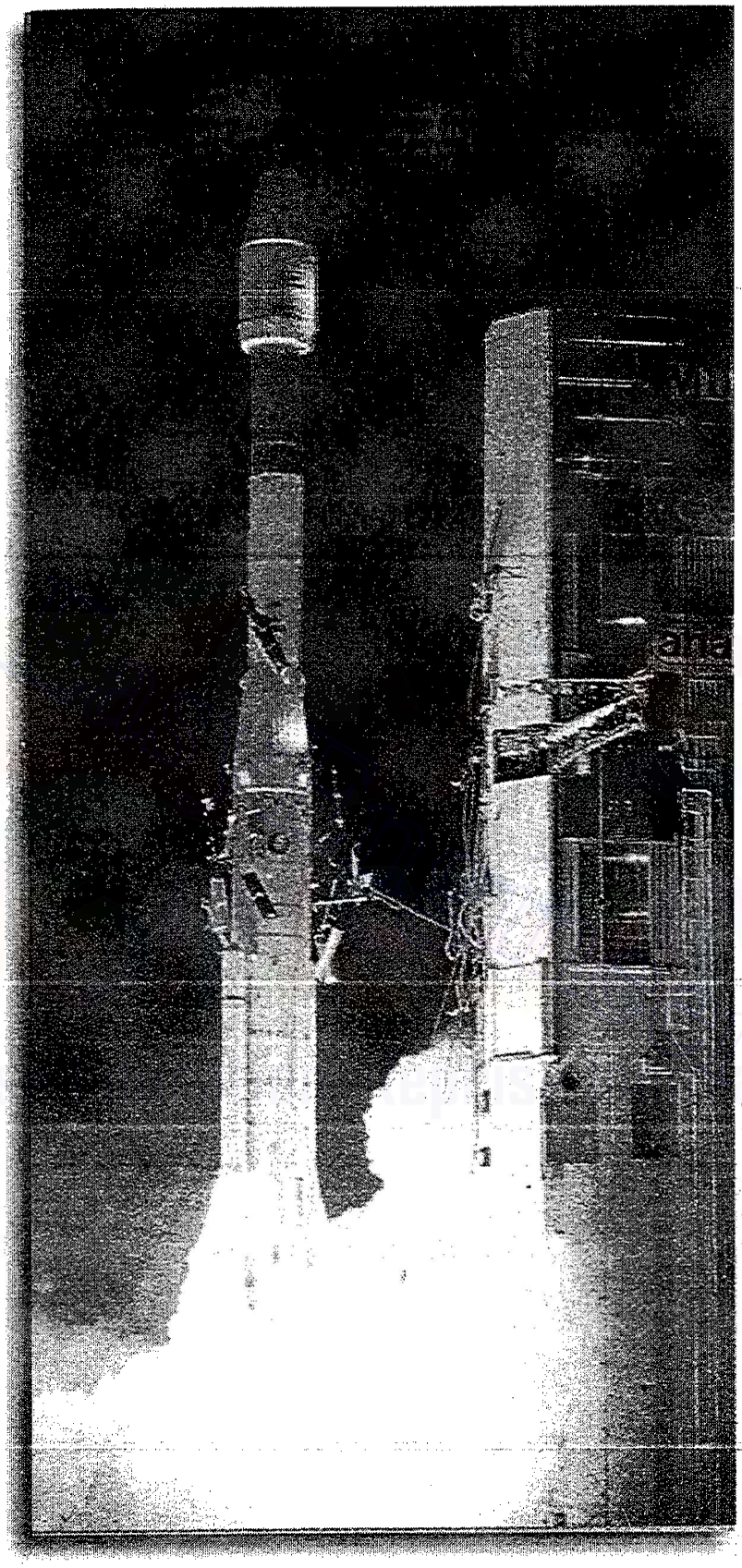

The European launcher Ariane 42P blasts off from Kourou in French Guiana on April 3, 1999 to deploy INSAT-2E. The multi-purpose satellite is perhaps the most advanced in the INSAT series so far in terms of technologies used in the hardware and in the payloads.

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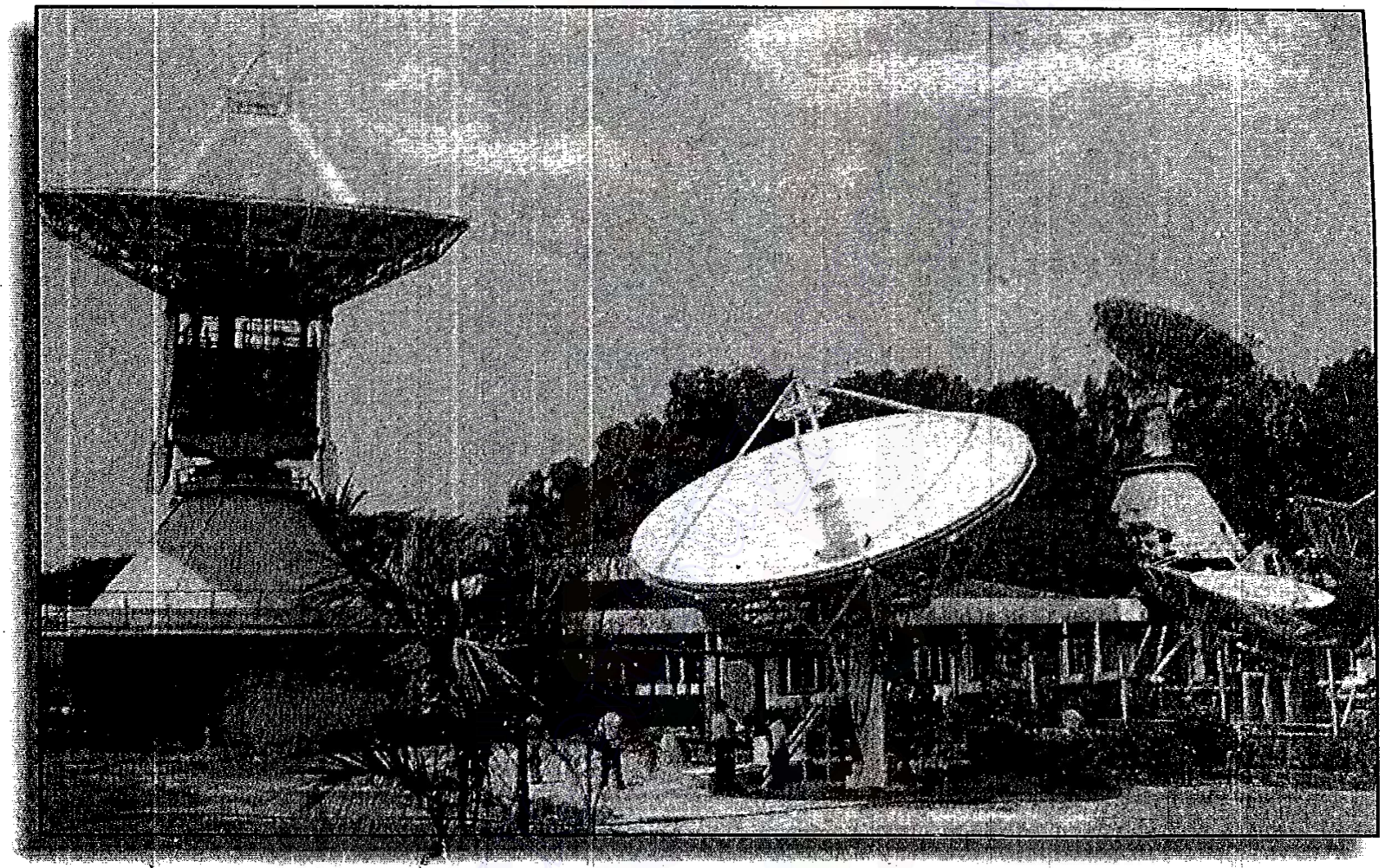

An array of full-motion antennae and limited-motion antennae at the Master Control Facility in Hassan (Karnataka), which controls, commands and tracks INSAT-2E.

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2A was its *search and rescue payload,* which could detect distress calls, eminating from emergency radio beacons, carried by ships, aircrafts, or group of mountaineers, etc. at 406 mega-hertz frequency. At present, such *distress call detection work* is being done by polar satellites of USA, orbitting in *low-Earth orbit,* around the world. Such a satellite, after detecting an emergency distress call, passes it on to a nearby *Local User Terminal Centre* (LUT)*, for helping in providing the necessary assistance to the distressed.

The detection and rescue operation carried out by USA some years back, in saving the lives of PLO chief–Mr. Arafat and others, whose hellicopter crashed in a desert, is an excellent testimony of the use of satellites in detecting such emergent needs.

Although *such distress-call detection work* is being done at present, by low earth polar satellites ; but near the equator, there could still be a delay of $1\frac{1}{2}$ to 3 hours, between the time an emergency radio beacon is activated by the distressed, and the time at which it is detected. Such a possible delay in detecting emergent signals, may prove disastrous to the distressed. This time gap can be eliminated by geostationary satellites, which will be in a position to detect such distress calls immediately, in their respective areas, being watched. The use of such rescue pay loads on geostationary satellites, is still experimented globally and the first Indian satellite equipped with this facility was INSAT-2A, which unfortunately has stopped working from April 1998.

A prospective view of INSAT–2A is shown in Fig. 7.7. Similarly, Fig. 7.8 shows a line eye view of the services, performed by it.

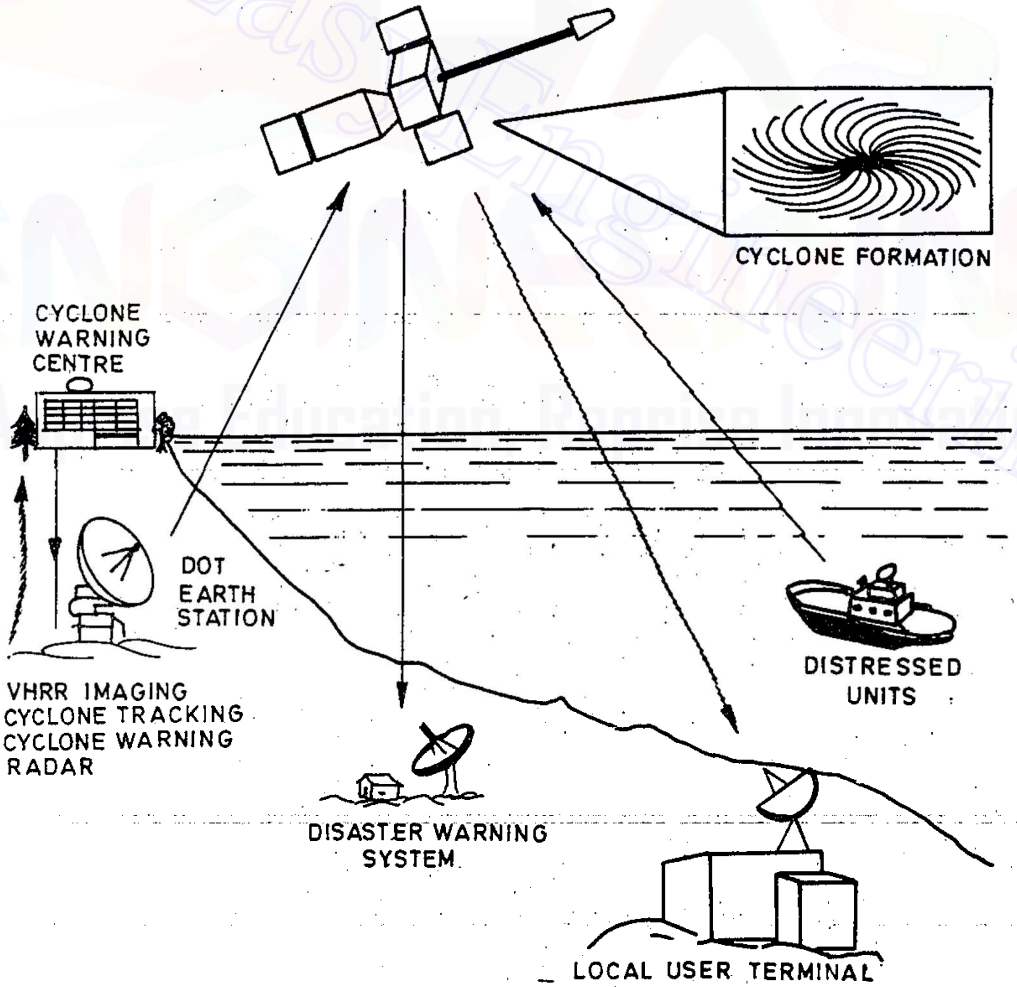

Fig. 7.8. Services performed by INSAT-2A.

INSAT programme in India is being conducted and controlled by *Indian Space Research Organisation* (ISRO). All the satellites in INSAT-1 series were, however, built

* In India, we have two LUTs ; one at Bangalore and one at Lucknow.

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by *Ford Aerospace* of USA, as per design specifications provided by ISRO. These satellites, after being brought from USA, were launched and parked in their parking slots, by American scientists stationed in Hassan near Bangalore. INSAT-2 series has however been indigenised to a large extent. Say for example, INSAT-2A was indigenised by about 40% and when once it was launched into the elliptical transfer orbit (200 km × 35786 km) by the Ariane rocket of *European Space Agency* from French Guyana, the raising and parking job was done by the Indian scientists and engineers of ISRO from the *Master Facilities Centre*, Hassan (Karnataka), from where commands are sent to INSATs. INSAT-2B, INSAT-2C, INSAT-2D as well as INSAT-2E were further indigenised and improved to give better performance on all fronts. *INSAT-2C and INSAT-2D (which had failed), however, were not provided with any meteorological transponders to relay weather data, and were aimed only at the enlarging and extending the television and telecom services even beyond Indian boundaries.* True to this aim, ISRO has leased nine transponders to INTELSAT (an international and inter-governmental space consortium created in association with 133 countries) under an agreement signed on 30.01.1995. This lease agreement fatches us $ 10 million (≈ Rs.42 crores), annually.

INSAT-2E has, however, been provided with sufficient improved meteorological pay loads, and is likely to enhance the meteorological services, which are being provided at present only by INSAT-1D, since these functions are now no longer being performed by INSAT-2A (which has already become inoperational due to depletion of on-board propellent), and INSAT-2B (due to burning of the electrical wiring of these parts), and since INSAT-2C and INSAT-2DT (replaced by INSAT-3E) do not have any meteorological pay loads.

From meteorological point of view, therefore, INSAT-2E (launched in June 1999) becomes the most important INSAT, as it has become the only INSAT providing such facilities after it has replaced INSAT-1D, which has been shut off, as stated earlier.

Considering the importance of INSAT-2E for continuing and enhancing the meteorological services, extra care has been taken for its success by removing the shortcomings of INSAT-2D and launching it in a larger elliptical orbit of 250 km × 36,155 km, thereby increasing perigee to 250 km from the earlier figure of 200 km to save on the on-board fuel, in raising it to the circular orbit of 35786 km.

INSAT-2E also carries improved meteorological pay loads, as it carries an improved version of the *Very High Resolution Radiometer* (VHRR) and a high resolution *Charge Coupled Device (CCD) Camera.* Besides the usual visible and thermal infra-red (I.R.) band, the VHRR will have a new *water vapour channel.* In addition to imaging cloud cover including the formation and movement of cyclones (as was being done earliear by INSAT-1D), the water vapour channel will aid in estimation of water vapour in the clouds, which will help in forecasting of rains in a better way. This will greatly help in precise and more accurate study of monsoons. Also, for the first time in India, this will enable the estimation of upper tropospheric winds, blowing at 8-12 km above the ground level. The CCD camera is an off shoot of the successful experience of working with such cameras in the Indian Remote Sensing (IRS) satellites. This camera operates in three bands — visible, near infra-red, and short wave infra-red ; and with its higher resolution of 1 km as against 2 km of VHRR, its data will compliment the VHRR data and assist in analysing cyclones, monitoring local severe storms, and in making better and reliable forecasts of heavy rainfall, etc.

In compliments to such expanded facilities in weather detection and warning made feasible by INSATs. IMD has developed a *main data utilisation centre* (at New Delhi); 20 subsidiary centres ; a network of nearly 100 unattended data collection platforms (DCP) ; and several *disaster warning systems* (DWS), spread around the country to suit local needs. The system has proved to be a great boon to forewarn and thus to forearm the public, especially settled in coastal areas of Andhra Pradesh and Tamil Nadu, to mitigate their miseries from perpetual cyclones. Several such DWS are being planned to be installed in

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coastal areas of Orissa, West Bental, Gujarat and Maharashtra, as pointed out earlier. Just as INSAT-2E is mainly being used for meteorological services, INSAT-3B on the other hand, is primarily to be used for business development & mobile communications. This has helped in improving VSAT* services, being used by about 300 corporate companies, Banks, stock markets, white goods sector, medium to heavy engineering companies, among others including telemedicinal & distant educational uses. INSAT-3E, on the other hand, is primarily meant for serving telecommunication and television services in the country.

The expenditure figures on INSATs are also quite high. The building cost of INSAT-2A and INSAT-2B was around Rs. 243 crores each, with an additional cost of around Rs. 161 crores for launching. Similarly, the total cost of INSAT-2E was about Rs. 550 crores, which included Rs. 286 crores paid for the launching to the Arianespace (French company) and the cost paid for insurance, etc. The cost of INSAT-3B is also approximated to be Rs. 500 crores including launching cost of Rs. 276 crores & insurance cost of Rs. 66 crore. *Be that as it may, no cost is high to modernize our systems, and to acquire such modern technologies, as it ultimately leads to enormous savings, in the form of prevention and protection of invaluable life as well as properties.*

PRECIPITATION GAUGES AND PRECIPITATION DATA

7.9. Types of precipitation

Although the moisture is always present in the atmosphere, but it is condensed only when the air is cooled, so that it becomes saturated with the same amount of water vapour. The usual mechanism by which the air is cooled to cause precipitation is the lifting of the air mass. There are three different methods by which the air mass gets lifted, so as to cause cooling and condensation of the atmospheric water vapour, and the subsequent precipitation mainly in the form of rain or sometimes under special conditions as snow, hail, sleet, etc. Depending upon the way in which the air is cooled, as to cause precipitation, we can have three kinds of precipitation, as given below :

7.9.1. Cyclonic Precipitation. It is caused by the lifting of an air mass due to the pressure difference. If low pressure occurs in an area, air will flow horizontally from the surrounding area, causing the air in the low pressure area to lift. The precipitation that results, is called **non-frontal cyclonic precipitation.** If one air mass lifts over another air mass, the precipitation is called **frontal cyclonic precipitation.** The boundary between these two air masses of different temperatures and densities (one warm air mass and the other colder) is known as a **front** *or frontal surface.* The large whirling mass of air, at the centre of which the barometric pressure is low, is known as a **cyclone.** The air that rushes horizontally into the low pressure area changes into a whirling mass because of the rotary motion of the Earth about its own axis. The cyclone is a very large mass of air ranging from 800 to 1600 km in diameter and moving with a velocity of about 50 km/hr. The central portion of this cyclone, where the pressure is low, acts like a chimney, through which the air gets lifted, expands, cools and finally gets condensed causing precipitation.

Cyclonic precipitation can occur in the form of drizzle, intermittent rain, or steady rain. If the precipitation is caused by a *cold front,* it is very intense and of short duration; while that caused by a *warm front* is more continuous. Third possibility is that of an occluded front. An *occluded front* occurs when a cold front over-takes a warm front. The precipitation pattern is a combination of both cold and warm frontal distributions. A cold front is that in which the warm air replaces the colder air, whereas in a warm front the case is opposite.

Cyclones are responsible for most of the rains in the central part of the United States, and for most of the winter rains in Haryana and Punjab in India.

7.9.2. Convective Precipitation. It is caused due to the upward movement of the air that is warmer than its surroundings. Generally this kind of precipitation occurs in

* Very small Aperture Terminal.

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tropics, where on a hot day, the ground surface gets heated unequally, causing the warmer air to lift up ; as the colder air comes to take its place. The vertical air currents develop tremendous velocities and are hazardous to aircrafts. Precipitation occurs in the form of showers of high intensity and short duration.

7.9.3. Orographic Precipitation. It is the most important precipitation and is responsible for most of the heavy rains in India. Orographic precipitation is caused by air masses which strike some natural topographic barriers like mountains, and cannot move forward and hence rise up, causing condensation and precipitation. The greatest amount of precipitation falls on the windward side, and the leeward side often has very little precipitation. Orographic barriers tend to increase both cyclonic and orographic precipitation because of the increased lifting involved. The rainfall is composed of showers and steady rainfall. A striking example of this kind of natural barriers is the southern slope of Himalayas in India. The winds, heavily laiden with moisture from the Bay of Bengal, strike the southern slope of Himalayas, causing intense rains ; so much so that in Cherrapunji, the average annual rainfall is of the order of 1270 cm. Similarly, the winds coming from the Pacific Ocean strike the western slopes of the Coastal ranges in Washington U.S.A., causing heavy rains there.

7.10. Measurement of Rainfall by Rain Gauges

In order to estimate the effects of precipitation, it is necessary to measure the precipitation, and to find out its distribution at various places on the earth.

All forms of precipitation are measured as the vertical depths of water that would accumulate on a level surface, if the entire precipitation remained where it fell. The total amount of precipitation falling on Earth in a given period, will hence be expressed as the depth to which it would accumulate on the *horizontal projection* of the Earth's surface, if there were no losses by evaporation or runoff and if any part of precipitation falling as snow or ice were also melted. The two important parts of precipitation (*i.e.* rain and snow) are measured separately by measuring devices, called rain gauges and snow gauges.

Since the amount of precipitation varies from place to place, it is necessary to install measuring devices at various key points. The simplest method of measuring precipitation is by setting up gauges with a horizontal circular aperture of known area, and collecting and measuring at regular intervals the precipitation collected in them. It is assumed that the amount of precipitation collected in the gauge is representative of a certain area around the point where the measurement is made.

7.10.1. Types of Rain-gauges. Any open receptacle with vertical sides can be used as a gauge for measuring rainfall. These refined receptacles, which are used for measuring rainfall, are called *rain gauges*. Two kinds of rain gauges, are generally used. First is the 'non-recording type' rain gauges, and the second is the 'recording type' rain gauges. These two types of gauges are described below :

(1) **Non-recording rain gauges.** Non-recording rain gauges are most widely used in India. They are known as 'non-recording type' because they do not record the rain, but only collect the rain. The collected rain is then measured by means of graduated cylinders, so as to directly represent the rainfall volume in cm of water depth.

i.e.
$$\left[\frac{\text{Volume of water collected in cm}^3}{\text{Area of the aperture of the gauge in cm}^2} = \text{cm depth of rain fallen}\right]$$

Various models have been designed for such gauges ; out of them, the **'Symon's type'** (Refer Fig. 7.9) was most widely used in India till 1969. However, since then, the

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Indian Meteorological Department has adopted another model called **Standard gauges** for their use at all rainfall stations in this country. The Symon's and other such types of rain gauges have, therefore, been replaced in India, and have become obsolete.

The complete installation of the new Standard type of non-recording rain gauge, which is now being used in India, is shown in Fig. 7.10. Such a rain gauge consists of a collector, with a gun metal rim, a base, and a polythene

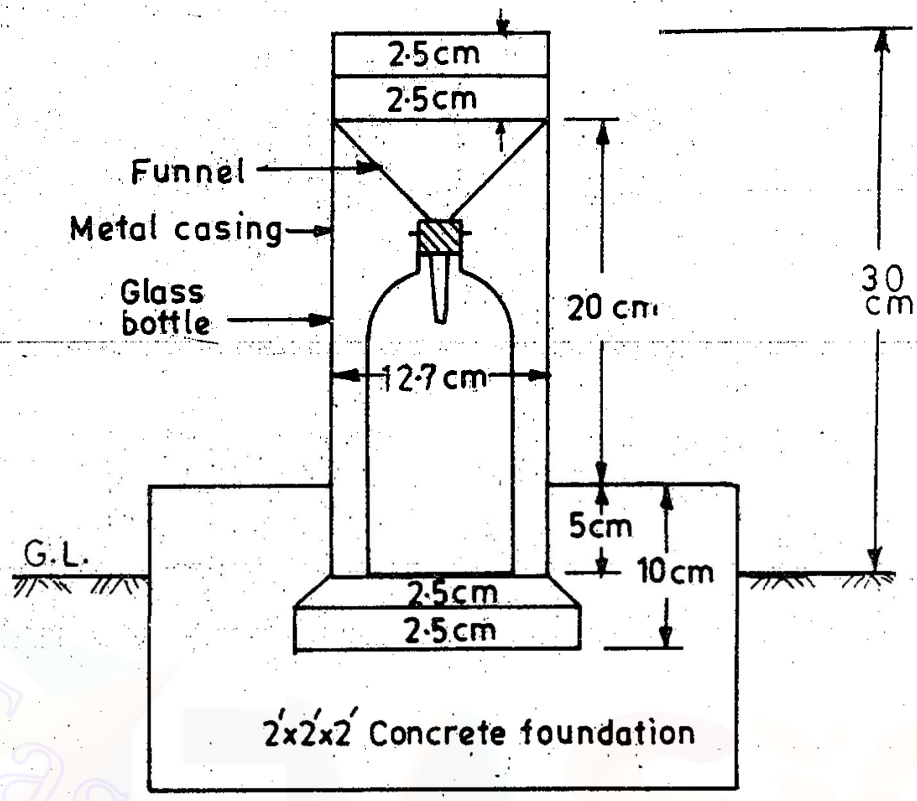

Fig. 7.9. Symon's non-recording rain gauge.

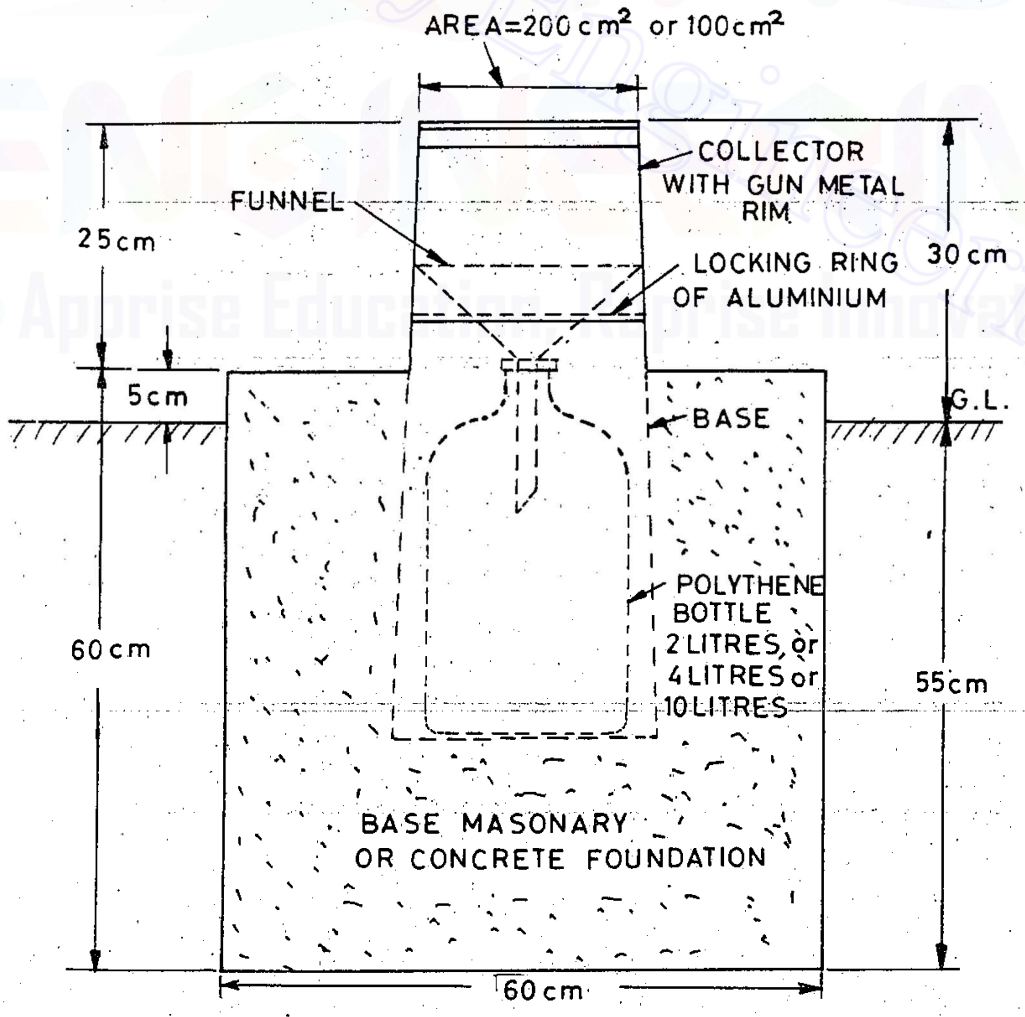

Fig. 7.10. Standard non-recording Rain-gauge of Indian Meteorological Department.

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bottle. Both the collector and the base are made of fibre glass reinforced polyester. The collector has a deep set funnel, and the complete rain gauge has a slight taper with the narrower portion at the top, as shown. The collectors have either apertures of 100 or 200 sq cm area (*i.e.* 112.9 mm or 159.6 mm dia) and are so made that they are interchangeable. There are also two types of interchangeable bases, the smaller base being used for all sizes of rain gauges except the largest. The polythene bottles are of three sizes having capacities of 2, 4 and 10 litres of water. The possible combinations of three essential components provide the following capacities of rain-gauges :

Table 7.4

S. No.	*Collector area in sq cm*	*Base size*	*Bottle capacity in litres*	*Normal capacity of rain gauge in cm of rainfall*
1	200	small	2	10
2	200	small	4	20
3	100	small	2	20
4	100	small	4	40
5	200	big	10	50
6	100	big	10	100

Of the above six possible combinations, those at serial numbers 1, 2, 4 and 6 are considered to be sufficient to meet the rainfall conditions in our country. Thus, the new standard rain gauges may have capacities of 10, 20, 40 and 100 cm of rainfall. It is worthwhile to note that the capacities of the rain gauges can be altered by changing either the collectors or the polythene bottles inside them. The 20 cm capacity rain gauge, with 200 sq cm collector and 4 litre bottle, is most widely used, and is sufficient to measue the 24 hours rainfall of most of the Indian stations. The 40 and 100 cm capacity gauges, being used only at few places, having heavy to very heavy rainfall, and 10 cm capacity gauges being used at places having very scanty rainfall.

All these standard rain gauges, irrespective of their capacities, can be installed at a standard height of 30 cm above the ground level, which was not possible with the old Symon's types.

The collector and the base are locked to each other by means of two complementary locking rings, fixed inside the collector at its lower end and the base at its top end.

When the rain falls, it is collected in the bottle. A man comes daily at 8.30 A.M. (in India) and measures and records the rainfall collected in the bottle by pouring it into a standard graduated measuring jar (two different types of jars for 200 cm^2 and 100 cm^2 collectors are available). *This process of measuring at 8.30 A.M. and recording the rainfall of the past 24 hours, is common throughout this country.* However, if the rainfall on a certain day is too much, and is likely to exceed the capacity of the bottle, then two or three intermediate readings may be taken, and their sum will then have to be recorded as the rainfall of the past 24 hours of the day, on which the final reading (*i.e.* at 8.30 A.M.) is taken. When the rainfall on a given day exceeds 2.5 mm, then that day is called a **rainy day.**

(2) **Recording type rain gauges.** Recording type rain gauges are those rain gauges, which can give us a permanent automatic rainfall record without any bottle reading. In this type of a gauge, a man has not to go to the gauge to measure or to read the amount of rain fallen. There is a mechanical arrangement by which the total amount of rain fallen, since the record was started, gets recorded automatically on a graph paper. The gauge, thus, produces a record of cumulative rain *vs.* time, in the form of a graph, which is known as the *mass curve* of rain fallen, and is shown in Fig. 7.11

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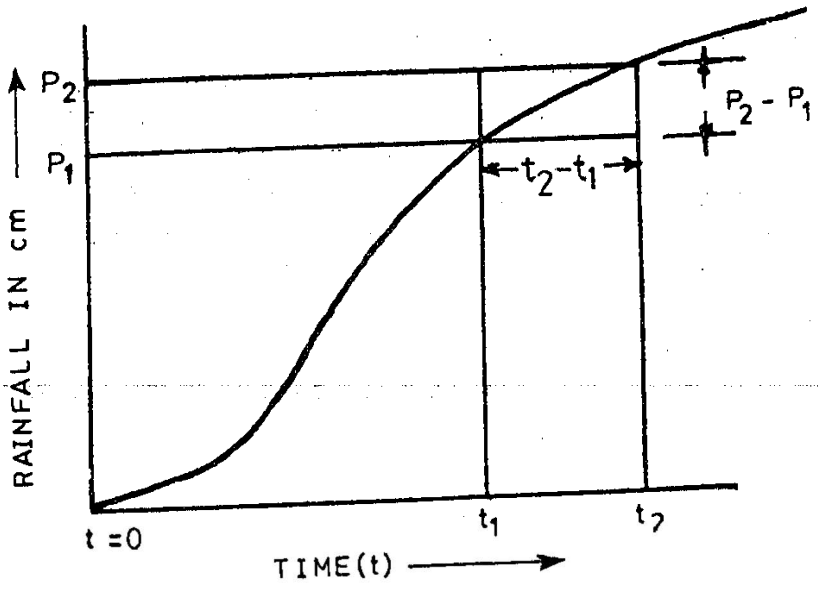
Besides giving the total amount of rain fallen at a station, such a curve will also help in indicating the times of onset and cessation of a rain, and, thereby indicating its duration. Moreover, the slope of the curve gives us the intensity of rainfall for any given period, as indicated in Fig. 7.11. Since such gauges represent the cumulative rain, they are sometimes called as *integrating rain gauges.* Moreover, such gagues can provide us with continuous recorded measurements for a number of days. They are of great utility in hilly and far off areas, where it is not practically feasible to daily visit the gauge station. Such gauges are, therefore, also sometimes known as *continuous rain gauges.*



$$\frac{P_2 - P_1}{t_2 - t_1} = \frac{dP}{dt} = \text{Rate of rainfall or Intensity of rainfall.}$$

Fig. 7.11. Mass curve of rain.

A recording rain gauge is generally installed in conjunction with an ordinary non-recording gauge exposed closeby. The nearby installed ordinary gauge will serve as an aid, and a standard for checking and adjusting the recordings of the recording gauge.

Various models have been designed for recording gauges ; on the basis of which, various gauges, such as "Tipping bucket type", "Weighing type", and "Floating type" etc. are

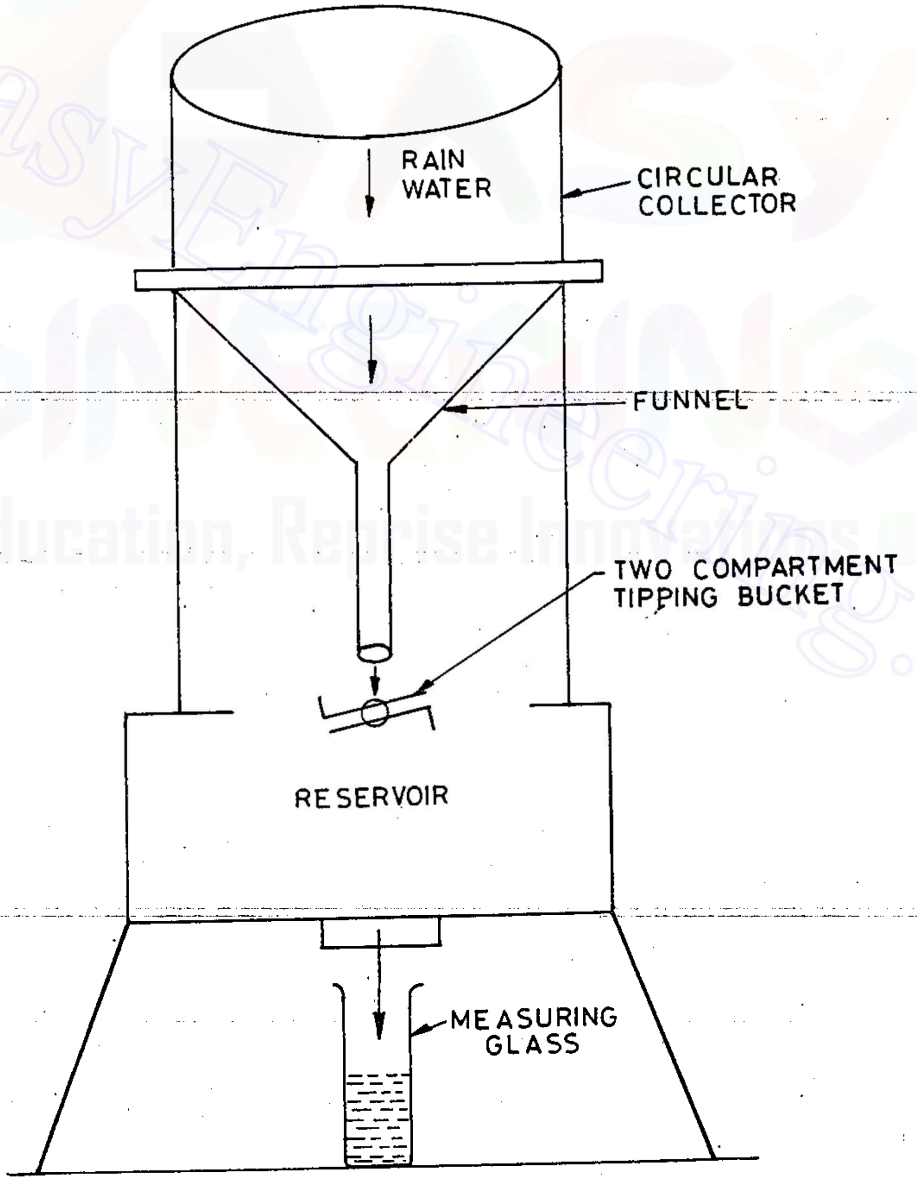


Fig. 7.12. (*a*) Indicating the recording mechanism of a "Tipping Bucket" type of a recording rain gauge.

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available in the market. In general, such a gauge consists of a *rotating drum* with a graph paper fixed around it. There is a pen point in contact with the graph paper, which moves up with the collected rain, and thus recording the cumulative rain, with the passage of time.

(*a*) In a *Tipping bucket gauge,* the rain water is first caught in a collector and then passed through a funnel. The funnel discharges the water into a two-compartment bucket. When 0.1. mm of rain water gets filled up in one compartment, the bucket tips, emptying into a reservoir, and moving the second compartment into place beneath the funnel, as shown in Fig. 7.12 (*a*) and (*b*). The tipping of the bucket completes an electric circuit, causing a pen to mark on a revolving drum. Since the movement of the tipping of the bucket can be transmitted electronically over distances, such gauges are generally installed in hilly and inaccessible areas, from where they can supply their measurements directly to the control room at meteorological station. No graph paper or drum etc. in this gauge is installed, and the rainfall measurements are directly recorded at the control room, and the gauges are left without any watch and ward, except periodic repairs and inspections.

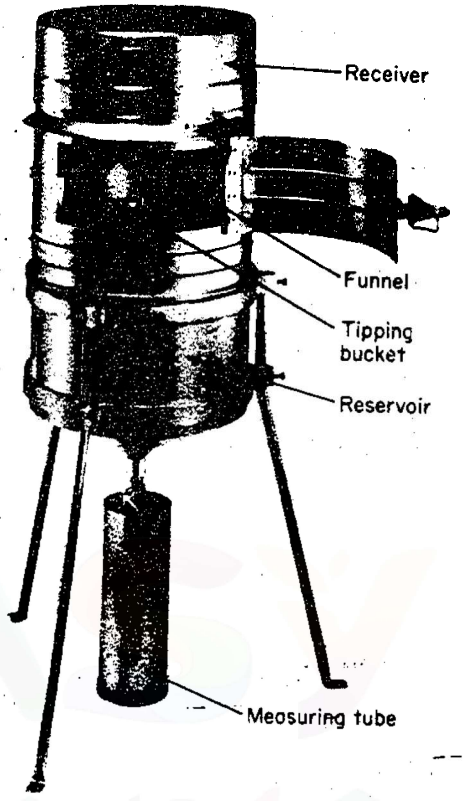

Fig. 7.12. (*b*) Photographic view of a Tipping Bucket type of a rain gauge.

(*b*) The *Weighing type gauge* weighs the rain which falls into a bucket placed on the platform of a spring or a lever balance, or any other weighing mechanism, as shown in Fig. 7.13. The increasing weight of the bucket helps in record-

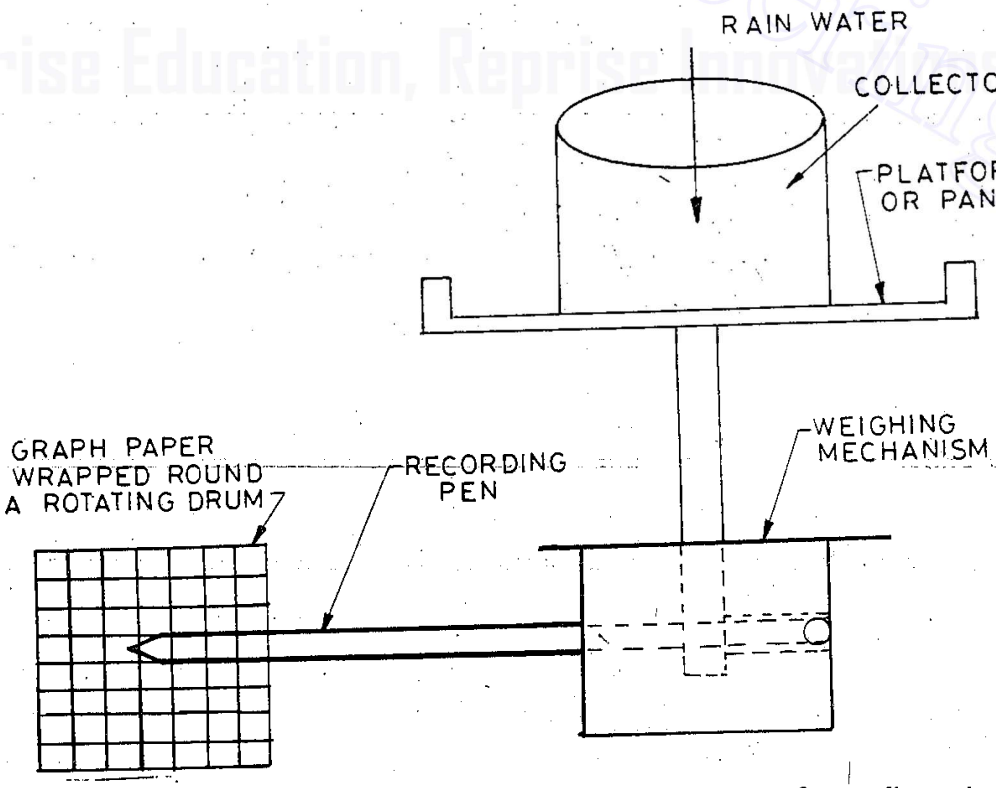

Fig. 7.13. Indicating the recording mechanism of a "Weighing" type of recording rain gauge.

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ing the increasing quantity of collected rain, with time, by moving a pen on a revolving drum. Such gauges are generally not used in India, though they are widely used in U.S.A.

(*c*) Various types of *Float recording gauges* are in use these days. In such a gauge, the rise of floating body due to the increasing rain catch, helps in lifting the pen point, which goes on recording the cumulative rain with time on a graph paper wrapped round a rotating drum. Such a gauge may be emptied either manually or by self-starting syphons. A float type gauge, provided with a self-starting syphoning arrangement, is most widely used in India ; and is popularly called *Natural Syphon Recording rain-gauge*. Such a rain gauge is described below in detail :

A natural syphon recording rain gauge is shown in Fig. 7.14 (*a*), (*b*) and (*c*). Different parts are properly named there.

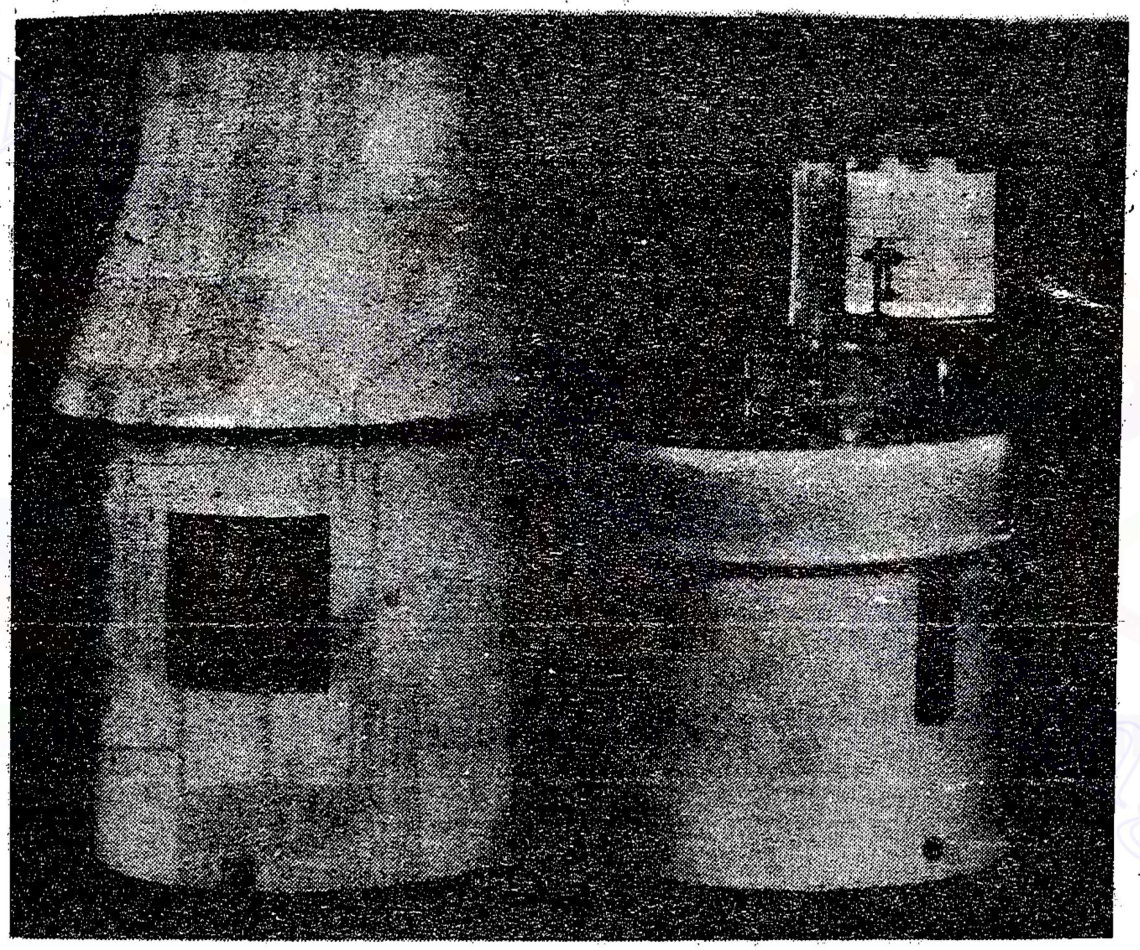

Fig. 7.14. (*a*) Photographic view of an automatic "Float recording" rain gauge popularly used in India.

The rain water entering the gauge at the top of the cover is led via the funnel to the receiver, consisting of a float chamber and a syphon chamber. A pen is mounted on the stem of the float, and as the water level rises in the receiver, the float rises, and the pen records on a chart, wrapped round a clockwise rotating drum, the amount of water in the receiver at any instant. The rotating drum completes one revolution in 24 hours, or sometimes in 7 days, depending upon the gauge, and the graph chart will have to be replaced after this much of time. Syphoning occurs, automatically when the pen reaches the top of the chart, and as the rain continues, the pen rises again from the zero line of the chart. If there is no rain, the pen traces a horizontal line from where it leaves off rising.

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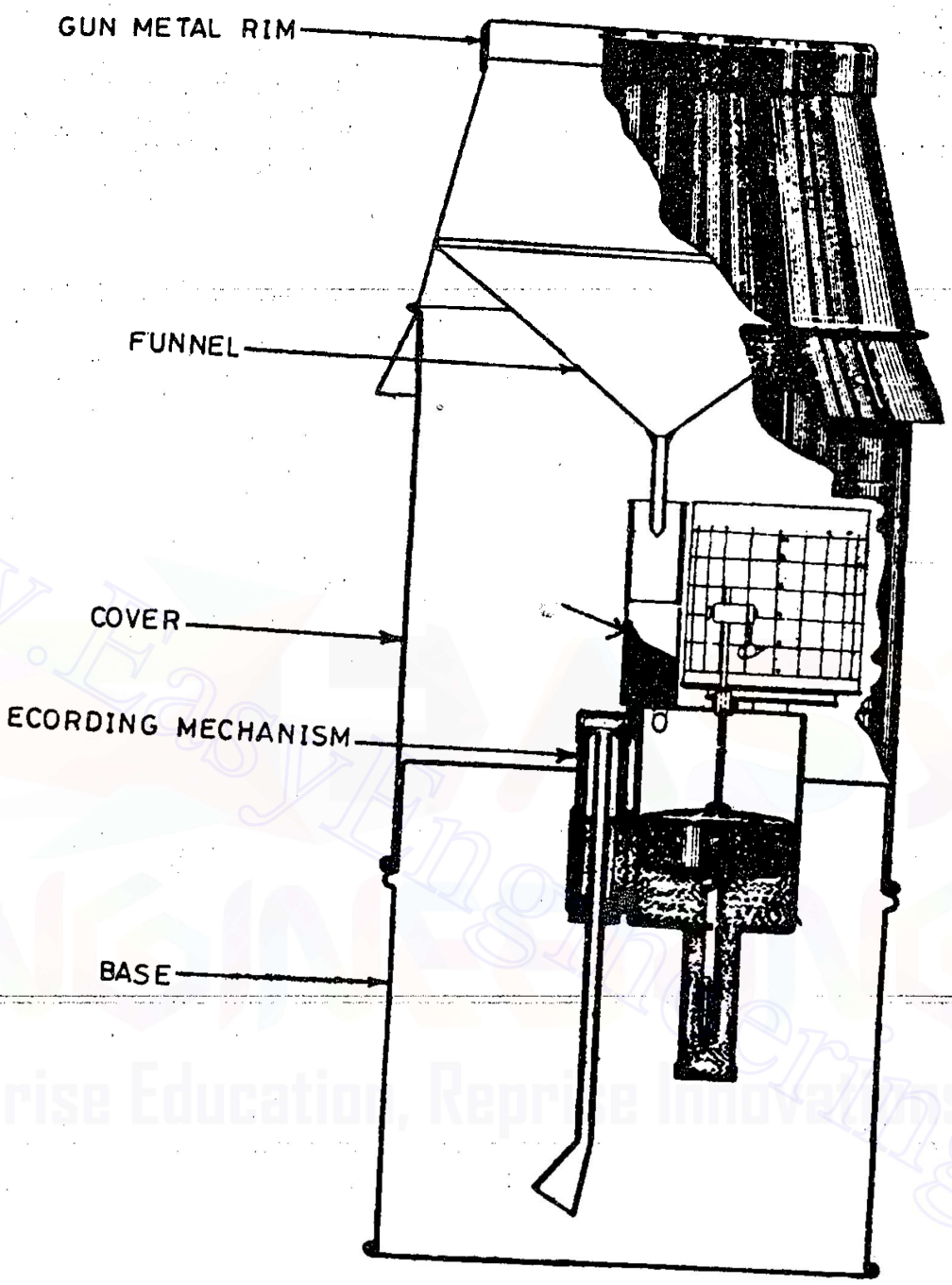

Fig. 7.14. (*b*) Section through an automatic rain gauge (Floating type).

The operation of the natural syphon arrangement is described below :

The syphon is arranged concentrically, the long discharge tube being surrounded by the shorter syphon chamber, which is directly connected to the float chamber. A glass piece is placed over the joint of these tubes. The passage connecting the two tubes at this joint is of almost capillary dimensions, but the sectional area is large enough to discharge the water collected in the receiver, with sufficient speed. The upper end of the water level falls to a certain depth. When syphon ceases to act, the water column is broken at a definite stage by a bubble of air which gets into the capillary, and freedom from dribbling is thus ensured. After syphoning, there is just enough water to float the float.

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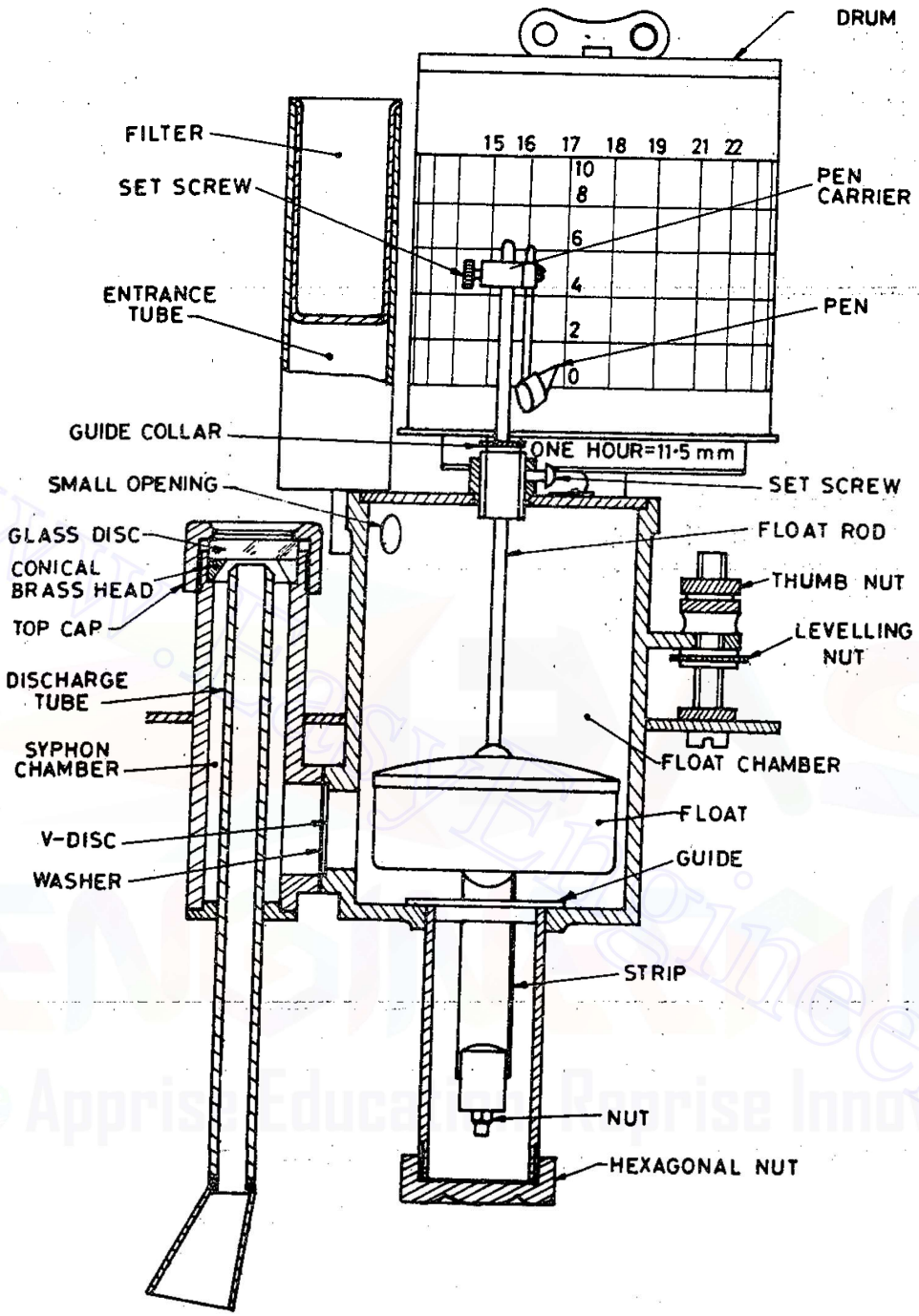

Fig. 7.14. (*c*) Recording mechanism details of Floating type automatic rain gauge.

7.11. Errors in Rain-gauge Measurement and Estimating True Rain-Catch

The amount of rain water collected and measured by a rain gauge may not always represent the exact amount, which would have been caught, if there were no factors trying to err the recorded amount. For example ; there may be instrumental errors in the gauges, or in their recording or measuring arrangements ; some rain water may get lost due to splash from the collector ; some water from an initial rain may get lost in moistening the gauge funnel and other inside surfaces ; blowing winds may tilt the rains from vertical, thus bringing lesser catch in the vertical gauge ; dents in the collector rim may change its receiving area, ; vertical upward air currents may impart upward acceleration to precipitation, thus bringing lesser catch in the gauge ; etc. All such factors try to introduce errors in the measured catches. Some of them may increase the catch,

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and some of them may decrease the catch. However, in general, it can be stated that almost all the errors that are introduced in rain catch measurements have a tendency to yield measurements which are too low. In other words, *the observed rain catch needs to be increased for the likely errors introduced in its measurement.*

Of all the possible errors, the most serious error is introduced by wind, which may result in a vertical acceleration of air, forced upward over the gauge. Such upward air current will impart an upward acceleration to the precipitation, about to enter the gauge, thus resulting in a deficient catch. The deficiency is greater for small rain drops than for large drops, and *is thus greater for lighter rains than for heavy rains.* It is not possible to precisely and reliably evaluate the wind error, because of problems involved in determining the true precipitation reaching the ground. Attempts at assessing the wind error, usually consist of comparing gauge measurements with weight changes in nearby lysimeters, or with changes in lake levels, or merely comparing measurements of shielded* and unshielded gauges. The error introduced by wind effects, increases with the wind velocity ; and since wind velocity is more at high attitudes, the gauge installed at higher altitude will definitely catch lesser rain catch. *Higher the gauge, greater will be the wind error, and hence more deficient will be the rain catch.*

Based upon the experimental works, it has now been possible to correlate the wind velocity and the gauge catch deficiency (*i.e.* so many per cent of lesser catch collected) as shown in Fig. 7.15.

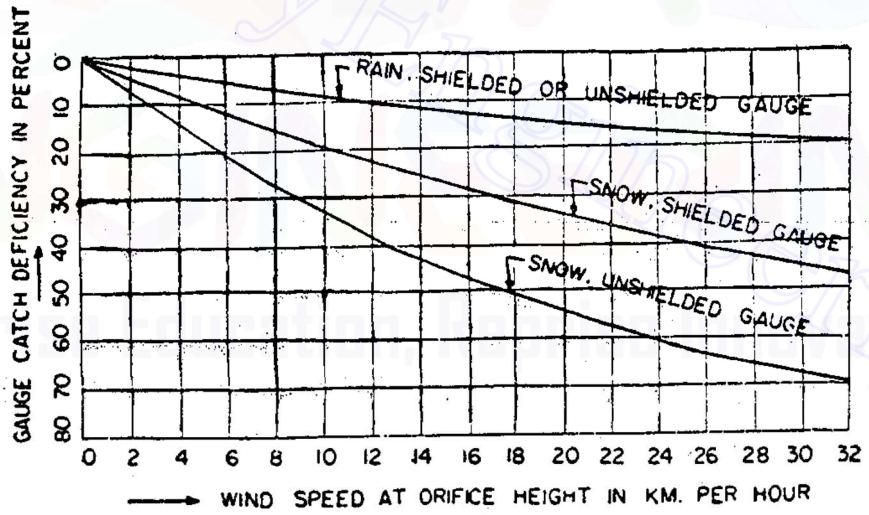

Fig. 7.15

The above chart has been provided by U.S. National Weather Service.

In order to get as correct measurements as possible, which may be free from wind errors, it may be worthwhile to install a gauge on a level ground with bushes or trees serving as *wind breakers,* provided they are not so close as to reduce the gauge catch. Trees or other obstacles serving as wind-breakers should subtend angles of 20° to 30° from the gauge orifice, with none greater than 45°, and should fairly well surround the gauge, so as to provide protection from all directions. An ideal site would be a *coniferous forest.*

Various methods and formulas have been put forward to estimate the true or actual amount of rainfall from the observed rain catch measurements. One of them is based on

* The gauges which are shielded from the effects of winds, etc.

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the premise that there is a relationship between the ratio of the unshielded gauge catch (P_u) to the actual precipitation (P), and the ratio of the unshielded-gauge catch (P_u) to the shielded gauge catch (P_s) ; or

$$\log \frac{P_u}{P} = b \cdot \log \frac{P_u}{P_s} \qquad ...(7.4)$$

where, b is a calibration coefficient, depending upon the type of the gauge.

Tilted gauges. When rain is falling vertically, a gauge inclined 10° from the vertical, will catch about 1.5 per cent less than it should. However, if a gauge on a level ground is inclined slightly towards the wind, it will catch more than the true amount.

7.12. Estimating Missing Rainfall Data

Sometimes, the rainfall amount at a certain rain gauge station for a certain day(s), may be missing due to the absence of some observer or instrumental failure. In such cases, it might be needed to estimate the missing rainfall amount by approximating the value from the data of the nearby rain gauge stations. The following methods are generally adopted for computing the missing rainfall data.

Three rain gauge stations as close to and as evenly spaced around the station with the missing record (*i.e.* station X) as possible, are, first of all, chosen. The rainfall data for these three stations (*i.e.* 1, 2 and 3) on the day for which the data at the station X is missing are now collected. The average annual rainfall values at all the four stations should also be known. Now, if the average annual rainfall at each of these three index stations differs within 10% of the average annual rainfall of the station X (*i.e.* the station with missing data), then a simple arithmetic average of the precipitations (corresponding to the missing precipitation) at the three index stations will give us the estimated quantity. Thus, if N_1, N_2, N_3 and N_x represent the average annual rainfalls at stations 1, 2, 3 and X respectively; and P_1, P_2, P_3 and P_x represent their respective precipitation data of the day for which the data is missing at station X; then we have

$$P_x = \frac{P_1 + P_2 + P_3}{3} \quad [\text{Provided } N_1, N_2 \text{ and } N_3 \text{ differ within 10\% of } N_x] \qquad ...(7.5)$$

However, when the average annual precipitation at any of the three stations differs from that at the station in question by more than 10%, the normal ratio method is used. In this method, the amounts at the three index stations are weighted by the ratios of their average annual precipitation values. Hence, the missing precipitation data P_x, in such a case, will be given by

$$P_x = \frac{1}{3}\left[P_1 \frac{N_x}{N_1} + P_2 \frac{N_x}{N_2} + P_3 \frac{N_x}{N_3}\right] \quad [\text{Provided any of } N_1, N_2 \text{ and } N_3 \text{ differs from } N_x \text{ by more than 10\%}] \qquad ...(7.6)$$

Example 7.1. *Precipitation station X was inoperative for part of a month during which a storm occurred. The respective storm totals at three surrounding stations A, B and C were 107, 89 and 122 mm. The normal annual precipitation amounts of stations X, A, B and C are respectively, 978, 1120, 935 and 1200 mm. Estimate the storm precipitation for station X.*

Solution. N_x = Average annual precipitation at X = 978 mm.

$\therefore$ 10% of N_x = 97.8 mm.

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Thus, maximum permissible annual ppt. at either of the three stations for taking ordinary mean = 978 + 97.8 mm = 1075.8 mm < 1120 and 1200 mm (given)

Hence, the annual ppt. at two of the three stations differ by more than 10% of N_x. Hence, weighted mean should be taken. Hence, using Eq. (7.6), we get

$$P_x = \frac{1}{3}\left[P_1 \frac{N_x}{N_1} + P_2 \frac{N_x}{N_2} + P_3 \frac{N_x}{N_3}\right]$$

or

$$P_x = \frac{1}{3}\left[P_a \frac{N_x}{N_a} + P_b \frac{N_x}{N_b} + P_c \frac{N_x}{N_c}\right]$$

Putting various given values, we get

$$P_x = \frac{1}{3}\left[107\left(\frac{978}{1120}\right) + 89\left(\frac{978}{935}\right) + 122\left(\frac{978}{1200}\right)\right]$$

$$= \frac{1}{3}[93.5 + 93.1 + 99.4] = \frac{1}{3}[286] = 95.3 \text{ mm}$$

Hence, the missing precipitation data = 95.3 mm **Ans.**

7.13. Checking the Consistency of Data of a Rain Gauge Station

Sometimes a significant change may occur in and around a particular rain gauge station. Such a change* occurring in a particular year, will start affecting the rain gauge data, being reported from that particular station. After a number of years, it may be felt that the data of that station is not giving consistent rainfall values. In order to detect any such inconsistency, and to correct and adjust the reported rainfall values, a technique, called **double mass curve method**, is generally adopted.

In this method, a group of 5 to 10 neighbouring stations are chosen, in the vicinity of the *doubtful station*. The yearly rainfall values, reported from this group of stations are serialled, and their mean yearly values are worked out for each consecutive year of available record. These mean yearly rainfall values (of the chosen group of stations) are serially arranged in a *reverse chronological order* (*i.e.* the latest year getting the first entry). Against these values, the recorded yearly rainfall values of the doubtful station are also serialled for each year. The cumulative values for both the columns are then worked out. The cumulative ppt. values of the doubtful station, X, say ΣP_x, and the cumulative values of the group averages say ΣP_{av} are then plotted on a graph paper, as shown in Fig. 7.16.

A decided change in the slope of the resulting curve (a st. line) persisting for more than 5 years, indicates a change in the precipitation regime of the station X.

The ppt. values at station X, prior to the period of change, are then corrected by using the relation :

*Common significant changes that may occur are :

(*i*) Location of raingauge station may undergo a change, which may go unreported; (*ii*) the neighbouring area of the raingauge station may undergo significant changes, such as new heavy built up coming in and around the station ; (*iii*) surrounding ecosystem may undergo significant changes due to calamities like slides, forest fires etc. ; observational errors may creep in the reported data.

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$$P_x' = P_x \frac{M'}{M} \qquad ...(7.7)$$

where P_x' = corrected ppt. at station X

P_x = original recorded ppt. at station X

M' = corrected slope of the double mass curve.

M = original slope of the double mass curve.

The method will become more clear on solving example 7.2.

Example 7.2. *The annual rainfall data, being reported from a station A, for 22 years are available, since 1969. In order to check the consistency of this data, six neighbouring stations have been chosen and the annual rainfall values of these stations have been averaged for all the years on record since 1969. These values are given in table 7.5.*

Table 7.5

Year	Yearly ppt. at station A in cm	6 station average yearly ppt. in cm
(1)	(2)	(3)
1969	177	143
1970	144	132
1971	178	146
1972	162	147
1973	194	161
1974	168	155
1975	196	152
1976	144	117
1977	160	128
1978	196	193
1979	141	156
1980	158	164
1981	145	155
1982	132	143
1983	95	115
1984	148	135
1985	142	163
1986	140	135
1987	130	143
1988	137	130
1989	130	146
1990	163	161

(a) Find out if any inconsistency in ppt. record of station A is indicated? And if yes, since when a change in the ppt. regime is indicated ?

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(b) Adjust the recorded data at station A and determine its mean annual precipitation.

Solution. The given data is arranged in a reverse chronological order, and their cumulative values are worked out, in table 7.6.

Table 7.6

Year (1)	*Yearly ppt. at station A* P_a *in cm* (2)	*Cumulative yearly ppt. at A* ΣP_a *in cm* (3)	*6 station Av. yearly ppt.* P_{av} *in cm* (4)	*Cumulative 6 station Av. yearly ppt.* ΣP_{av} *in cm* (5)
1990	163	163	161	161
1989	130	293	146	307
1988	137	430	130	437
1987	130	560	143	580
1986	140	700	135	715
1985	142	842	163	878
1984	148	990	135	1013
1983	95	1085	115	1128
1982	132	1217	143	1271
1981	145	1362	155	1426
1980	158	1520	164	1590
1979	141	1661	156	1746
1978	196	1857	193	1939
1977	160	2017	128	2067
1976	144	2161	117	2184
1975	196	2357	152	2336
1974	168	2525	155	2491
1973	194	2719	161	2652
1972	162	2881	147	2799
1971	178	3059	146	2945
1970	144	3203	132	3077
1969	177	3380	143	3220

A graph is now plotted between the values of col. (3) and (5) as shown in Fig. 7.16. Corresponding years from Col. (1) are also marked on the corresponding plotted points, as shown in this figure.

A perusal of this figure shows that the inconsistency has occurred from 1978. Hence, the present data, since 1978 to 1990 will be treated as correct, and the previous data prior to the year 1978, will be corrected. In other words, the ppt. values of years 1977 to 1969 will be corrected.

The correction ratio is read out from this figure,

$$= \frac{M'}{M} = \frac{\text{Corrected slope of double mass curve}}{\text{Original slope of double mass curve}}$$

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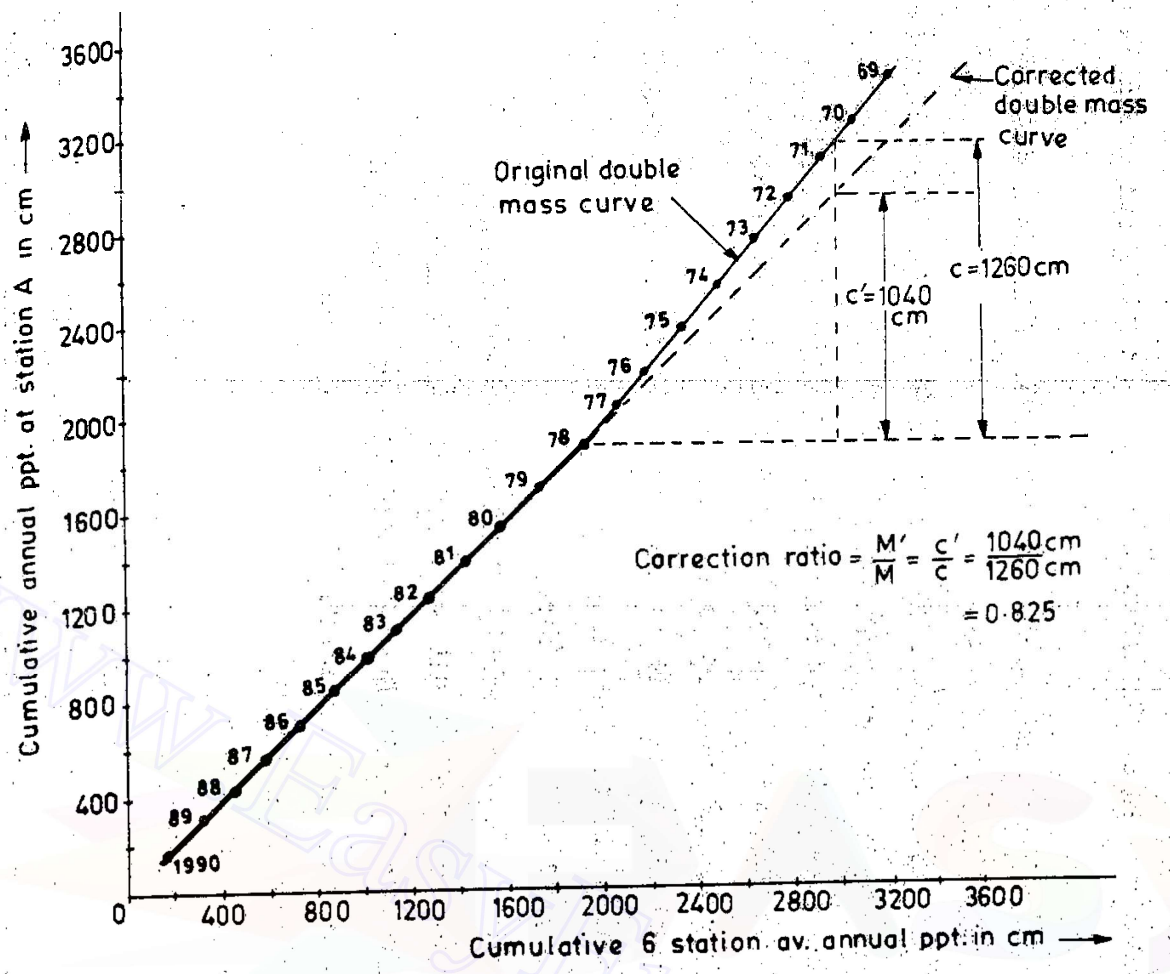

Fig. 7.16. Double mass curve plot for Example 7.2.

$$= \frac{C'}{C} = \frac{1040}{1260} = 0.825$$

The yearly ppt. values of station A (P_a) between the years 1977 to 1969 are thus, corrected by multiplying the original values by 0.825, as shown in table 7.7.

Table 7.7

Year	*Given ppt. values P_a in cm*	*Corrected ppt. values P_a' in cm = 0.825 × col. (2)*
(1)	*(2)*	*(3)*
1977	160	132
1976	144	119
1975	196	162
1974	168	139
1973	194	160
1972	162	134
1971	178	147
1970	144	119
1969	177	146

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The final adjusted values of annual ppt. at station *A* are serialled in table 7.8, and their mean value is determined as equal to **141.6 cm. Ans.**

Table 7.8

Year	*Corrected/adjusted annual ppt. values at station A in cm*
1969	146
1970	119
1971	147
1972	134
1973	160
1974	139
1975	162
1976	119
1977	132
1978	196
1979	141
1980	158
1981	145
1982	132
1983	95
1984	148
1985	142
1986	140
1987	130
1988	137
1989	130
1990	163
	$\Sigma = 3115$
	$av. = \frac{3115}{22} = 141.6$ cm. Ans.

7.14. Design of Rain Gauge Network

Rainfall records constitute the most important and fundamental data required for hydrological investigations. They are required for : analysing storms, fixing design flood, forecasting floods in a river, reservoir regulation, etc. For all such studies and investigations, a well distributed network of rain gauge stations within the catchment is essential.

To obtain reliable results, the various rain gauges should be evenly and uniformly distributed within a given catchment. Moreover, the total number of rain gauges installed within a given catchment area should neither be too many as to be costly, nor should be too less as to give unreliable results. However, this **Areal density of rain gauges*** may vary considerably from region to region in any country. The density of gauges may also depend upon the basic purpose behind their installation.

For example, a relatively sparse network of stations (*i.e.* lesser number of gauges per unit area), would be sufficient for studies of large general storms or for determining annual average rainfall over large areas of large terrain. On the other hand, however, a

* The number of gauges per unit of area in the catchment.

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dense network (*i.e.* more number of gauges per unit area) is required to determine the rainfall pattern in the thunder-storms.

If the network density is more (*i.e.* lesser area served per gauge), there is more probability that the storm centre will be recorded by the gauge. A network should also be planned to yield a representative picture of the areal distribution of rainfall. Hence, *there should be no concentration of gauges in heavy rainfall areas at the cost of dry areas, or vice-versa*. However, unfortunately, in actual practice, far flung and dry areas are generally left without gauges, because the cost of installing and maintaining a network, and accessibility of the gauge site to an observer are always very much taken into consideration, while deciding their locations.

In India, at present, there are about 5000 rain gauge stations spread over the vast area of about 31.5 lakh sq. km, which gives an average of about one station for every 630 sq km. This density is insufficient, and more gauges are required.

The minimum rain gauge density required for different types of catchments, as per the guidelines of World Meteorological Organisation, are given below :

(1) *For flat regions of temperate Mediterranean and Tropical zones* — one station per 600—900 sq. km. For countries with limited funds, population or other factors such as lack of communication, the density may be reduced to one station per 600—3000 sq. km.

(2) *For Mountainous regions of temperate Mediterranean and Tropical zones* — one station per 100—250 sq. km. For countries with problems mentioned above, the density may be reduced to one station per 250—1000 sq. km.

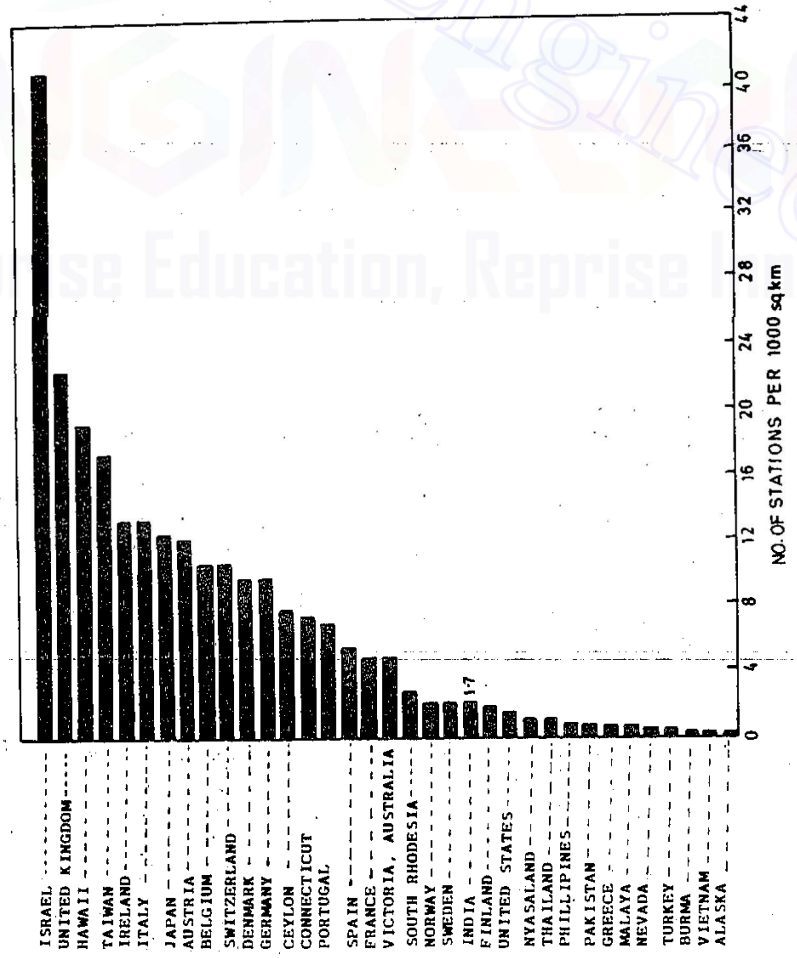

Fig. 7.17. Rain gauge density as adopted in different countries.

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(3) *For arid or polar zones* — the minimum density may be one station per 1500—10,000 sq. km. depending upon the feasibility.

In India, however, the standard prescribed by a High Level Committee on floods set up by the Govt. of India, recommends to establish minimum 1 station per 500 sq. km. (*i.e.* 1 station per 200 sq. miles).

Fig. 7.17 represents the rain gauge density values, as adopted in different countries of the world. This figure reflects that Israel has the maximum rain gauge density of the order of 37 gauges/1000 sq. km of area, as against the Indian figure of 1.7 gauge per 1000 sq km area.

7.14.1. Optimum Number of Rain Gauges. Statistics has been used in determining the optimum number of rain gauges required to be installed in a given catchment. The basis behind such statistical calculations is : that a certain number of rain gauge stations are necessary to give average rainfall with a certain percentage of error. If the allowable percentage error is more, lesser number of gauges would be required. Based upon this statistical principle, the optimum number of rain gauges (N) can be obtained by the following equation :

$$N = \left(\frac{C_v}{E}\right)^2 \qquad ...(7.8)$$

where C_v = Coefficient of variation of rainfall based on the existing rain gauge stations; and

E = Allowable percentage error in the estimate of basic mean rainfall.

C_v can be computed as follows.

(*i*) Calculate mean average annual rainfall value as :

$$\overline{P} = \frac{\Sigma P}{n} \qquad ...(7.9)$$

where n is the number of rain gauges existing, and ΣP is the total rainfall, *i.e.* summation of their normal annual rainfall values.

(*ii*) Calculate mean of squares, *i.e.*

$$\overline{P^2} = \frac{\Sigma P^2}{n} \qquad ...(7.10)$$

(*iii*) Calculate Sample Standard Deviation, *i.e.*

$$\sigma = \sqrt{\frac{n}{n-1}\left[\overline{P^2} - (\overline{P})^2\right]} \qquad ...(7.11)$$

(*iv*) Calculate the coefficient of variation, *i.e.*

$$C_v = \frac{100\,\sigma}{\overline{P}} \qquad ...(7.12)$$

Example 7.3. *There are four rain gauge stations existing in the catchment of a river. The average annual rainfall values at these stations are 800, 620, 400 and 540 mm respectively. (a) Determine the optimum number of rain gauges in the catchment,*

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if it is desired to limit the error in the mean value of rainfall in the catchment to 10%.
(b) How many more gauges will then be required to be installed.

Solution. (*i*) Mean rainfall $= \bar{P} = \frac{\Sigma P}{n} = \frac{800 + 620 + 400 + 540}{4} = \frac{2360}{4} = 590$ mm

(*ii*) $\Sigma P^2 = (800)^2 + (620)^2 + (400)^2 + (540)^2$

$= 6,40,000 + 3,84,400 + 1,60,000 + 2,91,600 = 14,76,000$

$\overline{P^2} = \frac{\Sigma P^2}{n} = \frac{14,76,000}{4} = 3,69,000$

(*iii*) Standard Deviation σ is given by

$$\sigma = \sqrt{\frac{n}{n-1}\left[\overline{P^2} - (\bar{P})^2\right]}$$

$$= \sqrt{\frac{4}{4-1}[3,69,000 - (590)^2]} = \sqrt{\frac{4}{3}[3,69,000 - 3,48,100]}$$

$$= \sqrt{27,867} = \mathbf{166.93}$$

(*iv*) The co-efficient of variation C_v is given by

$$C_v = \frac{100\,\sigma}{\bar{P}} = \frac{100 \times 166.93}{590} = 28.29$$

(*v*) Now, optimum number of rain gauges (N) is given by

$$N = \left(\frac{C_v}{E}\right)^2$$ where $C_v = 28.29$ (Calculated above)

$E = 10\%$ (*i.e.* Per cent error allowed)

$\therefore$ $$N = \left(\frac{28.29}{10}\right)^2 = \mathbf{8.004}$$; Say 8 Nos. **Ans.**

(*b*) Additional gauges required to be installed

$= 8 -$ Existing 4 gauges $= 8 - 4 = 4$ Nos. **Ans.**

7.15. Average Annual Rainfall and Index of Wetness

The amount of rain collected by a given rain gauge in 24 hours is known as daily rainfall, and the amount collected in one year is known as *annual rainfall.* This annual rainfall at a given station should be recorded over a number of years, say 35 to 40 years or so. In India, this rainfall cycle period is taken to be about 35 years. The mean of the annual rainfalls over a period of 35 years or so, is therefore, known as *average annual rainfall or normal annual rainfall* of the given station.

Whenever, we talk of the rainfall at any given place, we generally refer to the average annual rainfall of that place. Thus, when we speak that Cherrapunji gets 1270 cm of yearly rain, it means that this figure has been averaged over a long period of about 35 years. This is known as average rainfall. But in any given particular year, the rain may not fall equal to this amount. It may be less than this average value or may exceed this average value. The ratio of the actual rainfall in a given particular year at a given place to the normal rainfall of that place, is known as *index of wetness.*

Index of wetness

$$= \frac{\text{Actual rainfall in a given year at a given place}}{\text{Normal annual rainfall of that place}} \qquad ...(7.13)$$

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Index of wetness, thus, gives us an idea of the wetness of the year, and hence, it indicates the deficiency of rain. 60% index of wetness means a rain deficiency of 40%. A deficiency of about 30—45% is known as large deficiency, 45 — 60% as serious deficiency, and more than 60% as disastrous deficiency.

The year in which the rainfall is less than the average annual rainfall, is called a bad or a sub-normal year; and if the rainfall in a given year is more than the average, then it is known as a good year. If the rainfall in a particular year is approximately equal to the annual average value, then it is known as a normal year or an average year.

7.16. Indian Rainfall

India is a tropical country, and its precipitation varies considerably in frequency, intensity, as well as in duration. The total precipitation occurs either as rainfall or as snowfall (in higher reaches). The largest amount of precipitation (75% or so) falls as rain during the *monsoons* (May—Oct.); whereas, moderate amount (25% or so) mainly as snow, falls during the winter season (Dec.—Feb.) as *winter precipitation.*

Besides these two major precipitation periods, *minor* quantity of rain falls, either as *post monsoon* (*i.e. retreating monsoon*) *rain* or as *pre-monsoon rain.*

Post monsoon rain (Oct.—Nov.) may fall in Tamil Nadu State and other nearby coastal areas; whereas, **Pre-monsoon rain** (March—May) may occur in Kerala, West Bengal, Assam, and other east coast areas.

7.16.1. Monsoon Precipitation. The *monsoon*, strictly speaking, the ***south-west monsoon*** (or *summer monsoon*), causes about 75% of annual precipitation, covering almost the entire country, except Jammu & Kashmir, and south eastern parts of South India (peninsula). Monsoon winds, infact, originate in the Indian Ocean, and thus appear first in Kerala, by the end of May. During such a summer season, south-westerly monsoon winds blow at speeds as high as 30 km/hr. These winds get divided into two branches, due to the relief of our country. It is the Indian peninsula, which rather deflects the monsoon, into the two branches, *viz.* :

(*i*) the Arabian Sea branch; and

(*ii*) the Bay of Bengal branch.

The Arabian Sea branch of monsoon, first of all, appears as early as middle or end of May, in Kerala. These moisture laiden monsoon winds strike the western ghats straight, thus causing heavy rain, of the order of 250 cm of so, between western ghats and west coast (Malabar coast). These winds, containing the residual moisture, then travel as far as east coast, causing smaller rain in Madras (30 cm or so). Another current of these winds move northwards, striking the hills of Chhota Nagpur, causing moderate to heavy rain there, of the order of 150 cm or so. A third part of these winds, moves over to Rajasthan, passing the parallel Aravali ranges without striking and without causing rain, till reaching and striking the hills of Punjab, causing moderate to heavy rain in Simla, Kangra, Dharamsala, etc. This Arabian Sea monsoon reaches Delhi by June end, causing low to moderate rain.

The *Bay of Bengal branch of monsoon* arises from the Bay of Bengal, and is deflected into India by the Arakan mountains of Burma, and by Himalayas.

A part of these monsoon winds move towards the Assam hills (Garo, Khasi and Jaintia). The winds, thus strike these high hills, causing very heavy rainfall here. So much so, that the place of maximum rainfall of India (*i.e.* Cherrapunji) is also situated here in Khasi hills. A part of these winds also advances westwards, towards Bihar and U.P., ultimately reaching Delhi by about June end *i.e. by* about the same time when the Arabian Sea branch reaches Delhi. But as the winds advance, they go on becoming less and less laiden with moisture, and hence the rainfall goes on reducing from Calcutta (160 cm) to Benares (100 cm) and to Delhi (60 cm).

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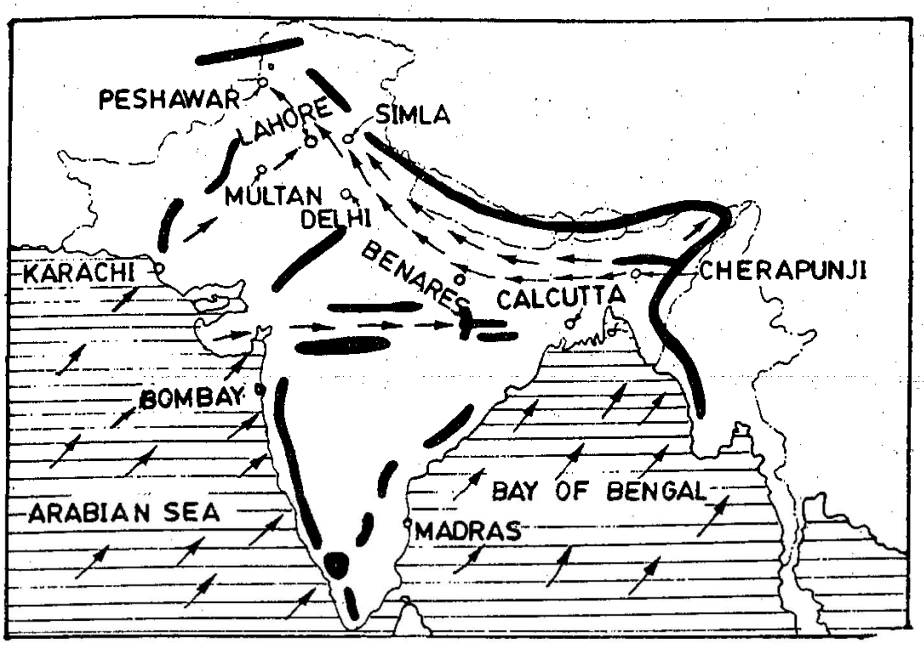

Fig. 7.18. Summer monsoons.

Infact, a low pressure region, called *monsoon trough*, is formed between these two monsoon branches, which extends from the Bay of Bengal to Rajasthan, and its position decides the precipitation pattern over the country. During the monsoon period, cloudy weather persists, giving relief from oppressive heat, and large amounts of rain falls in frequent spells almost over the entire country, owing to the passage of low pressure regions. Depressions formed in Bay of Bengal @ 2 to 3 Nos. per month, move along the trough, causing heavy rain up to as high an intensity as 20 cm/day or so. The heaviest rain falls in Assam and north-eastern States with annual value being of the order of 200—400 cm. West coasts and Western ghats (200—300 cm), West Bengal (120—160 cm), U.P., Haryana and Punjab (100—120 cm), then receive decreasing amounts of rain.

7.16.2. Winter Precipitation. By about mid December, *western disturbances* of extra tropical origin travel eastward to India, crossing Afghanistan and Pakistan, causing heavy rainfall and snowfall in Himalayas and Jammu & Kashmir. Some light rainfall also occurs in northern plains.

Lighter rains (10—12 cm) during winter are also caused in southern parts of Tamil Nadu, by *winter monsoon* (*north-east monsoon*), passing over Bay of Bengal and striking the eastern ghats.

Standard charts, showing the Average annual rainfall and normal Monsoon rainfall values all over India, are given in author's another book titled "Hydrology and Water Resources Engineering".

7.17. The Mean Rainfall Over a Drainage Basin

A given drainage basin is divided into various parts or sub- basins, and rain gauge stations are evenly distributed over that basin. The rain-catch at one station in a basin may be different from that of a second station in the same basin. An average value of these rain-catches is worked out, so as to get an idea of the average precipitation on the entire basin. The following methods are generally used to work out the mean rainfall on an area.

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7.17.1. Arithmetical Mean Method. This is the simplest method that can be used. It consists of averaging all the amounts that have been recorded at the various stations in the area, as given below :

$$P = \frac{P_1 + P_2 + P_3 + P_4 + \ldots + P_n}{n} \qquad \ldots(7.14)$$

where P = the mean precipitation on the basin. $P_1, P_2, P_3, \ldots, P_n$ are respective precipitations at stations 1, 2, 3, ..., n ; and n is the total number of stations.

If the stations are uniformly distributed over the area, and the rainfall rate does not differ much at various stations, then this method is quite satisfactory. But if there are many topographic differences, and the rainfall rates vary considerably, then this method may lead to large errors.

7.17.2. Thiessen's Mean Method. In this method, adjacent stations are joined by straight lines, thus dividing the entire area into a series of triangles. Perpendicular bisectors are erected on each of these lines, thus forming a series of polygons, each containing one and only one rainfall station. It is assumed that the entire area within any polygon is nearer to the rainfall station that is included in the polygon than to any other rainfall station. The rainfall recorded at that station is, therefore, assigned to that polygon. If P is the mean rainfall on the basin, and area of the basin is A, then

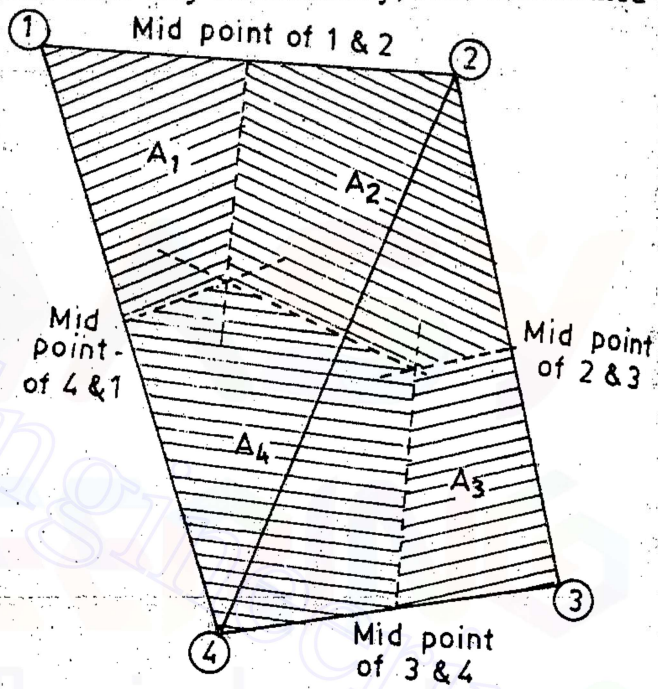

Fig. 7.19. Thiessen's Polygon method.

$$P = \frac{A_1P_1 + A_2P_2 + A_3P_3 + \ldots + A_nP_n}{A} \qquad \ldots(7.15)$$

where $P_1, P_2, P_3, \ldots, P_n$ represent rainfalls at the respective stations, whose surrounding polygons have the areas $A_1, A_2, A_3, \ldots, A_n$ respectively.

A graphical representation of this method is shown in Fig. 7.19 by taking 4 stations.

To divide the area into triangles, most frequently the shorter of the two diagonals is to be joined. Sometimes, the reverse is to be adopted when the clear polygons are not formed and they overlap.

So in Fig. 7.19, let us join the various stations namely 1, 2, 3 and 4. And then divide it into triangles by joining the shorter diagonal, *i.e.* 2 and 4, then draw perpendicular bisectors (shown by dotted lines) so as to divide the area into polygons. The figure is self-explanatory.

The method will become more clear on solving the following numerical example.

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Example 7.4. *Find the mean precipitation for the area sketched in Fig. 7.20 by Thiessen's method. The area is composed of a square plus an equilateral triangular plot of side 4 kilometres. Rainfall readings in centimetres at the various stations are also given in the Fig. 7.20.*

(U.P.S.C., Engg. Services, 1968)

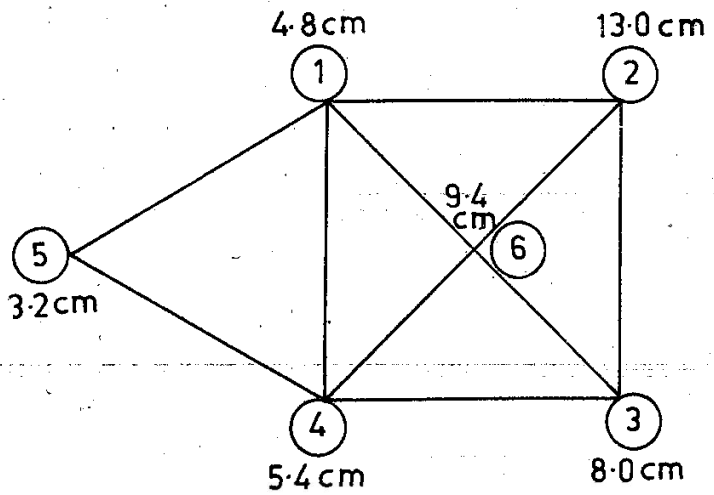


Fig. 7.20

Solution. First of all, divide the entire area into triangles by joining the various consecutive stations, *i.e.* 1, 2, 3, 4, 5, and 1, 6, 3, and 1, 4. Now, draw perpendicular bisectors on these lines to divide the area into polygons, as shown in Fig. 7.21.

In triangles *ABC* and *DBE*

Side *DE* is the join of the mid-points of *AB* and *BC*

Therefore $DE = \frac{1}{2} AC$

If a = side of the square

then $AC = \sqrt{2} \cdot a$

$\therefore \quad DE = \frac{\sqrt{2} \cdot a}{2} = \frac{a}{\sqrt{2}}$

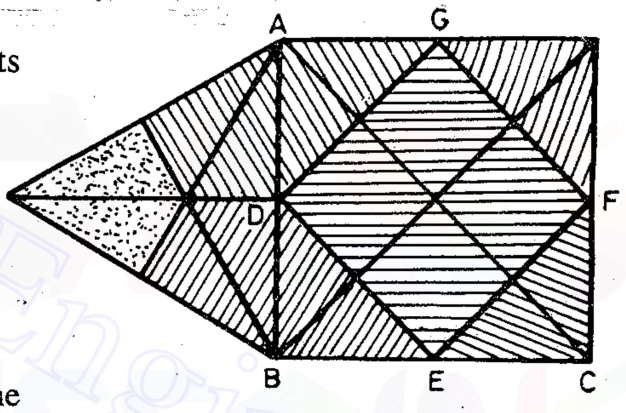


Fig. 7.21.

Area of the square *DEFG* feeding the station No. 6

$$= \frac{a}{\sqrt{2}} \times \frac{a}{\sqrt{2}} = \frac{a^2}{2} \qquad ...(1)$$

Area of square plot contributing to each corner station

$$= \frac{1}{2} \cdot \frac{a}{2} \cdot \frac{a}{2} = \frac{a^2}{8}$$

In equilateral triangle, the three vertices stations will be fed equally, *i.e.* equal to 1/3 × area of triangle

$$= \frac{1}{3} \times \left(\frac{1}{2} a \cdot \frac{\sqrt{3}}{2} a \right) = \frac{1}{3} \times \frac{\sqrt{3}}{4} a^2 = \frac{a^2}{4\sqrt{3}}$$

Thus, the areas feeding various stations can now be stipulated as below :

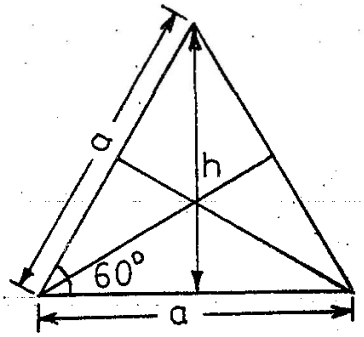
A_1 = Area due to triangular plot + Area due to square plot

$$= \frac{a^2}{4\sqrt{3}} + \frac{a^2}{8} \text{ ; where } a = 4 \text{ km}$$

$$= 0.26925\, a^2 = 0.26925 \times 16 = 4.308 \text{ km}^2$$

$$A_2 = \frac{a^2}{8} = \frac{16}{8} = 2 \text{ km}^2$$

$$A_3 = \frac{a^2}{8} = 2 \text{ km}^2$$



$\frac{h}{a/2} = \tan 60° = \sqrt{3}$; $h = \frac{\sqrt{3}}{2} a$

Fig. 7.22.

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$$A_4 = A_1 = 4.308 \text{ km}^2$$

$$A_5 = \frac{a^2}{4\sqrt{3}} = 2.309 \text{ km}^2$$

$$A_6 = \frac{a^2}{2} = 8 \text{ km}^2$$

Using $P = \dfrac{P_1A_1 + P_2A_2 + \ldots + P_nA_n}{A}$, we have

$$P = \frac{4.8 \times 4.308 + 13 \times 2 + 8 \times 2 + 5.4 \times 4.308 + 3.2 \times 2.309 + 9.4 \times 8}{4.308 + 2 + 2 + 4.308 + 2.309 + 8} = 7.36 \text{ cm}$$

Hence, the mean precipitation over the area = 7.36 cm. **Ans.**

7.17.3. The Isohyetal Method. Isohyets are the contours of equal rainfall. They are drawn on the map by using common sense, after the rainfall at each station is plotted.

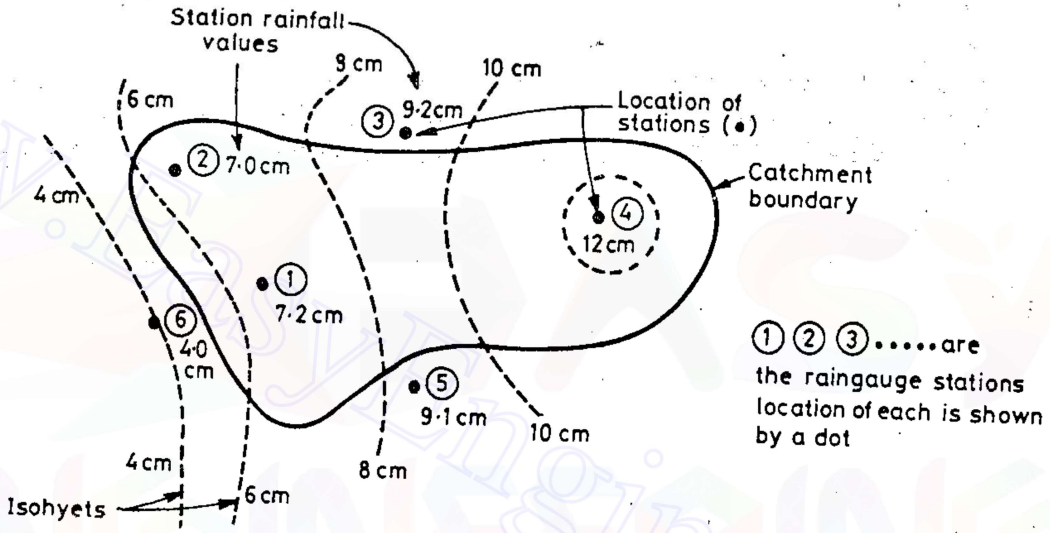

Fig. 7.23. Isohyetal method for computing average rainfall.

The area between the adjacent isohyets is either estimated on the graph paper or measured by planimeter. Let them be $A_1, A_2, A_3, \ldots, A_n$; and let the average precipitation for these areas be $P_1, P_2, P_3, \ldots, P_n$; then the mean precipitation P on the basin is given by

$$P = \frac{P_1A_1 + P_2A_2 + P_3A_3 + \ldots + P_n \cdot A_n}{A_1 + A_2 + A_3 + \ldots + A_n} \quad \ldots(7.16)$$

If A_1 is the area between the contours of 6 cm and 8 cm (Fig. 7.23).

then $P_1 = 7$ cm

Similarly, here in figure, we have

$P_2 = 9$ cm

Note : Since the area between the two adjacent Isohyets is assumed to be having a rainfall equal to the arithmetic average of rainfall values represented by the two isohyets, the method evidently assumes uniform arial distribution of rainfall in between two isohyets. Such an assumption may not always be true, as orological considerations of the ground below might affect this assumption.

Example 7.5. *The isohyets for annual rainfall over a catchment basin were drawn. The areas of strips between the isohyets are indicated below in table 7.9. Find the average depth of annual precipitation over the basin.*

Table 7.9.

Isohyets cm	*Area in sq. km.*	*Isohyets cm*	*Area in sq. km.*
75—85	*580*	*105—115*	*1,000*
85—95	*2,960*	*115—135*	*610*
95—105	*2,850*	*135—155*	*160*

(U.P.S.C. Engg. Services, 1970)

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Solution. $$P = \frac{580\times 80 + 2960\times 90 + 2850\times 100 + 1000\times 110 + 610\times 125 + 160\times 145}{580 + 2960 + 2850 + 1000 + 610 + 160}$$

$$= \frac{46{,}400 + 2{,}66{,}400 + 2{,}85{,}000 + 1{,}10{,}000 + 76{,}250 + 23{,}200}{8{,}160}$$

$$= \frac{8{,}08{,}250}{8{,}160} = \mathbf{99.05\ cm}\quad \textbf{Ans.}$$

7.17.4. Merits and Demerits of Three Methods. As is quite evident, the 1st, and 2nd methods, *i.e.* 'arithmetic mean' and 'Thiessen's mean' are only mechanical and mathematical processes and do not require any special skill, but on the contrary, the third method, *i.e.* 'the Isohyetal method' requires a lot of extra special judgement for drawing the contours' (isohyets). The third method is the most accurate method, if the completion of contours (isohyets) is best done. But the accuracy depends upon the judgement of the person completing the contours.

The chief advantage possessed by the 'Thiessen's' and 'Isohyetal' methods over the 'arithmetic mean method' is the fact that in these methods, suitable importance is attached to the various stations, which is not so in 'arithmetic mean' method. Also in these methods, the stations which are situated beyond the boundary of the drainage basin, are also used to determine the mean rainfall on the basin, but their influence reduces as their distance from the boundary increases.

Example 7.6. *The isohyetal map for 24 hour storm gave the areas enclosed between different isohyets, as follows :*

Isohyets in mm	*21*	*20*	*19*	*18*	*17*	*16*	*15*	*14*	*13*	*12*
Enclosed area in sq. km.	*543*	*1345*	*2030*	*2545*	*2955*	*3280*	*3535*	*3710*	*3880*	*3915*

Determine the average depth of rainfall over the basin.

Solution. The computations for calculating average rainfall by using the formula $P = \frac{P_1A_1 + P_2A_2 + \ldots}{A_1 + A_2 + \ldots}$ are easily done in a tabular form, as shown in Table 7.10.

Table 7.10

Isohyet mm (1)	*Area between each isohyet and basin boundary in km²* (2)	*Net incremental area between two isohyets i.e. A_1, A_2, A_3, ... in km²* (3)	*Av. rainfall on the incremental areas i.e. av. of two isohyets; i.e., P_1, P_2, P_3, ... in mm* (4)	*Rainfall volume on incremental areas;, i.e., P_1A_1, P_2A_2, P_3A_3, ... col. (3) × col. (4) in mm. km²* (5)
21	543	543*	21**	11,403
20	1345	802	20.5	16,441
19	2030	685	19.5	13,357
18	2545	515	18.5	9,527
17	2955	410	17.5	7,175
16	3280	325	16.5	5,362
15	3535	255	15.5	3,952
14	3710	175	14.5	2,537
13	3880	170	13.5	2,295
12	3915	35	12.5	437
	Σ	3915		72,486

$$\text{Av. Ppt over the basin} = \frac{\Sigma PA}{\Sigma A} = \frac{72486}{3915}\ \text{mm} = \mathbf{18.51\ mm.}\quad \textbf{Ans.}$$

*for the end contour, this is the area between the end isohyet and the basin boundary.

**The average value being the same as the value of the end isohyet.

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ANALYSING PRECIPITATION DATA

7.18. Characteristics of a Rain-storm

A rainfall at a place can be completely described, if its intensity, duration, and frequency, are known. The intensity of a rain is the rate at which it is falling; the duration is the time for which it is falling with that given intensity, and frequency is the number of times it falls.

7.18.1. Rain Hyetograph. The **intensity of a rain** is expressed in cm/hr, but this rate at which the rain falls, changes continuously throughout the storm. It may rain 5 cm in one hour giving an average rainfall rate of 5 cm/hr. However, during this particular hour, sometimes the rainfall rate will greatly exceed 5 cm/hr, and sometimes it may be much less than 5 cm/hr. An idealized curve showing the intensity variation with time is shown in Fig. 7.24 (*a*), and is known as the *time-intensity pattern.*

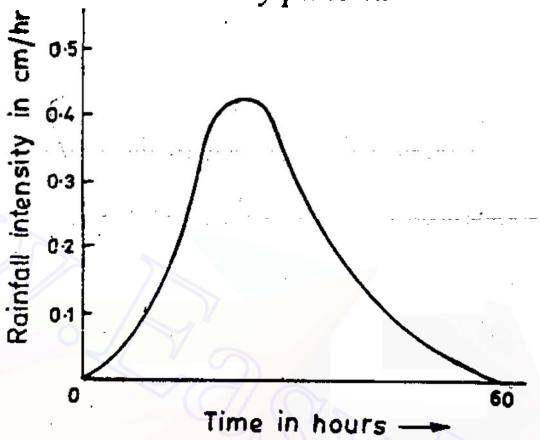

Fig. 7.24 (*a*) Time intensity pattern

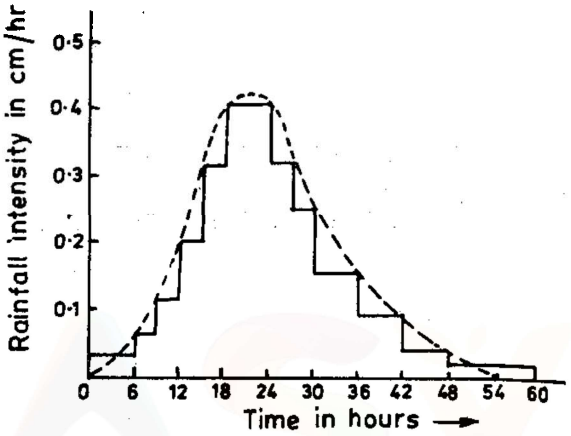

Fig. 7.24 (*b*) Rain hyetograph.

The above time-intensity curve is generally assumed to be made up of straight lines, as shown in Fig. 7.24 (*b*), and is called a *rain hyetograph.* A **hyetograph** can, therefore, be defined as a plot of intensity of rainfall against the time interval, represented as a bar chart.

The area under a hyetograph represents the total rainfall received in that period. This chart is very useful in representing the characteristics of a storm, and is particularly important in developing the design storms to predict extreme floods.

The time interval used, depends on the purpose; say for example, in urban drainage problems, small durations are used ; while in flood flow computations in larger catchments, the intervals are of about 6 hour.

7.18.2. Mass Curve of Rainfall. If the total accumulated precipitation is plotted against time, the curve obtained is known as the *mass curve of the storm*, and is shown in Fig. 7.25. The curve rises steeply in the beginning and then tends to become constant.

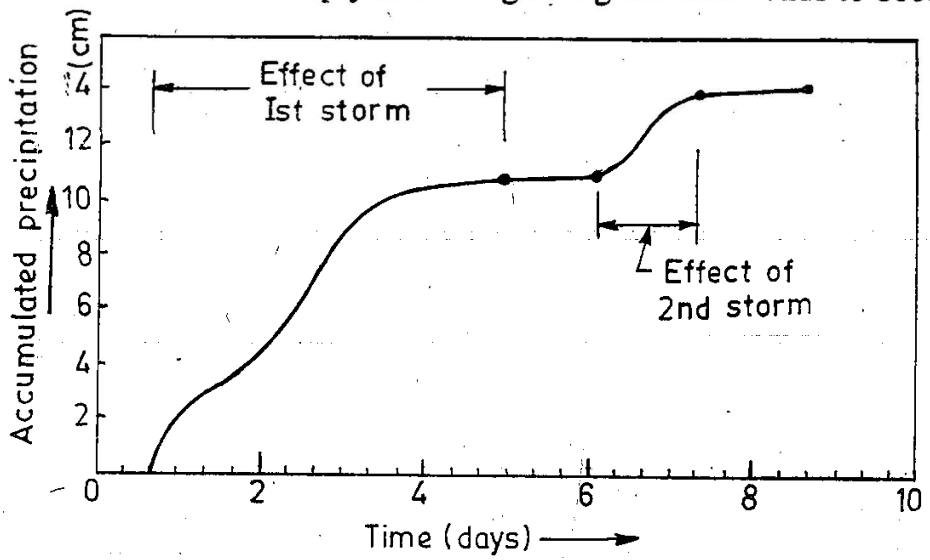

Fig. 7.25. Mass curve of rainfall.

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MAXIMUM PRECIPITATION

7.18.3. Moving Average Curve of Annual Rainfalls. The rainfall records of various years can be graphically represented by plotting the annual rainfall values on Y-axis and the respective years on X-axis, as to get a bar graph, as shown in Fig. 7.26. Such a bar graph, however, does not reflect any clear trends or patterns in the rainfall due to abrupt variations in individual years. In order to suppress these and to bring out the general trend of the rainfall over say 3 years or 5 years, the averages of the three (or five) consecutive years, are computed progressively, and are then plotted, as shown in Fig. 7.26.

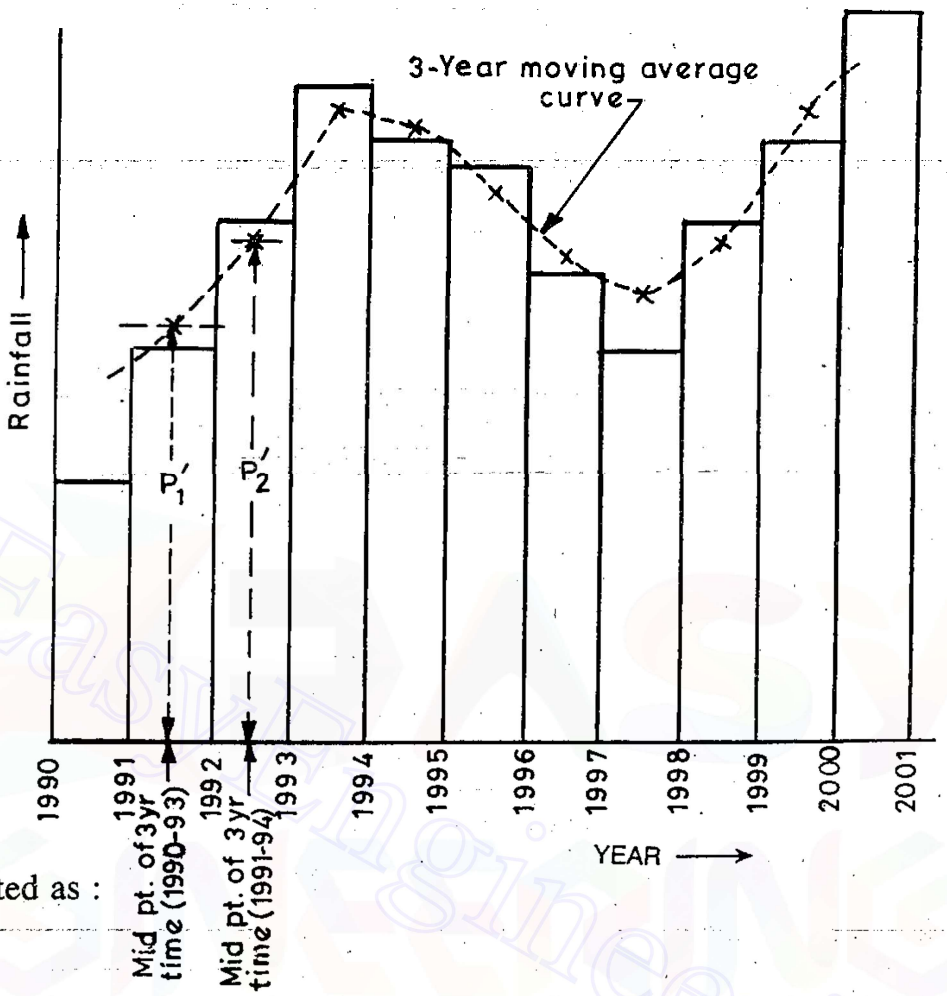

Fig. 7.26. Moving average curve of annual precipitation.

Thus, for a **three year moving mean**, the various values of rainfall to be plotted against time (mean of 3 years) shall be calculated as :

$$P_1' = \frac{P_1 + P_2 + P_3}{3} \qquad P_2' = \frac{P_2 + P_3 + P_4}{3} \qquad P_3' = \frac{P_3 + P_4 + P_5}{3}$$

The values of P_1', P_2', P_3' ..... are then plotted against the corresponding times (mid pts of respective 3 years). The plot so obtained will be known as the *3 year moving average curve of annual rainfall*, and will indicate the *general trends regarding increase or decrease of annual precipitation with time.*

7.18.4. Depth-Area-Duration (DAD) Relationship. The **areal characteristics of a storm** of a given duration is reflected in its depth-area relationship. Such a relation is represented by a curve of the type shown in Fig. 7.27. *Different such curves will be obtained for different storms of a given duration*, as shown in Fig. 7.28; from which maximum observed DAD curve of a particular duration can be determined.

A depth-area duration curve expresses graphically the relation between progressively decreasing average depth of rainfall over a progressively increasing area (from the centre of the storm outward to its edges) for a given duration of rainfall.

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The immediate purpose of DAD analysis of a particular storm is to determine the largest average depth of rainfall that fell over various sizes of area during the standard passage of time in hours or days; such as the largest average depth of rainfall falling over 500 sq. km in 1 day. Such values when determined for each of the past storms occurring on the area, as well as for each of the *transposable storms**, provide the basic data to estimate the **probable maximum precipitation (PMP)** for the basin in question.

A *depth-area curve* can be constructed from any isohyetal map of rainfall of the given duration, by measuring the areas associated with each isohyet of rainfall depth, and then plotting the progressively decreasing average depth on *Y*-axis, and the corresponding areas on *X*-axis. Different such curves can be plotted for different duration storms, to obtain a composite DAD chart, for a common areal base. Incremental isohyetal maps for storms of selected durations, such as 6 hr. 12 hr. 1 day, 2 day, 3 day, etc. are required in this procedure, which, will become more clear when we solve Example 7.7.

Example 7.7. *Estimate the average depth of precipitation (from depth area curve) that may be expected over an area of 2400 sq. km, due to the storm of 27th Sept. 1978, lasting for 24 hours, assuming the storm centre to be located at the centre of the area. The isohyetal map for the storm gave the areas enclosed between different isohyets as follows :*

Isohyets in mm	*21*	*20*	*19*	*18*	*17*	*16*	*15*	*14*	*13*	*12*
Enclosed area (in sq. km.)	*543*	*1345*	*2030*	*2545*	*2955*	*3280*	*3535*	*3710*	*3880*	*3915*

(U.P.S.C. Engg. Services, 1986)

Solution. In order to plot the depth-area-duration curve, we have to, first of all, analyse the given data to determine the **equivalent-uniform depths** (EUD)** of rainfall, as computed in column (7) of Table 7.11, which is otherwise self-explanatory.

Table 7.11

Isohyet (given) mm	*Area enclosed between isohyet and basin boundary in km^2*	*Net incremental area between isohyetes km^2*	*Av. rainfall on incremental areas mm*	*Rainfall volume on incremental areas ($mm.km^2$) i.e. col. (3) × (4)*	*Cumulative Volume ($mm.km^2$) Σ (5)*	*Equivalent Uniform Depths E.U.D. (mm) Col. (6) / Col. (2)*
(1)	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*	*(7)*
21	543	543	21	11,403	11,403	21.00
20	1345	802	20.5	16,441	27,844	20.70
19	2030	685	19.5	13,357	41,201	20.30
18	2545	515	18.5	9,527	50,728	19.93
17	2955	410	17.5	7,175	57,903	19.59
16	3280	325	16.5	5,362	63,265	19.29
15	3535	255	15.5	3,952	67,217	19.01
14	3710	175	14.5	2,537	69,754	18.80
13	3880	170	13.5	2,295	72,049	18.57
12	3915	35	12.5	437	72,486	18.51

* A storm occurring on a similar basin, when considered to be equivalent to a likely storm that can occur on the basin under consideration, is called a transposable storm. Storm transposition, therefore, means the application of a storm, from one area to another area, within the same meteorological homogeneous region.

** mean depths of rainfall over increasing areas of a basin

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A curve is then plotted between equivalent uniform depths (*EUD*) of column (7), and the areas enclosed of column (2), as shown in Fig. 7.27.

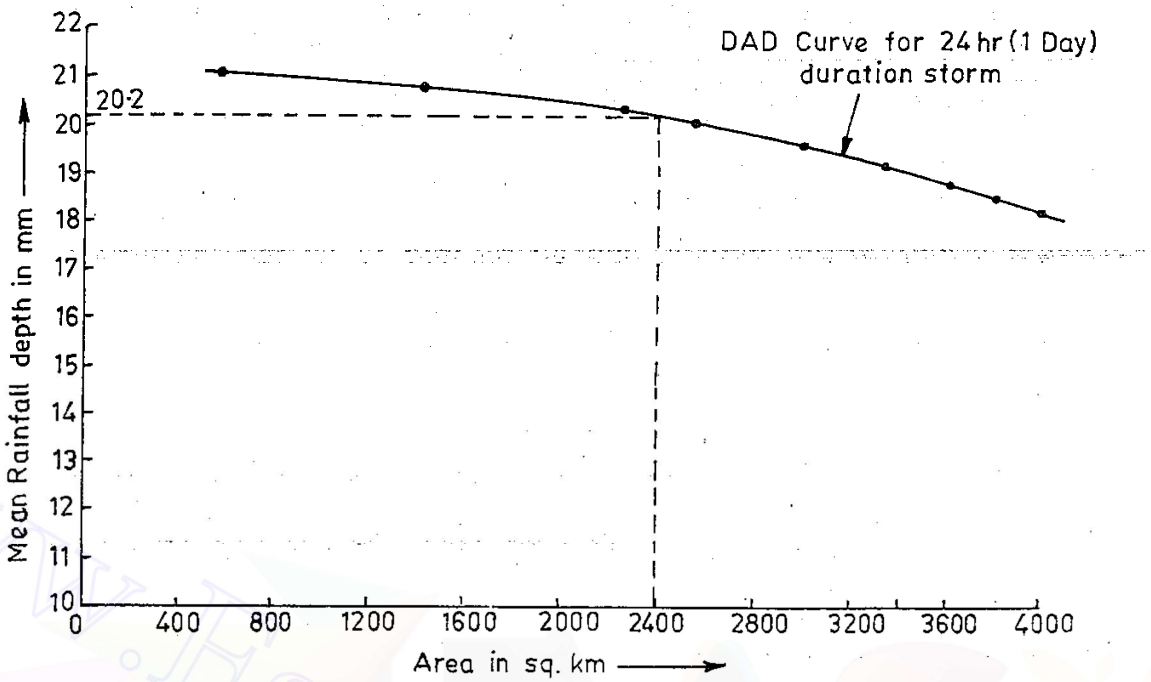

Fig. 7.27. DAD curve of 1 day rainfall for example 7.7.

From this curve, the average depth of precipitation for 2400 sq. km area is read out as 20.2 mm. **Ans.**

7.18.4.1. Equation of depth area-duration curve. The depth area curve for a given rain of a particular duration is of the type shown in Fig. 7.27. Such a curve can be expressed in a mathematical form, as :

$$P = P_0 (e)^{-K/A^n} \qquad ...(7.17)$$

where P = average depth in *cm* over an area A km^2

P_0 = Highest amount of rainfall in cm at the storm centre

K and n = constants for a given region.

On the basis of the study of a number of severest storms in North India, Dhar and Bhattacharya (1975) have computed the following values of constants K and n for storms of different durations.

Table 7.12

Duration in day	*K*	*n*
1	0.0008526	0.6614
2	0.0009877	0.6306
3	0.001745	0.5961

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Since the storm centre may not coincide with a rain-gauge station, the value of P_0 may not be exactly known. Hence, in the analysis of large area storms, the highest station rainfall is taken as the average depth (P) over an area of 25 km^2, to workout P_0. The above equation (7.17) is useful in extrapolating an existing storm data over an area.

7.18.4.2. Maximised depth-area-duration curves. The depth-area curve (for a given duration) can be worked out, as explained above, for any given storm. All severer storms can be used to compute such curves, and the curve which gives *maximum precipitation values*, can be refered to as the **'observed maximum DAD curve'** of a particular duration (Refer Fig. 7.28 (*a*); DAD curve at the top is the observed maximum 3 day curve for Tajewala catchment). Such maximum observed precipitation curves of different durations are usually considered to project the *standard project storms* of different durations. These values are, however, further enhanced for worst and rare possible combinations of rain causing hydrometeorological factors, to obtain *absolute maximum DAD curves*, called *maximised DAD curves*, which represent the *limiting estimated values, or probable maximum values.*

Such maximised DAD curves at various durations are plotted and kept ready in the design office for subsequent use. A set of such **maximised DAD curves** for Damodar basin is shown in Fig. 7.28 (*b*).

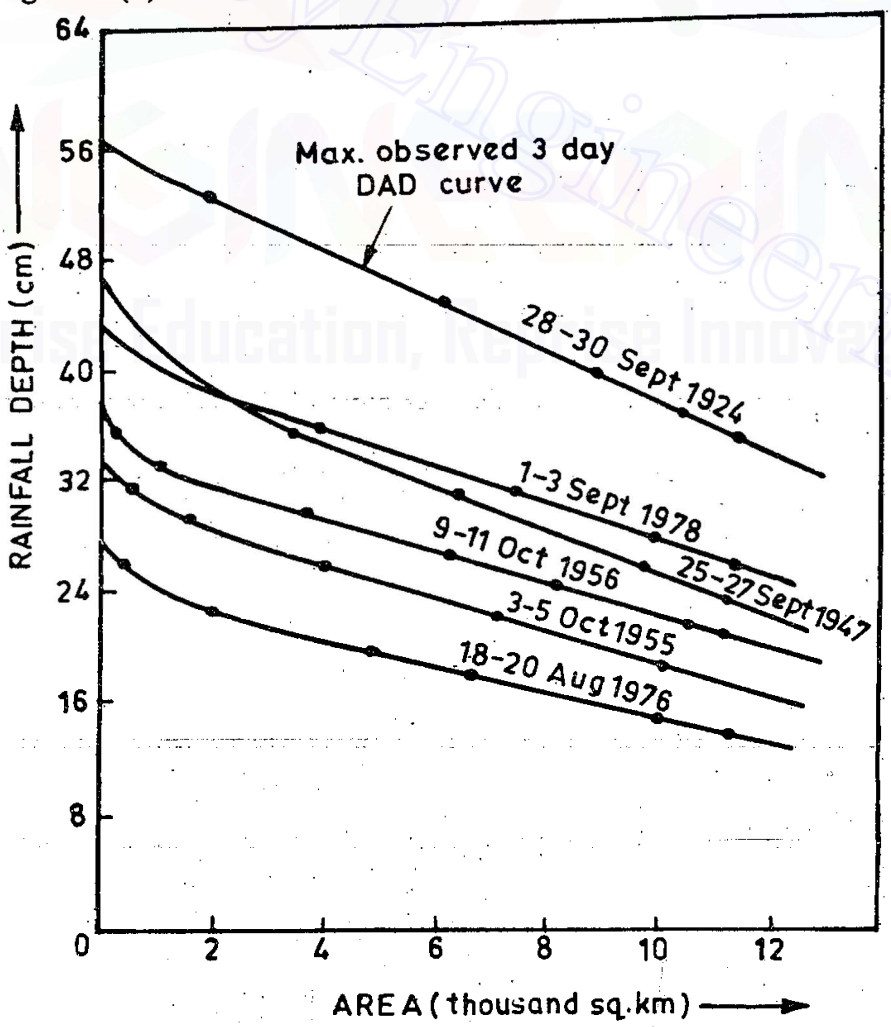

Fig. 7.28 (*a*). DAD curves of 3 day duration storms for Yamuna catchment upto Tajewala.

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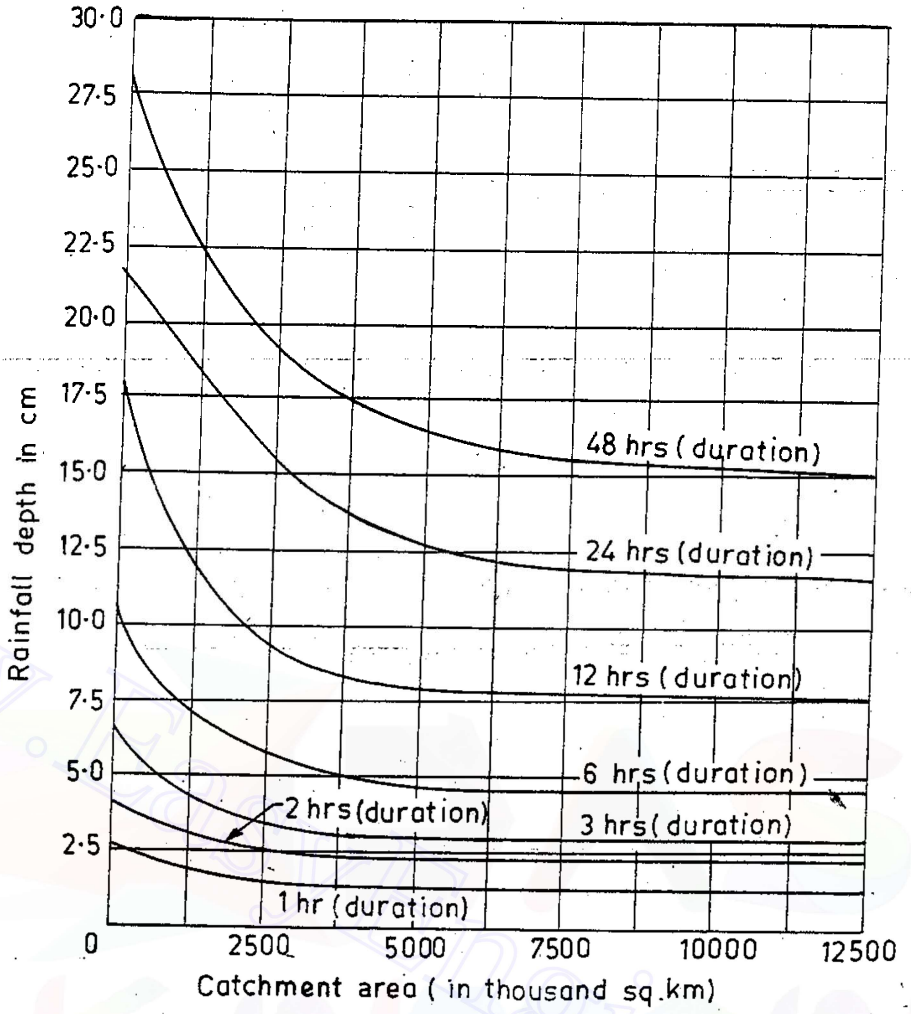

Fig. 7.28 (*b*). Typical maximised DAD curves for Damodar Basin (as supplied by CWC).

7.18.5. Frequency of a Storm or Recurrence Interval of a Storm. By recurrence interval (R.I.) of a given storm is meant the time interval during which the given storm is likely to be equalled or exceeded. Thus, if we say that at a given station, the maximum precipitation of 20 cm has got a recurrence interval of 5 years, we mean that on this station, the chances of rainfall are such that once in five years, rain is likely to equal or exceed 20 cm. This does not mean that a rain equal to 20 cm or more, will occur after every five years. It may occur twice in one set of five years and may not occur at all for consecutive eight years or so. What it means is, that during 100 years or so, such a rain will occur 20 times or so.

7.18.5.1. California probability method to determine recurrence interval. (T) If N is the total number of years of record, and T is the recurrence interval, then it can be easily stated that

$$N = T \cdot m \qquad ...(7.18)$$

where m is the number of times the given rain is equalled or exceeded, and is known as the **ranking of the storm** or *ranking number.*

As an example, suppose, the rainfall records at a given station have been kept for 52 years, and it is desired to pick up the storm that is expected to appear once in eight

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years; then $T = 8$, $N = 52$, and we have $m = \frac{52}{8} = 6.5$. Since the ranking of the storm is 6.5, we would pick up the 6th most severe storm. To make the things very clear, we solve the following example.

Example 7.8. *The values of annual precipitation at a rain-gauge station expressed in cm per year in chronological sequence from 1956 to 1967 are indicated below :*

36.5	*29.0*	*56.2*	*82.0*	*27.8*	*23.4*
71.2	*48.3*	*31.4*	*18.1*	*29.0*	*65.6*

Estimate the value of precipitation which has a recurrence interval of 5 years, using the probability or statistical method. (*U.P.S.C. Engg. Services, 1969*)

Solution. Total No. of years on record (N) = 12 years

Recurrence interval (T) = 5 years.

$\therefore$ m = Ranking of the storm $= \frac{N}{T} = \frac{12}{5} = 2.4$

Therefore, we should pick up the 2nd severest storm, *i.e.* 71.2 cm. (First most severe storm is 82.0 cm, and next severe storm is 71.2 cm).

Hence, the value of precipitation, which has a recurrence interval of five years, is 71.2 cm. **Ans.**

Limitations of Simple Probability Method. The recurrence interval or frequency can be determined easily by this simple probability method. However, a difficulty arises in determining the recurrence interval of very severe storms. Because, if there is a record of 40 years at a place, then the most severe storm will have its ranking equal to 1, and therefore, $T = \frac{N}{m} = \frac{40}{1} = 40$. The recurrence interval for this storm will come out to be 40 years. Moreover, this value is based on only one sample of 40 years of records, if no such strong rain equal to or more than this, falls during the next 40 years; then the frequency would be calculated at 80 years. On the other hand, if two such strong rains fall during the next 40 years, then the frequency would be calculated as $\frac{80}{3} = 26.7$ years.

Thus, it can be concluded that the frequency of the largest rain as determined from the equation $N = T.m$, cannot be relied upon, and it should be considered only as a very rough & approximate value. Similarly, the frequency of the next severest storm as calculated from the equation $N = T.m$, will also be approximate, but the approximation in this case will be lesser than that in the first case. Hence, for severe rains, the method gives only approximate values. It is, thus, very much possible, to calculate for a given storm, a frequency of 2 years from a ten years record, and to calculate 5 years frequency for the same storm on the basis of 60 years of records.

Secondly, if we want to determine a storm having recurrence interval of more than the total number of years of which the records are available, it is not possible to find it out by using the simple probability method *i.e.* $N = T.m$ (Because, in that case, T is greater than N, and

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therefore, m is less than one, *i.e.* ranking of storm is less than 1, which is not known). These are the two *limitations of this method.*

In order to make use of the above method to determine the storm of such a high recurrence interval, the available records are sometimes plotted graphically. The curve is extended back, so as to determine the storm of the desired recurrence interval.

In this method, a graph is plotted between the recurrence interval (R.I.) on X-axis and precipitation in cm on Y-axis, as shown in Fig. 7.29.

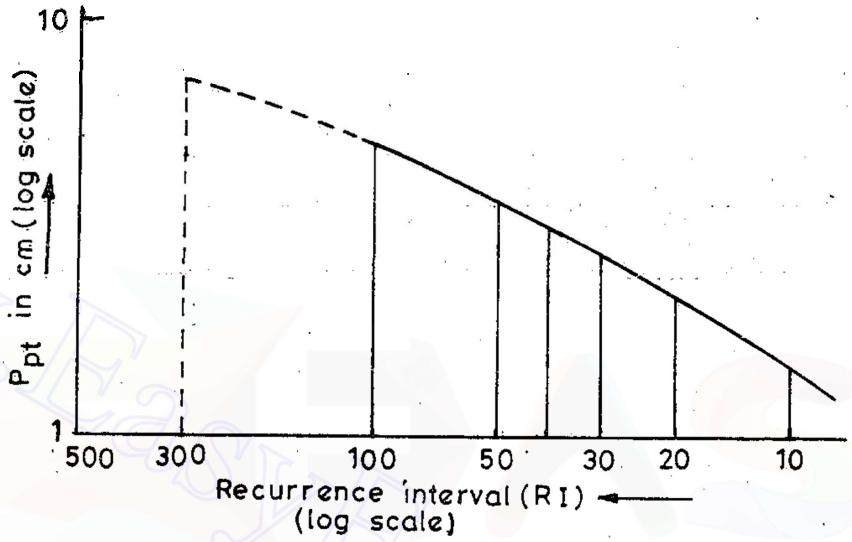

Fig. 7.29. Extrapolating the recurrence interval.

The curve can be extended back, so as to determine the precipitation of a storm having a higher recurrence interval. The method can be easily understood on solving the following example.

Example 7.9. *The maximum values of 24 hours summer precipitation at a rain gauge station expressed in cm from 1940 to 1969 are indicated below :*

10.7	*11.2*	*10.8*	*9.6*	*14.9*	*15.2*	*12.0*	*12.2*	*15.8*	*16.8*
9.7	*11.3*	*11.6*	*11.9*	*12.4*	*12.7*	*6.4*	*10.5*	*9.2*	*18.7*
17.1	*17.7*	*16.9*	*15.4*	*13.8*	*12.9*	*16.3*	*15.7*	*16.6*	*17.2*

Estimate the maximum precipitation having a recurrence interval of (a) 5 years (b) 10 years (c) 20 years (d) 40 years.

Solution. Let us first arrange the values of precipitation in the decreasing order with their corresponding years, as shown in Table 7.13. Value of recurrence interval (T) are then calculated in col. (4) of this table, which is otherwise self explanatory.

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Table 7.13

Year (1)	*Precipitation in cm* (2)	*Ranking of storm m* (3)	*Recurrence interval in years* $T=\frac{N}{m}=\frac{30}{m}=\frac{30}{(3)}$ (4)
1959	18.7	1	30.0
1961	17.7	2	15.0
1969	17.2	3	10.0
1960	17.1	4	7.5
1962	16.9	5	6.0
1949	16.8	6	5.0
1968	16.6	7	4.3
1966	16.3	8	3.8
1948	15.8	9	3.3
1967	15.7	10	3.0
1963	15.4	11	2.7
1945	15.2	12	2.5
1944	14.9	13	2.3
1964	13.8	14	2.1
1965	12.9	15	2.0
1955	12.7	16	1.9
1954	12.4	17	1.8
1947	12.2	18	1.7
1946	12.0	19	1.6
1953	11.9	20	1.5
1952	11.6	21	1.4
1951	11.3	22	1.4
1941	11.2	23	1.30
1942	10.8	24	1.25
1940	10.7	25	1.20
1957	10.5	26	1.15
1950	9.7	27	1.10
1943	9.6	28	1.07
1958	9.2	29	1.03
1956	6.4	30	1.00

Note. Column 4 of Table 7.13, shows the values of R.I. for all precipitations. Hence, the amount of ppt. for any R.I. can be directly read from this Table.

Case (*a*) When the recurrence interval or frequency is 5 years, $T = 5$ years, and the total number of years for which the records are available $N = 30$ years. Since $N = T.m$, therefore, $30 = 5 \times m$, or $m = 6$, *i.e.* ranking of the required storm = 6; so choose the 6th severest storm = 16.8 cm. **Ans.**

Case (*b*) When $T = 10$ years, $N = T.m$, therefore $30 = 10 \cdot m$ or $m = 3$, so choose the third severest storm = 17.2 cm. **Ans.**

Case (*c*) When $T = 20$ years, $N = T.m$, therefore, $30 = 20 \cdot m$, or $m = 1.5$; hence choose the first severest storm, *i.e.* 18.7 cm. **Ans.**

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Case (*d*) When $T = 40$ years, $N = T.m$, therefore, $30 = 40 \cdot m$. Hence $m = \frac{30}{40} = 0.75$.

The storm ranking does not lie within the available records. So let us plot a graph between frequency (or recurrence interval) and maximum precipitation on log-log graph paper, as shown in Fig. 7.30. Then extend the curve, so as to determine the maximum

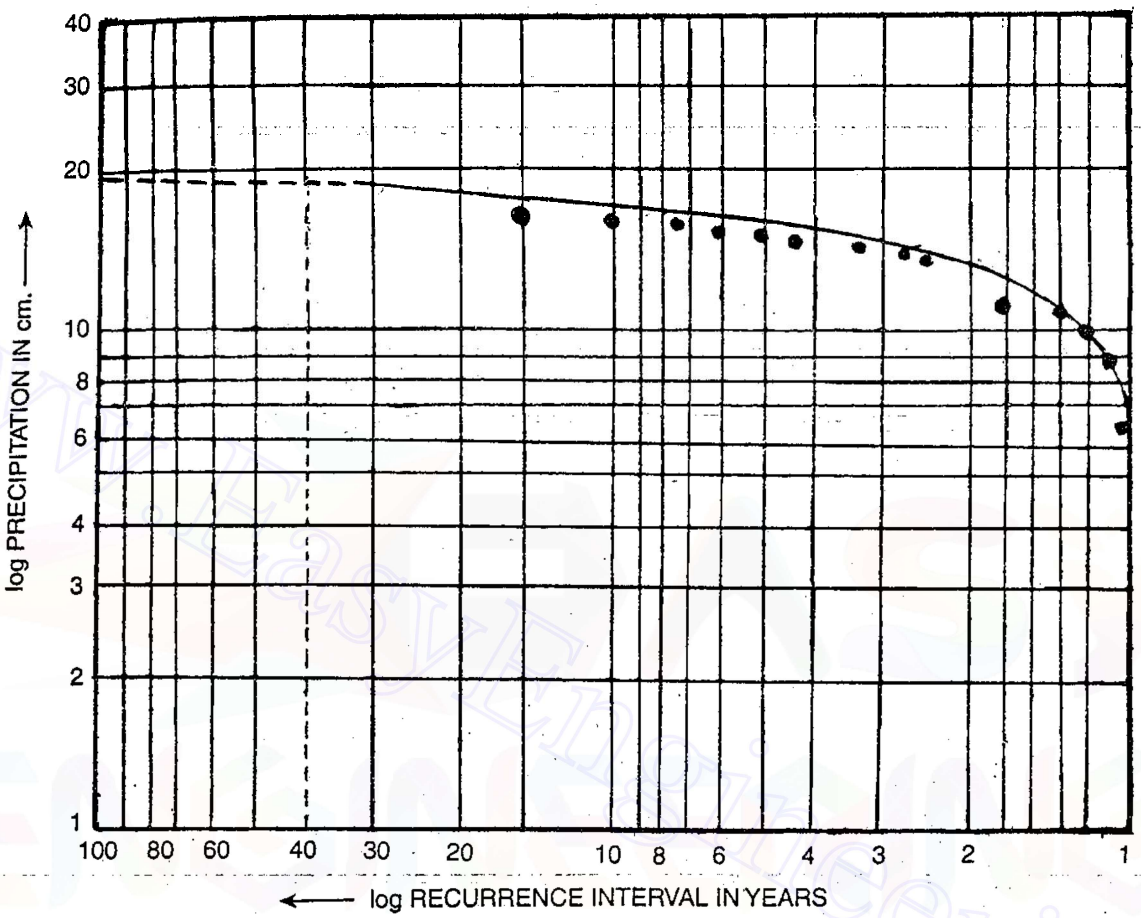

Fig. 7.30

precipitation as equal to 19.5 cm (from graph) for a recurrence interval of 40 years, as shown in Fig. 7.30 . **Ans.**

7.18.5.2. Station year method to determine the recurrence interval. Station year method is used to determine recurrence intervals which are longer, and hence, cannot be computed from the available records of a single station. This method is an extension of the method $N = T.m$, (described earlier) with the only difference that the records of various stations are combined together. All the records which are available from the various stations situated in a *meteorologically homogeneous* area are combined together, so as to represent the record of an imaginary representative station, called the *station year*. This method is, thus based upon the assumption, that all the stations situated in a meteorologically homogeneous area will experience similar rainfall frequencies, and that, if these stations are widely enough separated, the total of rainfall experienced at all of the stations will be similar to the experience of any one station, over a period of record equal to the sum of periods at the individual stations.

The limitations of this method are the same as that of the method $N = T \cdot m$, plus an assumption that all the stations should lie in a meteorologically homogeneous area.

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7.19. Meteorologically Homogeneous Areas

If the probability of occurrence of a storm of a given intensity is the same throughout an area, then the area is said to be meteorologically homogeneous. Strictly speaking, the rainfall, therefore, is the same at all points in such an area. Thus, if the frequencies as well as the intensities of all the rains are the same throughout, the total annual rain or the average annual rain will also be the same at all places in such an area. Hence, if there are two places which are meteorologically homogeneous, their average annual precipitations must also be the same. The converse is not true, as the two places having the same mean annual rainfalls may not be meteorologically homogeneous, because the intensities of storms constituting the total mean annual rainfall may not be the same in both the cases.

However, if the two places have the same mean annual rainfall and the same average number of days of rainfall per year, they are regarded as meteorologically homogeneous.

7.20. Intensity-Duration-Frequency Curves (IDF curves)

Rainfall of a place can be completely defined if the intensities, durations and frequencies of the various storms occurring at that place are known. Whenever, an intense rain occurs, its magnitude and duration is generally known from the meteorological readings. Thus, at a given station, the magnitudes of the isolated rains of various durations, such as 5 minutes, 10 minutes, 15 minutes, 30 minutes, 60 minutes, 120 minutes, etc., are generally known. This available data can be used to determine the frequencies of the various rains, as described earlier (by using the equation $N = T \cdot m$).

This frequency-data for storms of various durations, so obtained, can be represented by *intensity-duration frequency curves*. An intensity duration frequency curve is a plot of average rainfall intensity* in cm/h and the duration in minutes. A typical intensity-duration curve of a particular frequency is shown in Fig. 7.31. Different such curves can

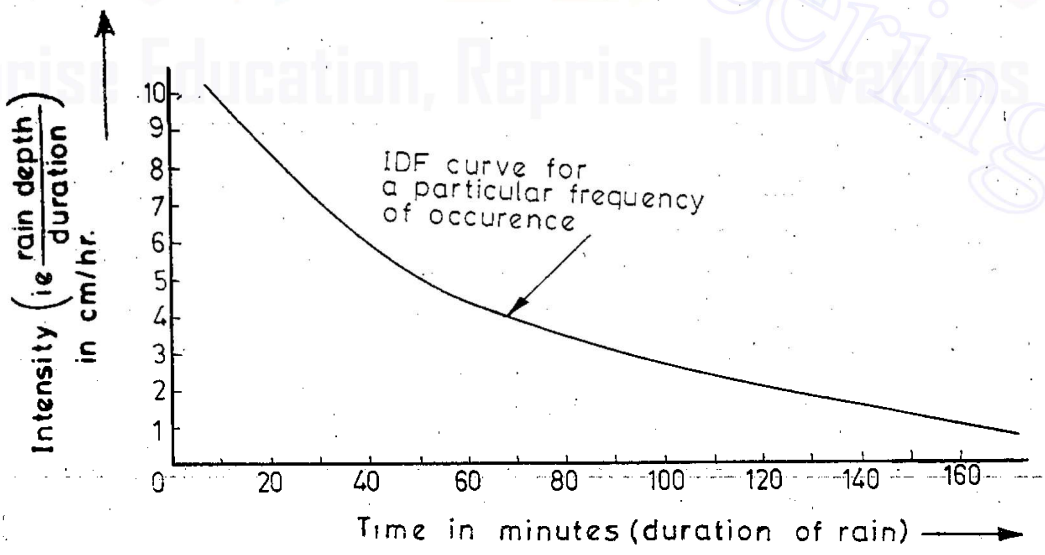

Fig. 7.31. A typical IDF curve.

be plotted for different frequencies, as shown in Fig. 7.32 and 7.33.

*rainfall depth is averaged over the duration i.e. $\frac{\text{rainfall depth}}{\text{duration}}$ = av. rainfall intensity in that duration.

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Example 7.10. *The various data were obtained for rains of various durations at a station for 31 years. The records were analysed and eleven worst storms of various durations have been stipulated in their decreasing order, as shown in Table 7.14.*

Table 7.14

5 minutes		10 minutes		15 minutes		30 minutes		60 minutes		90 minutes		120 minutes	
Year	Ppt. in cm	Year	Ppt. in cm	Year	Ppt. in cm	Year	Ppt. in cm	Year	Ppt. in cm	Year	Ppt. in cm	Year	Ppt. in cm
1908	0.85	1908	1.20	1908	1.40	1908	1.74	1908	2.15	1908	2.46	1915	2.97
1921	0.76	1915	1.04	1915	1.18	1904	1.55	1904	1.92	1915	2.38	1908	2.63
1915	0.73	1921	0.93	1904	1.11	1915	1.36	1915	1.70	1904	2.14	1904	2.34
1934	0.72	1904	0.88	1921	1.03	1921	1.22	1926	1.45	1921	1.81	1921	2.12
1929	0.66	1926	0.84	1926	0.97	1926	1.18	1921	1.40	1926	1.65	1926	1.83
1926	0.62	1934	0.80	1934	0.92	1931	1.10	1914	1.33	1914	1.50	1917	1.64
1931	0.51	1929	0.78	1929	0.90	1934	1.05	1931	1.25	1931	1.40	1914	1.55
1904	0.45	1931	0.68	1931	0.82	1929	1.01	1934	1.20	1917	1.36	1931	1.51
1917	0.36	1911	0.52	1911	0.67	1911	0.95	1929	1.14	1934	1.34	1934	1.46
1914	0.28	1917	0.51	1917	0.62	1917	0.83	1911	1.11	1929	1.27	1929	1.41
1911	0.21	1914	0.39	1914	0.50	1914	0.79	1917	1.09	1911	1.23	1911	1.34

Plot the intensity-duration curves for storms of frequencies (a) 11; (b) 1.4; (c) 1.0.

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Solution. We will, first of all, calculate the frequencies of the various storms by using the equation $N = T \cdot m$, as shown in Table 7.15, using total number of years $N = 11$, and $T = \frac{11}{m}$.

Table 7.15
Calculation Table for Example 7.10

S. No.	*5 minutes ppt*	*10 minutes ppt*	*15 minutes ppt*	*30 minutes ppt*	*60 minutes ppt*	*90 minutes ppt*	*120 minutes ppt*	*m = ranking of storm*	*T = frequency* $= \frac{N}{m} = \frac{11}{(9)}$
(1)	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*	*(7)*	*(8)*	*(9)*	*(10)*
1.	0.85	1.20	1.40	1.74	2.15	2.46	2.97	1	11
2.	0.76	1.04	1.18	1.55	1.92	2.38	2.63	2	5.5
3.	0.73	0.93	1.11	1.36	1.70	2.14	2.34	3	3.7
4.	0.72	0.88	1.03	1.22	1.45	1.81	2.12	4	2.8
5.	0.66	0.84	0.97	1.18	1.40	1.65	1.83	5	2.2
6.	0.62	0.80	0.92	1.10	1.33	1.50	1.64	6	1.8
7.	0.51	0.78	0.90	1.05	1.25	1.40	1.55	7	1.6
8.	0.45	0.68	0.82	1.01	1.20	1.36	1.51	8	1.4
9.	0.36	0.52	0.67	0.95	1.14	1.34	1.46	9	1.2
10.	0.28	0.51	0.62	0.83	1.11	1.27	1.41	10	1.1
11.	0.21	0.39	0.50	0.79	1.09	1.23	1.34	11	1.0

Average intensities for different durations such as 5, 10, 15, 30, 60, 90 and 120 minutes are then worked out for three different frequencies of 11, 1.4 and 1.0 years, as desired in the question, and shown in Table 7.16.

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Table 7.16

Duration in minutes (1)	*Ppt. in cm from Table 7.15* (2)	*Av. intensity in cm/hr; Col. (2)* $\times \frac{60}{Col.\ (1)}$ (3)
Part (*a*) Frequency = 11 years		
5	0.85	12.20 (*i.e.* 0.85 × 60/5)
10	1.20	7.20 (*i.e.* 1.2 × 60/10)
15	1.40	5.60 (*i.e.* 1.4 × 60/15)
30	1.74	3.88 (*i.e.* 1.74 × 60/30)
60	2.15	2.15 (*i.e.* 2.15 × 60/60)
90	2.46	1.64 (*i.e.* 2.46 × 60/90)
120	2.97	1.42 (*i.e.* 2.97 × 60/120)
Part (*b*) Frequency = 1.4 years		
5	0.45	5.40
10	0.68	4.01
15	0.82	3.28
30	1.01	2.02
60	1.20	1.20
90	1.36	0.90
120	1.51	0.75
Part (*c*) Frequency = 1.0 year		
5	0.21	2.52
10	0.39	2.34
15	0.50	2.00
30	0.79	1.58
60	1.09	1.00
90	1.23	0.82
120	1.34	0.67

The three curves corresponding to three frequencies can then be plotted by using the above values, as shown in Fig. 7.32.

Thus, for a given storm and of a given frequency of occurrence, an intensity-duration curve can be plotted easily, as described in the above example. Highest curve in Fig. 7.32 represents intensity duration for a storm of 11 years frequency, and such a curve can be re-used to determine the average intensity of any other 11 years frequency rain for any desired duration, which when multiplied by that duration will directly indicate the rainfall depth, which could help in calculating the excess rainfall depth, and hydrograph of DRH by application on unit hydrograph, as discussed in article 7.50.4. IDF curves can, thus, prove to be an important tool in computation of runoff from a basin for a rainfall of a particular frequency and duration.

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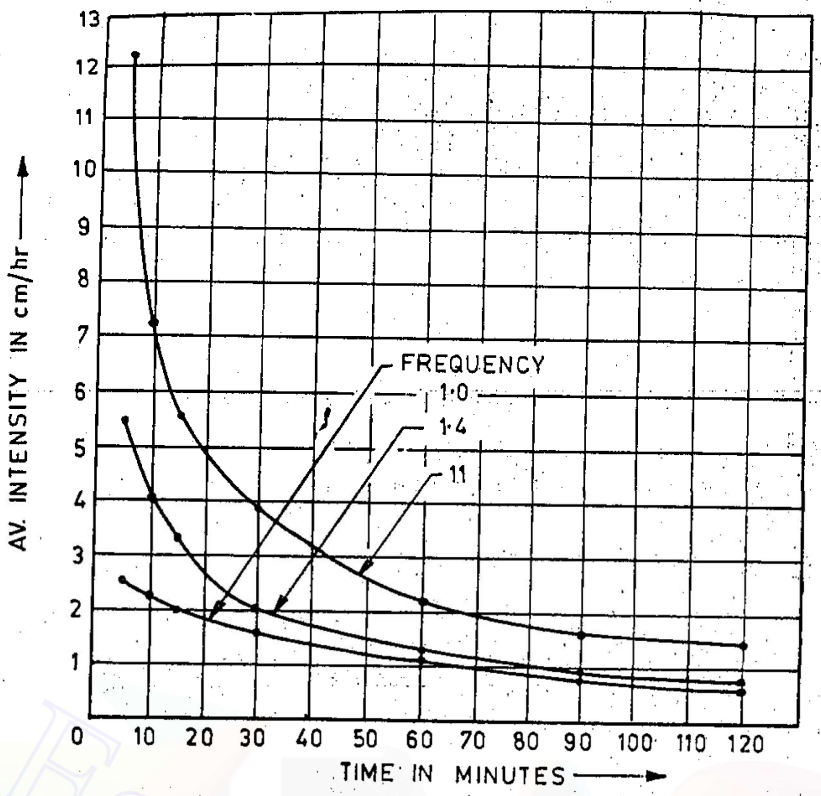

Fig. 7.32. Intensity-duration curves for different frequencies. (Example 7.10)

7.20.1. Computation of IDF curves from Long Duration Storms. When several *isolated storms* of different durations are not available on record, an IDF curve can be plotted with the data of a single long duration storm. Say for example, an IDF curve was plotted in Example 7.10 from the known rainfall depths of 7 individual storms, respectively having duration of 5 min, 10 min, 15 min, 60 min, 90 min, and 120 min. This job can also be done if the mass rainfall curve of a single storm of 120 min duration is known. In such a case, the entire duration of 120 min shall be divided into several intervals, say of 10 min each, and the rainfall depths falling in each successive interval of 10 min shall be computed. *The maximum water depth falling in any of these intervals shall be chosen as the requisite maximum depth of rainfall for 10 min duration.* When this maximum depth is divided by the above duration (10 min), the *maximum average intensity* corresponding to 10 min duration shall be obtained. The above procedure shall then be used to compute maximum depth of rain that can fall in any interval of 20 min., during the entire 120 min, duration. Proceeding in this manner, maximum average* intensities for different durations can be computed and plotted to obtain a maximum I.D. curve for the given storm, as explained in Example 7.11.

However, the frequency of this curve cannot be marked, since the frequency of this storm is not known. This can of course be found out by carrying out frequency analysis of known severer storms of the past years**. All these storms shall first be analysed in the above manner to compute maximum depths or max. average intensities (max. av. intensity = max. depth/duration) for different durations. Several values of max. depths (or max. average intensities) will thus be obtained from different storms for a particular duration. This data will be subject to frequency analysis $\left(T = \frac{N}{m}\right)$ to determine their

* averaged over the duration.

** one storm is usually chosen for each year and this gives max. depth among the several storms occurring in that year.

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respective frequencies. Similar procedure can be repeated for other durations also, to ultimately compute max. depths (or max. av. intensities) for different durations at different frequencies. These values can finally be plotted to obtain **standard IDF curves** of the given basin, as shown in Fig 7.33.

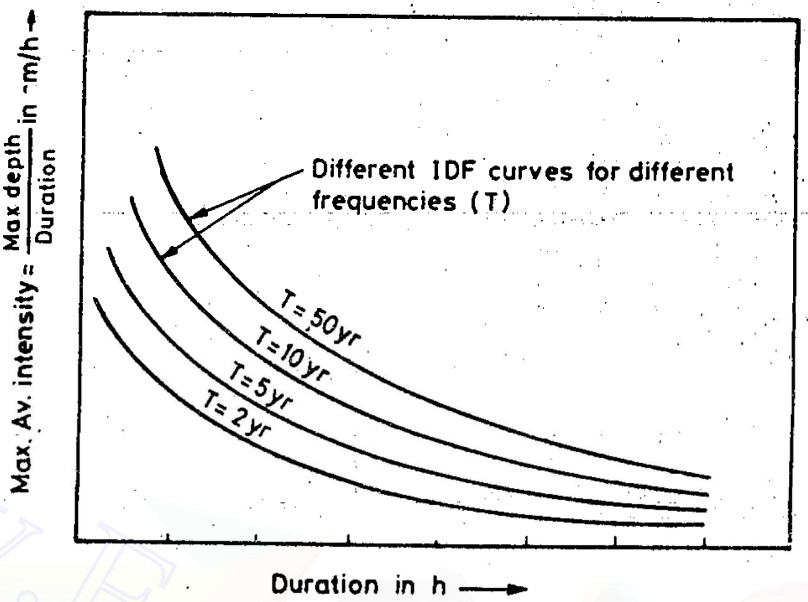

Fig. 7.33. Standard IDF curves of a basin.

Example 7.11. *The mass curve of precipitation resulted from the storm of 14th August, 1983 gave the following values :*

Hour min.	*Accumulated depth at the end of periods (in mm)*
22.00	*0.0 (Beginning of storm)*
22.05	*10.2*
22.10	*20.8*
22.15	*33.0*
22.20	*47.2*
22.25	*55.8*
22.30	*64.0*
22.35	*71.6*
22.40	*78.8*
22.45	*85.4*
22.50	*91.4 (end of storm)*

From the above storm, construct hyetograph and draw maximum intensity-duration curve. *(U.P.S.C. Engg. Services, 1984)*

Solution. To draw the requisite hyetograph, the intensities of rainfall are calculated, using uniform time interval of 5 minutes, as shown in Table 7.17. The table is otherwise self-explanatory.

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Table 7.17

Time hr.min. (1)	*Cumulative rainfall in mm* (2)	*Rainfall in successive 5 min. intervals in mm* (3)	*Rainfall intensity in mm/hr* col. (3) $\times \frac{60}{5}$ = (3) × 12 (4)	*Rainfall intensity in cm/hr* (5)
22.00	0.0			
22.05	10.2	10.2	122.4	12.24
22.10	20.8	10.6	127.4	12.72
22.15	33.0	12.2	146.4	14.64
22.20	47.2	14.2	170.4	17.04
22.25	55.8	8.6	103.2	10.32
22.30	64.0	8.2	98.4	9.84
22.35	71.6	7.6	91.2	9.12
22.40	78.8	7.2	86.4	8.64
22.45	85.4	6.6	79.2	7.92
22.50	91.4	6.0	72.0	7.20

A graph is now plotted between the rainfall intensities in cm/hr obtained in col. (5) of this table and the time interval of 5 min. each, as shown in Fig. 7.34, so as to obtain the requisite hyetograph.

To draw maximum intensity duration curve, we have to calculate the maximum rainfall values for durations of 5 min, 10 min, 15 min, 20 min, ..., 50 min, as shown in Table 7.18. These maximum rainfall values are shown in dark figures in col. (3) to

Table 7.18

Time in hr-min (1)	*Accumulated rainfall in mm* (2)	*Rainfall in any possible time interval equal to*									
		5 min (3)	*10 min* (4)	*15 min* (5)	*20 min* (6)	*25 min* (7)	*30 min* (8)	*35 min* (9)	*40 min* (10)	*45 min* (11)	*50 min* (12)
22.00	0.0										
22.05	10.2	10.2									
22.10	20.8	10.6	**20.8**								
22.15	33.0	12.2	**22.8**	33.0							
22.20	47.2	**14.2**	**26.4**	**37.0**	**47.2**						
22.25	55.8	8.6	22.8	35.0	45.6	**55.8**					
22.30	64.0	8.2	16.8	31.0	43.2	53.8	**64.0**				
22.35	71.6	7.6	15.8	24.4	38.6	50.8	61.4	**71.6**			
22.40	78.8	7.2	14.8	23.0	31.2	45.8	58.0	68.6	**78.8**		
22.45	85.4	6.6	13.8	21.4	29.6	38.2	52.4	64.6	75.2	**85.4**	
22.50	91.4 (end of storm)	6.0	12.6	19.8	27.4	35.6	44.2	58.4	70.6	81.2	**91.4**

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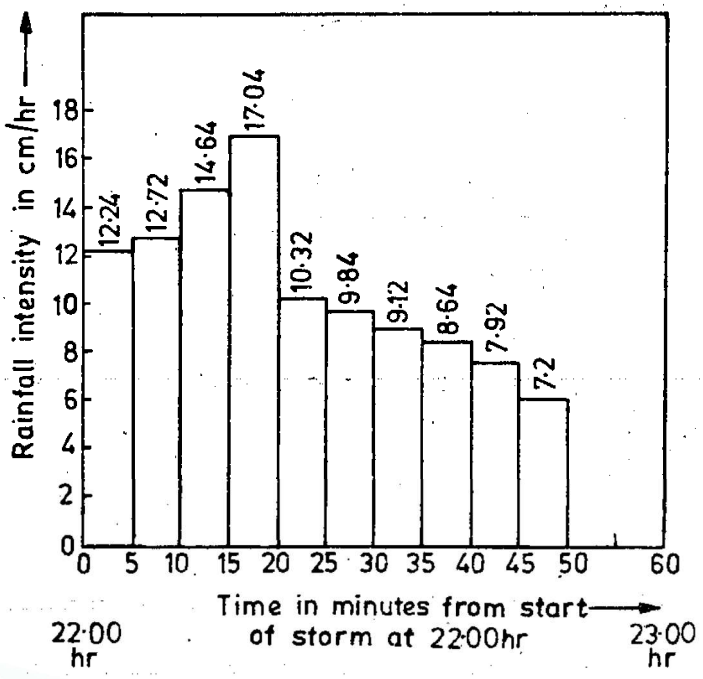

Fig. 7.34. Rain Hyetograph for Example 7.11

col. (12) of this Table. These maximum values are then used to calculate their respective maximum intensities, as shown in Table 7.19.

Table 7.19

Duration in min	*Max. rainfall in mm, from Table 7.18*	*Max. rainfall intensities*	
		in mm/hr $\text{col. (2)} \times \frac{60}{\text{Col. (1)}}$	*in cm/hr*
(1)	*(2)*	*(3)*	*(4)*
05	14.2	170.4	17.04
10	26.4	158.4	15.84
15	37.0	148.0	14.80
20	47.2	141.6	14.16
25	55.8	133.9	13.39
30	64.0	128.0	12.80
35	71.6	122.7	12.27
40	78.8	118.2	11.82
45	85.4	113.9	11.39
50	91.4	109.7	10.97

A curve is now plotted between values of col. (1) and (4) of the above table, so as to obtain the max. I.D. curve, as shown in Fig. 7.35.

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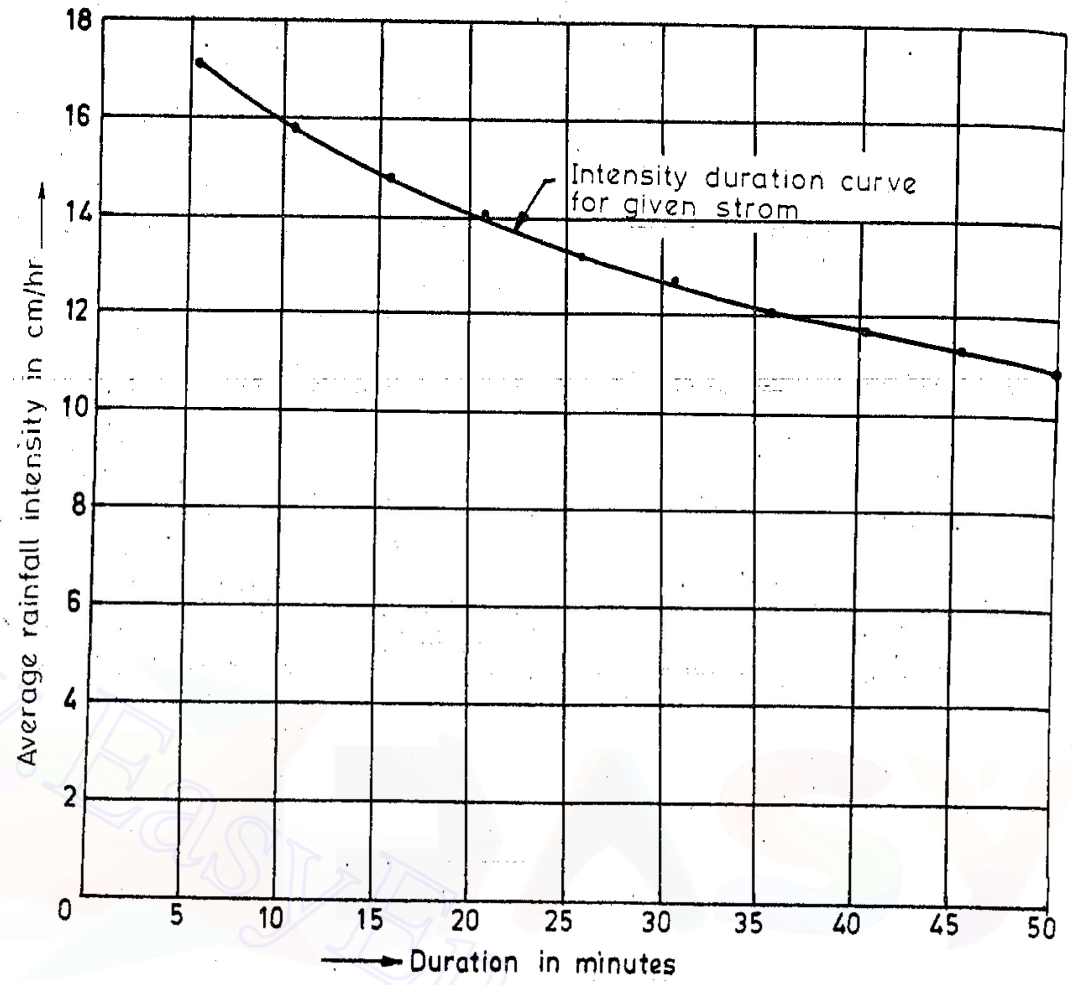

Fig. 7.35. Max. ID curve for Example 7.11.

7.20.2. Equation of an IDF curve. An IDF curve can be represented by an equation of the type :

$$p = \frac{A \cdot T^x}{(t_r + B)^n} \qquad ...(7.19)$$

where p = intensity of rain in cm/h

t_r = duration of rain in h

T = Return period or Frequency of rain in years.

A, B, n and x are all constants for a given catchment, and their typical values for a few places in India are shown in table 7.20

Table 7.20. Typical Values of Constants in eqn. (7.19)*

S. No.	Place in India	Values of constants			
		A	B	n	x
1	Bellary	6.16	0.50	0.694	0.972
2	Bhopal	6.93	0.50	0.189	0.878
3	Chandigarh	5.82	0.40	0.160	0.750
4	Nagpur	11.45	1.25	0.156	1.032
5	Raipur	4.68	0.15	0.139	0.928

* Based on the studies conducted by the Central Soil and Water Conservation Research and Training Institute, Dehradun.

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Isopluvial maps. Instead of providing the standard IDF curves, indicating extreme rainfall intensities of different durations for different frequencies (T), IMD has standardised *isopluvial maps*- indicating extreme point rainfall values of different durations and for different frequencies over the whole country. On an Indian geographical map, therefore, *lines of equal rainfall* (called *isopluvial lines*) have been drawn, each for a particular duration and a particular frequency. Hence, different isopluvial charts indicating extreme rainfall depths for different durations of 15 min, 30 min, 45 min, 1 hour, 3 hour, 6 hour, 9 hour, 12 hour, 15 hour, etc. each for 2, 5, 10, 25 and 50 years of return period are available with IMD as well as CWC, and can be used for specific needs. 1 hour isopluvial charts of 2, 5, and 50 years return period are also enclosed in this book as plates 7.1, 7.2 and 7.3, respectively.

7.21. Probable Maximum Precipitation (PMP) Curves

The probable maximum precipitaion (PMP) may be defined as the *maximum* depth of precipitation for a given duration that may *possibly occur* on a given catchment at any time of the year. Hence, such a precipitation would result from the possible severest storm, that may result from the worst possible combinations of hydrological conditions in the area. Such a worst possible storm is called **probable maximum storm** (PMS), and will be used for designing large hydraulic structures, as discussed in article 7.51.

As is evident from its very definition, the PMP can be estimated by maximising the different parameters, such as *wind velocity, humidity*, etc. of an observed severe storm, over the given basin. Occasionally, when enough storm data for a given basin is not available, PMP can be estimated by adopting a severe storm over a neighbouring catchment, and transposing it to the catchment under consideration. Detailed discussions of such exercises is beyond the scope of this book, as the same falls in the domain of Hydro meteorology, and any standard book on that subject may by referred to, if needed.

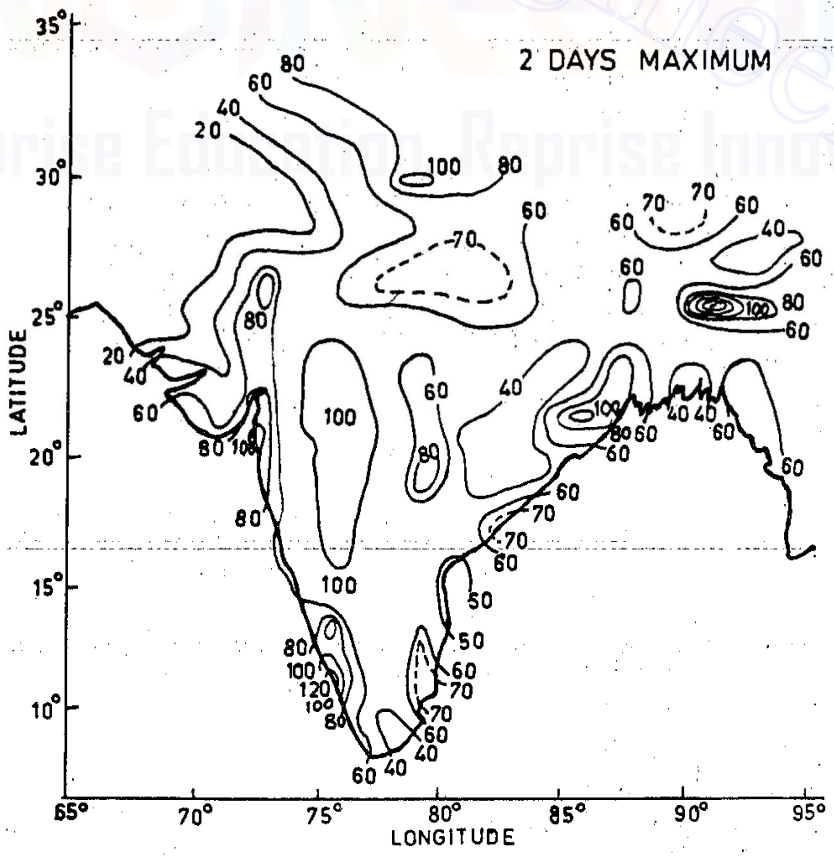

Fig. 7.36. (*a*) Heaviest rainfall in cm – 2 days duration.

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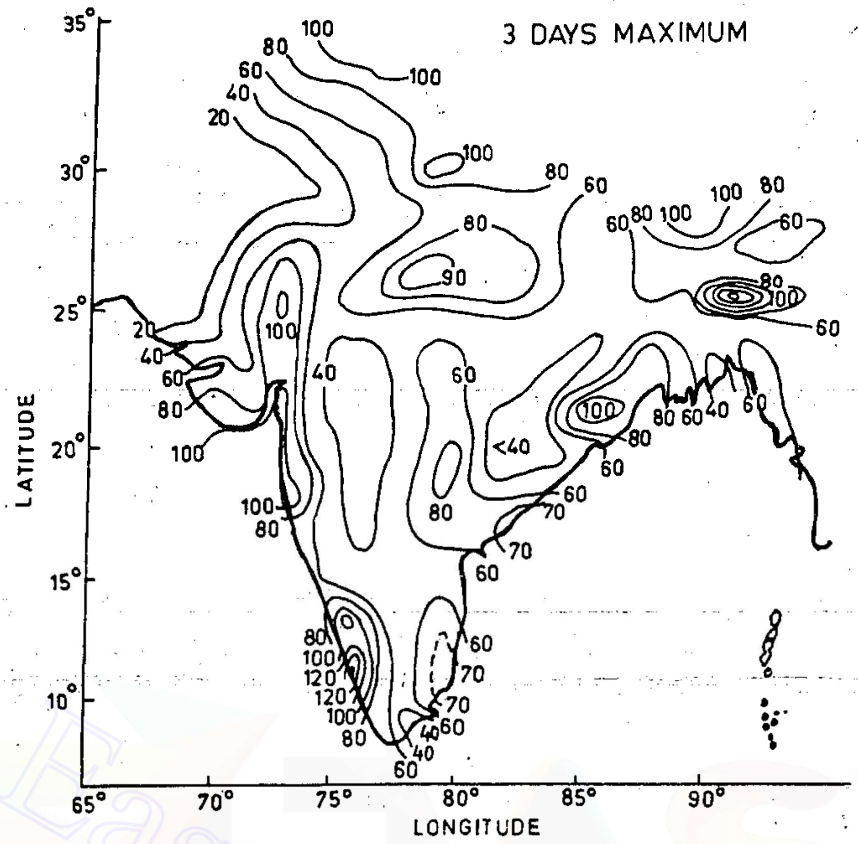

Fig. 7.36. (*b*) Heaviest rainfall in cm – 3 days duration.

Indian Meteorological Department (IMD) has prepared guiding curves, showing isohyetal lines for maximum storm depths (absolute for any frequency), that may result from the severest storm of the given duration. Fig. 7.36 (*a*) and (*b*) shows such charts for 2 days and 3 days maximum rainfall, respectively. A possible value of maximum rainfall at any given city, having 2 days or 3 days duration, can, thus be readout from these charts, easily.

SNOWFALL AND SNOW MELT

7.22. Snowfall and its Measurement

Snow is the second important part of precipitation, after rainfall. It usually occurs at high altitudes, and particularly during the winter season. It may or may not be accompanied by rainfall, depending upon the coldness of the area. In India, heavy snowfalls occur in the upper reaches of the *Himalayan rivers*, such as Ganga, Yamuna, Brahmaputra, etc., during December, January and February, as *winter precipitation.*

Snowfalls are generally responsible for giving continuous delayed supplies in the rivers, rather than giving immediate flood flows, as produced by intense rain storms. As a matter of fact, snow generally goes on falling and accumulating over the river valley and the basin, during the winter months, without giving any immediate streamflow; but later, with the onset of spring season in late March or April, when the temperature of the region starts increasing, the snow starts melting, leading to runoff in the river below. Depending upon the available snow depth, runoff continues during the summer season, and possibly till the arrival of the next winter, when snow again starts falling and accumulating. The process proves a great boon, as it ensures perennial river supplies, so very necessary for us.

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Snow usually falls in the form of very light *snow flakes*, having density as low as 0.004. When such snow flakes fall and get deposited over the ground, one over the other, they get compressed, leading to increase in its density. As the depth of accumulated snow increases with the passage of time, it gets consolidated more and more, with increase in its density. Freshly fallen snow (last 24 hours), therefore usually has a density of 0.1, while the heavy snow accumulations may have as high a density as 0.4 to 0.6. When the accumulated snow depth in the river valley becomes quite high, say more than 60 to 75 m or so, it begins to move down the valley, like a flow in the stream. This snow-body is called a **glacier**, and may have as high a density as 0.9. Such a glacier may, sometimes, cause severe floods in the river below, if by chance, it happens to burst. As for example, in August 1959, a glacier burst in the remote mountain valley in the *Karokorum* mountain ranges (K_2) in Kashmir, causing a flood rise of 30 m at a distance of more than 40 km from the point of outburst. Such sudden **glacier floods**, locally called *Jokulhlaup*, may cause heavy loss of life and other valuables, and needs to be predicted well in advance. The discharge hydrograph from such a glacier flood resembles a normal storm unit hydrograph, except that the flood rise here is rather slow, and the fall is abrupt.

Since the snowfall plays an important role in supplying continuous flow in a river, and also chance floods, it is necessary to measure the snowfall, and, thus, to estimate its accumulated depths and runoff potentials.

Snowfall measurements can be divided into two categories, viz.

(*i*) *measurement of snowfall*, caused by each individual snow storm or daily snow-fall; and

(*ii*) *snow surveys* to measure the accumulated snow depth deposited during the entire winter season, or at smaller intervals and to determine its density ; and hence to compute the water equivalent of the accumulated snow for estimating runoff.

Both the above types of measurements are discussed below :

7.22.1. Snow Gauging. The *snow gauging* can be carried out in the following two ways :

(*i*) by measuring the depth of snow, actually collected on an open *level platform* ; and

(*ii*) by measuring the depth of snow, caught in a *shielded gauge vessel*, after levelling the accumulated snow without compressing.

The above snow gauging is recorded every day, during the entire snowfall season, usually at 8.30 A.M., as is done at a rain gauge station.

(*i*) The **open platform method** is more suitable in the absence of strong winds, and in areas where snowfalls occur unaccompanied by rainfalls. The method consists of constructing a permanent level pucca platform, 2 m in square, at a suitably located gauge site. The top of this cement concrete platform (75 mm thick C.C.) is kept at the ground level, and a graduated wooden square rod, 50 mm in section, 4 m long is usually installed at the centre of this platform. Lower 1 m length of this wooden rod is embedded into the ground and the pucca platform ; while, the upper 3 m length of the rod is exposed and graduated on all the four sides, for taking *stake* readings. Evidently, the reading at

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